CS 225

Data Structures

Volodymyr Kindratenko

Slides courtesy of Wade Fagen-Ulmschneider

Hashing

Goals:

We want to define a **keyspace**, a (mathematical) description of the keys for a set of data.

…use a function to map the **keyspace** into a small set of integers.

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K

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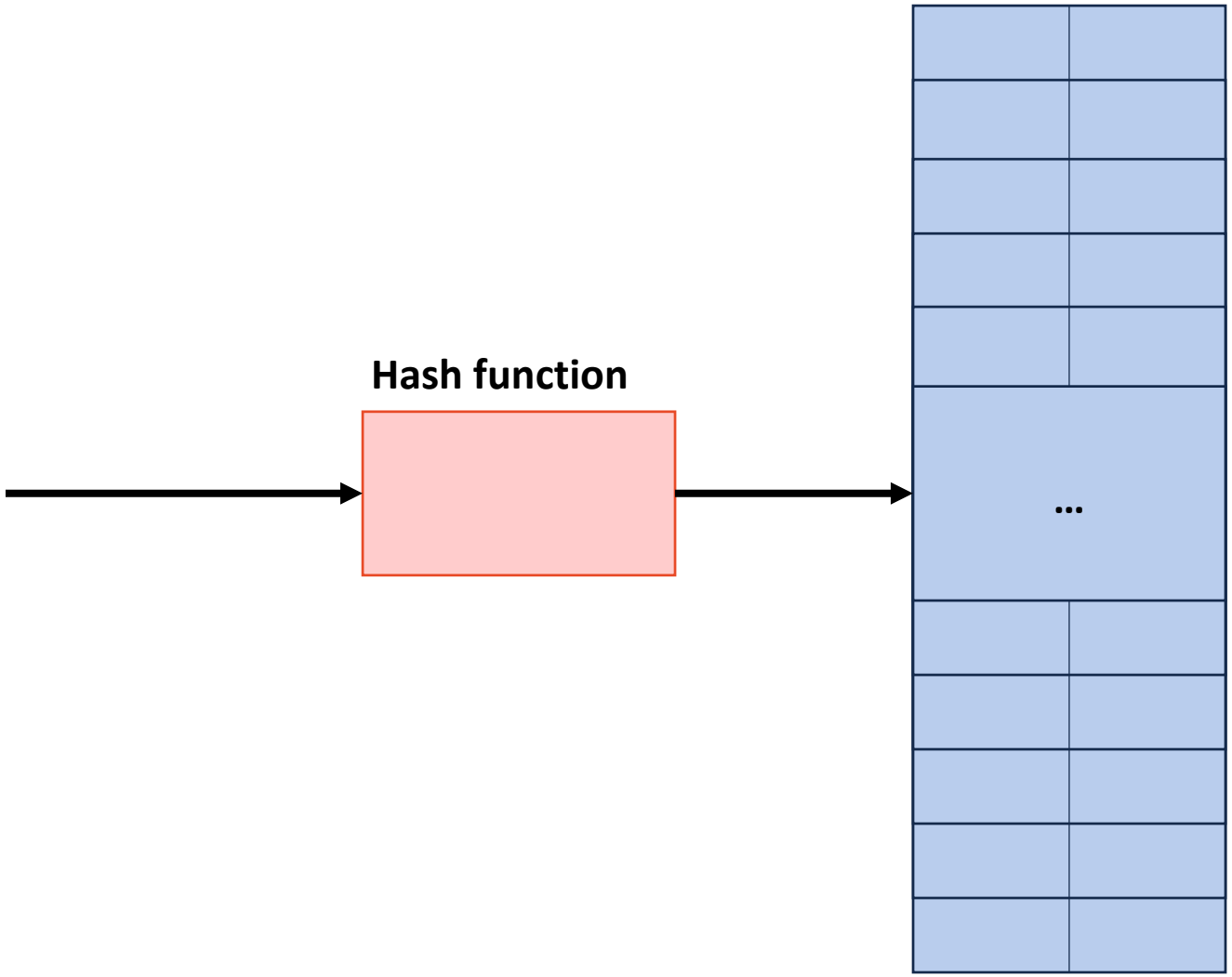
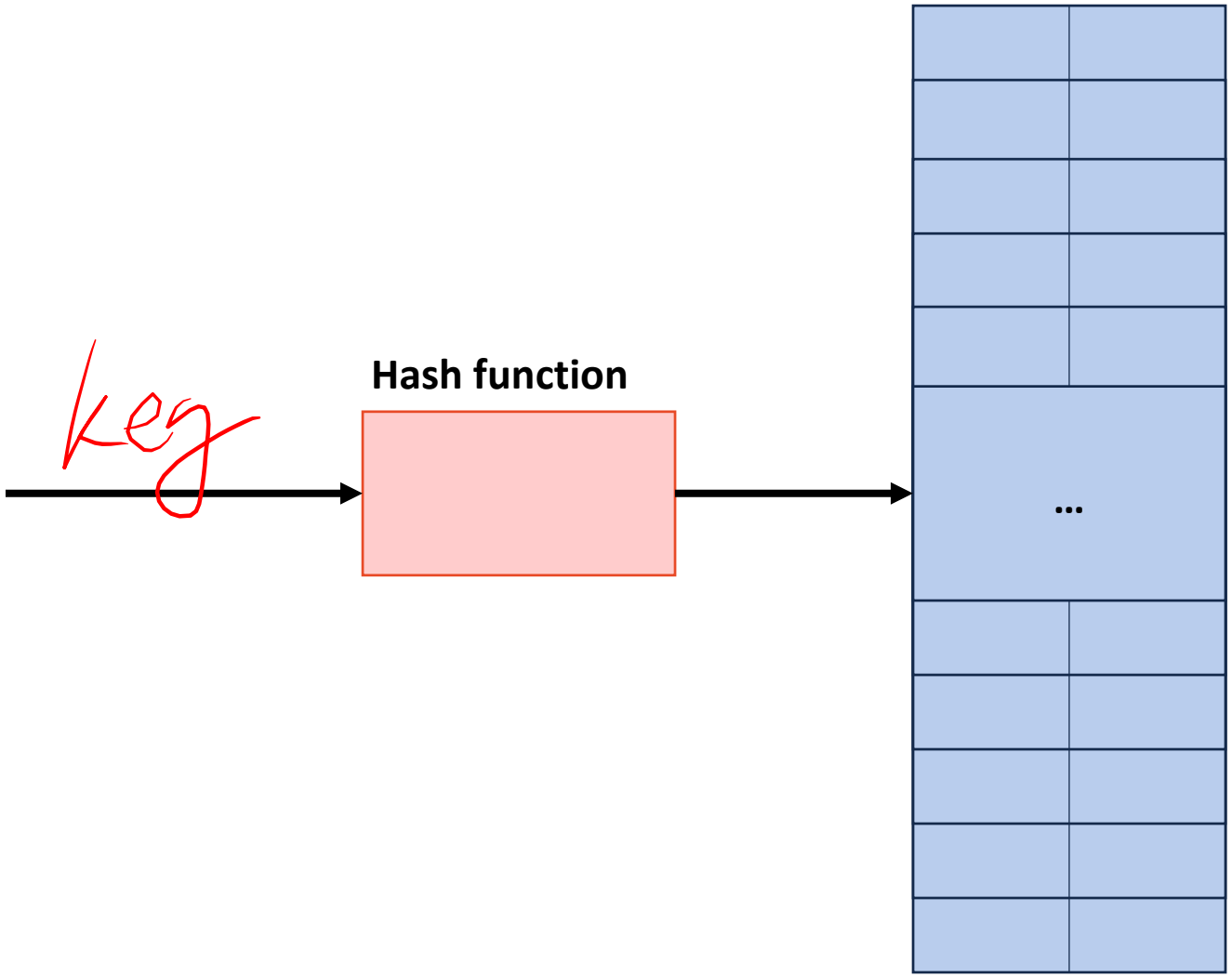
h(key) : key → N

Hashing



Hashing



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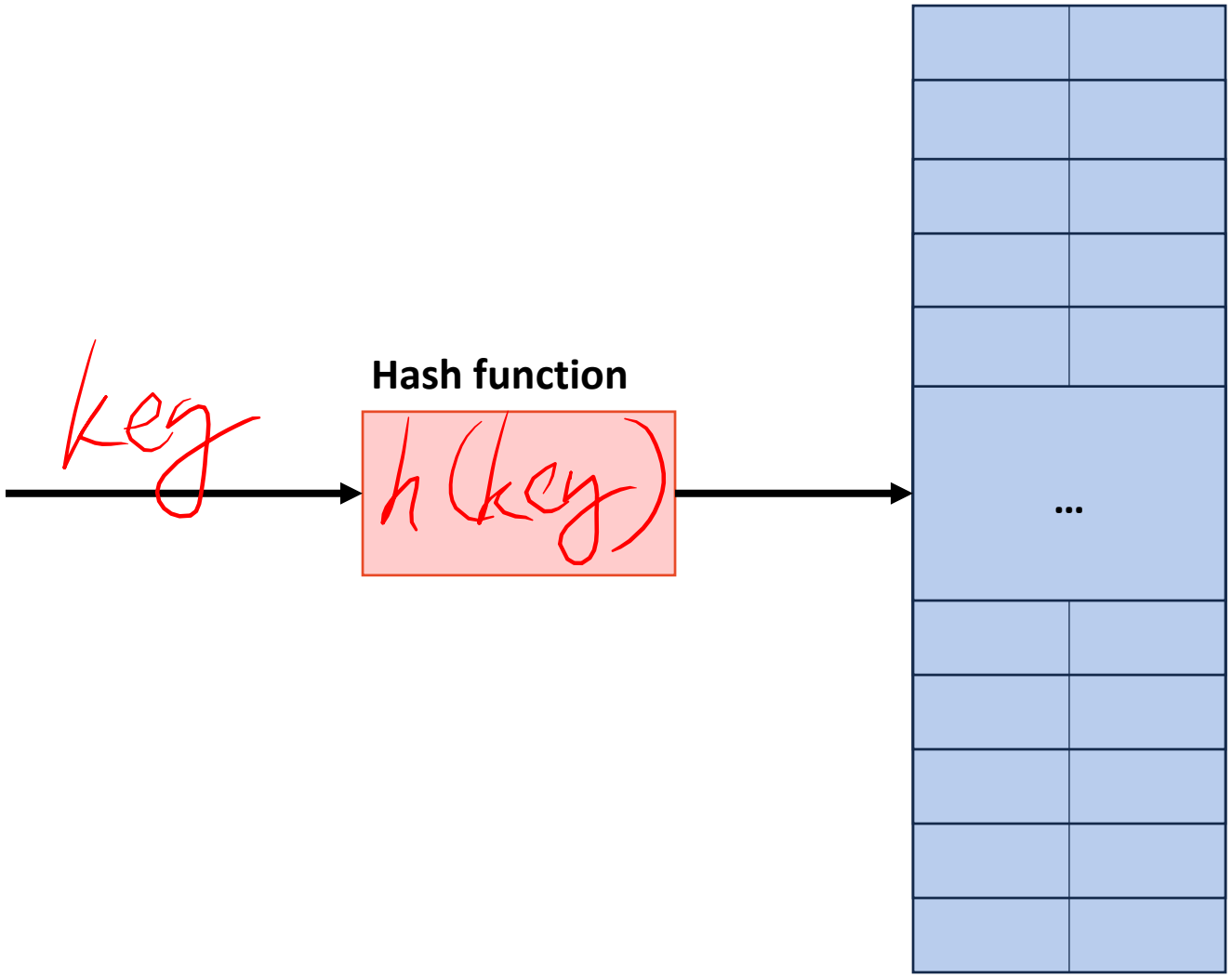

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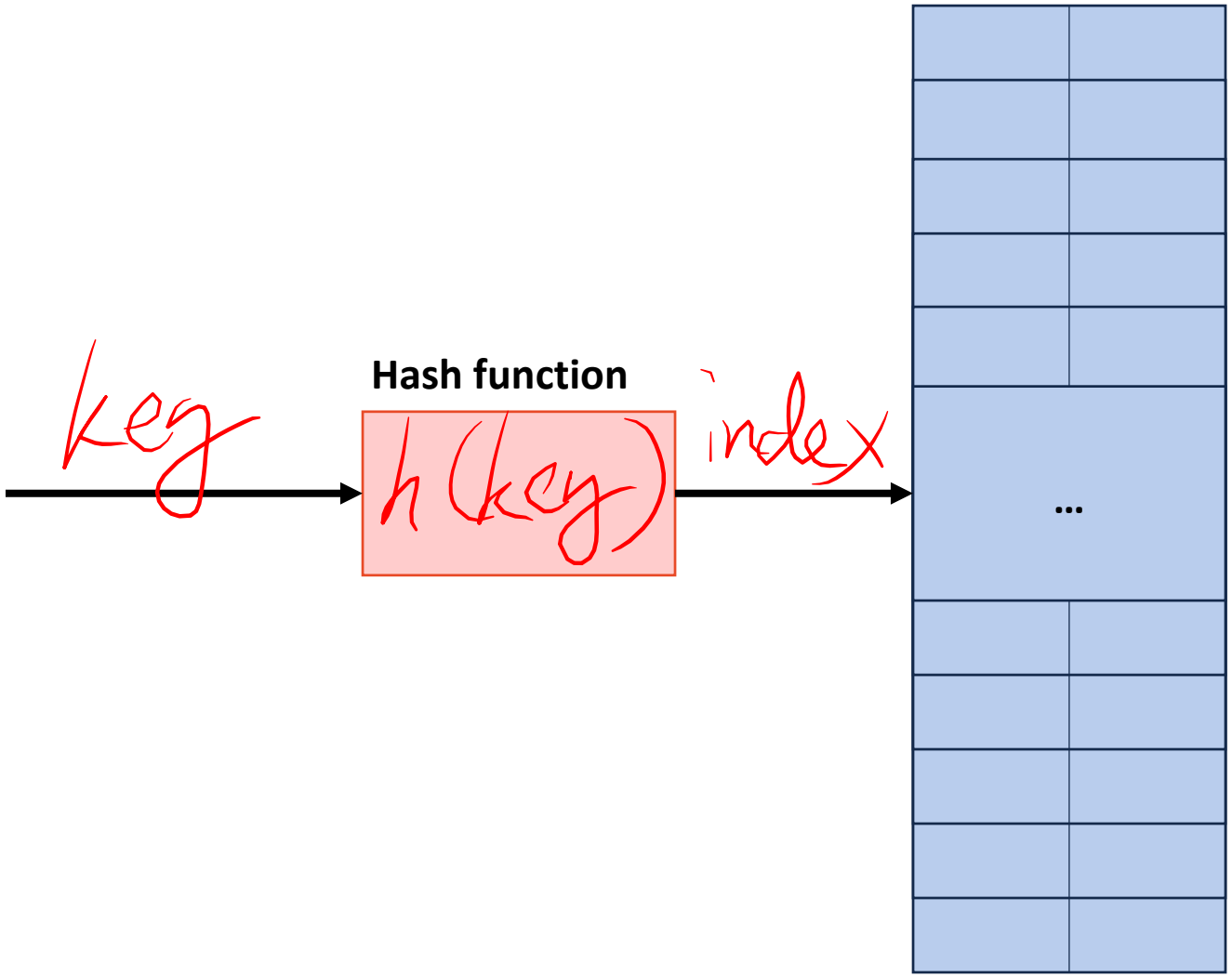

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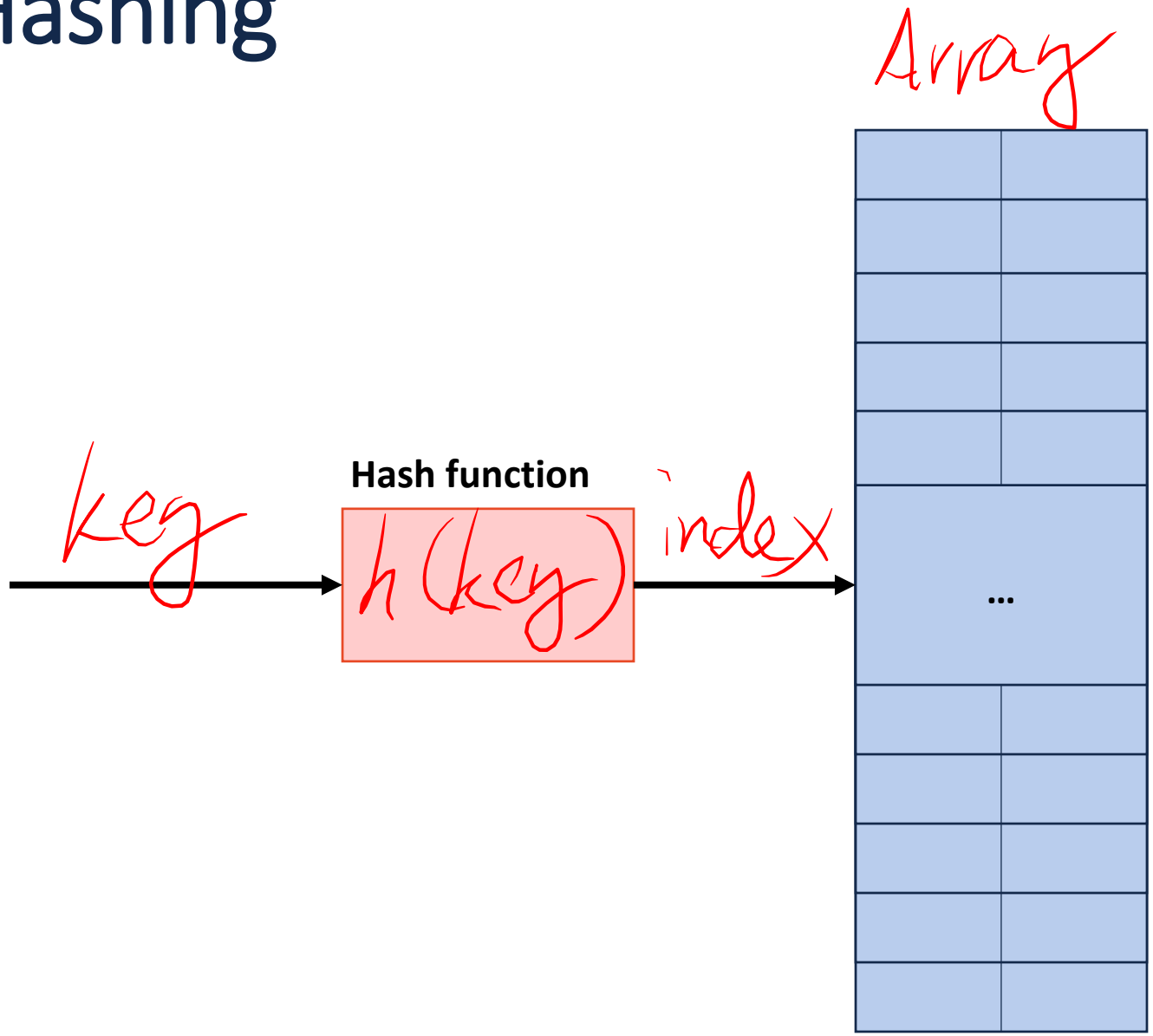

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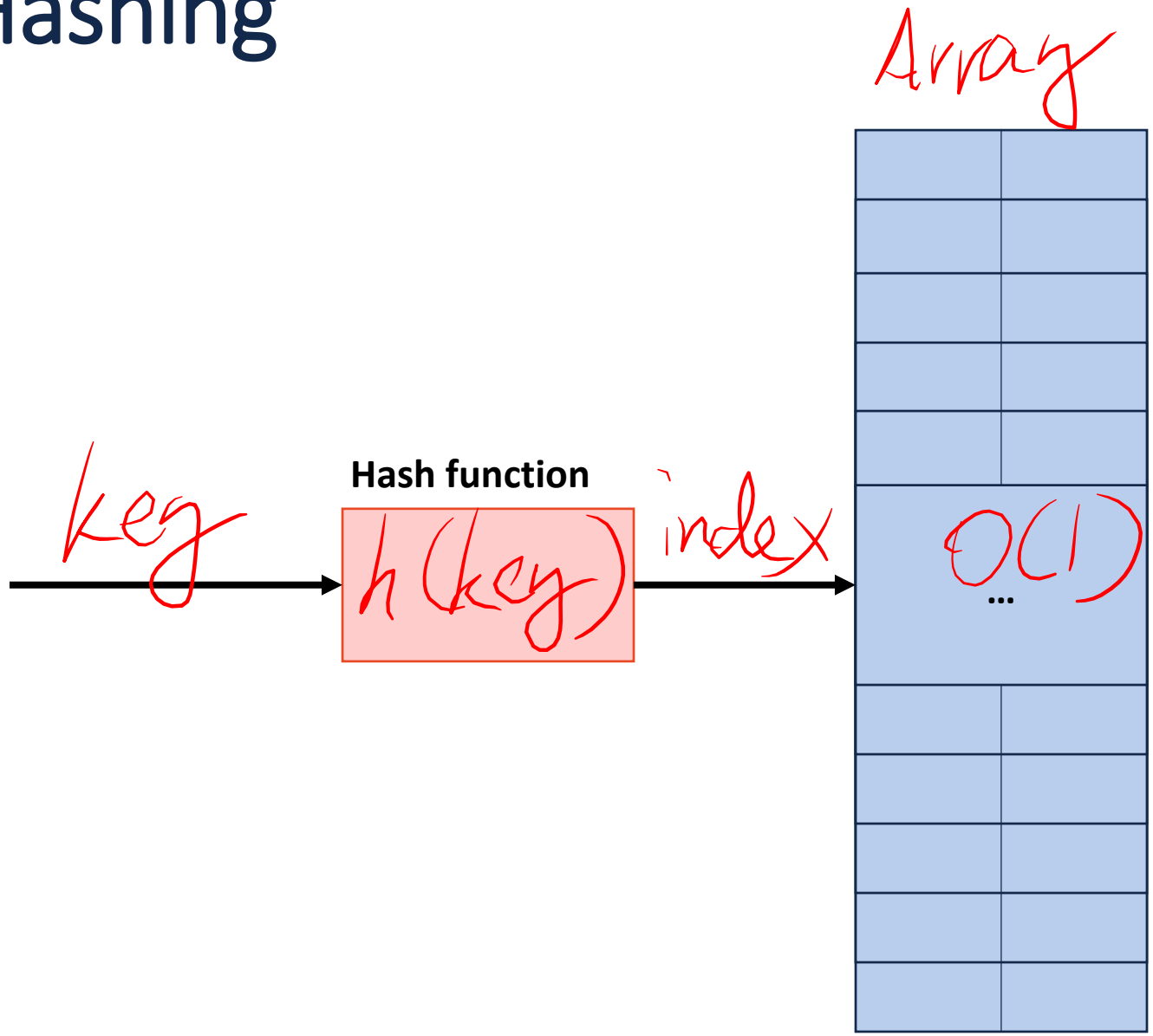

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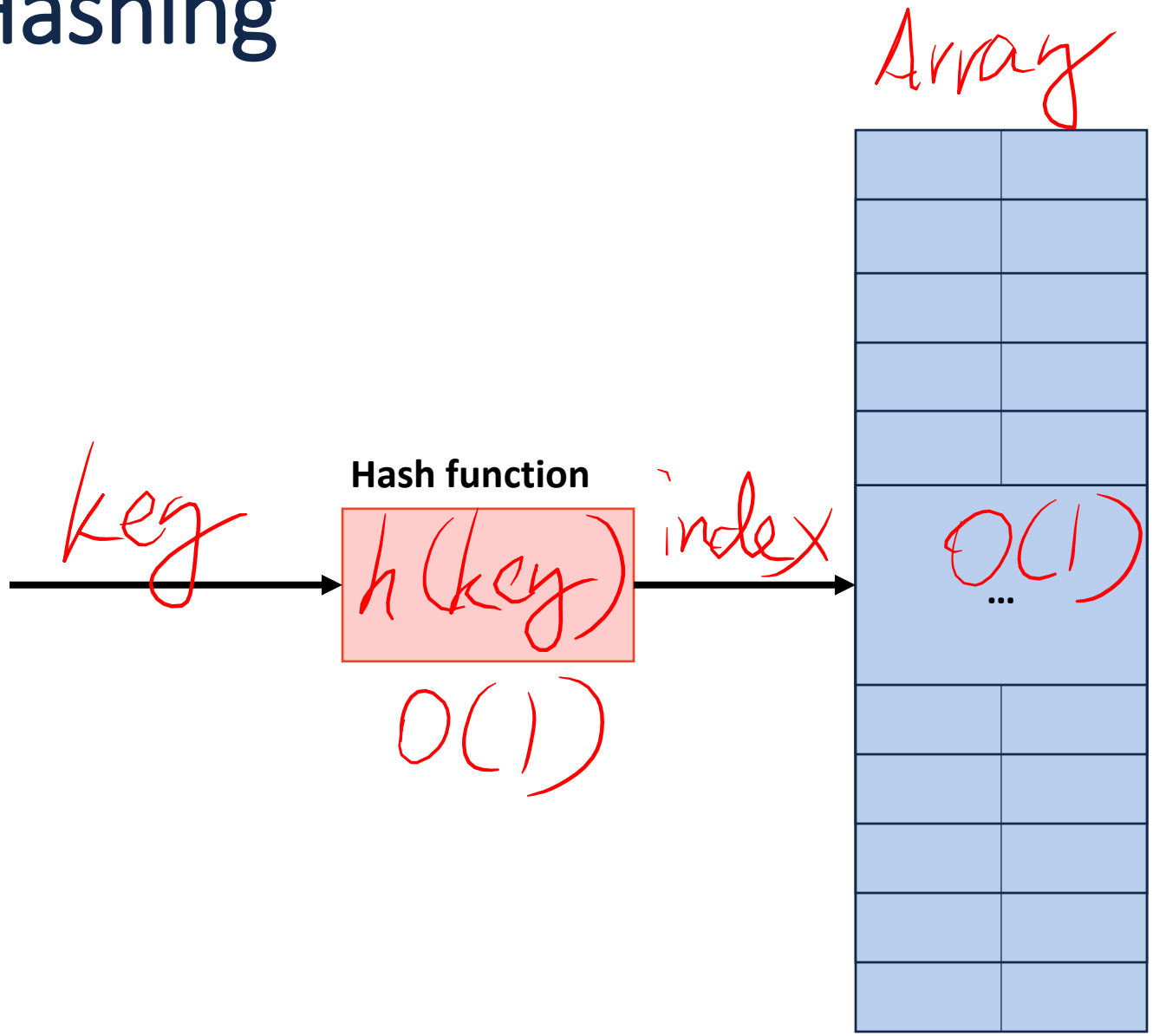

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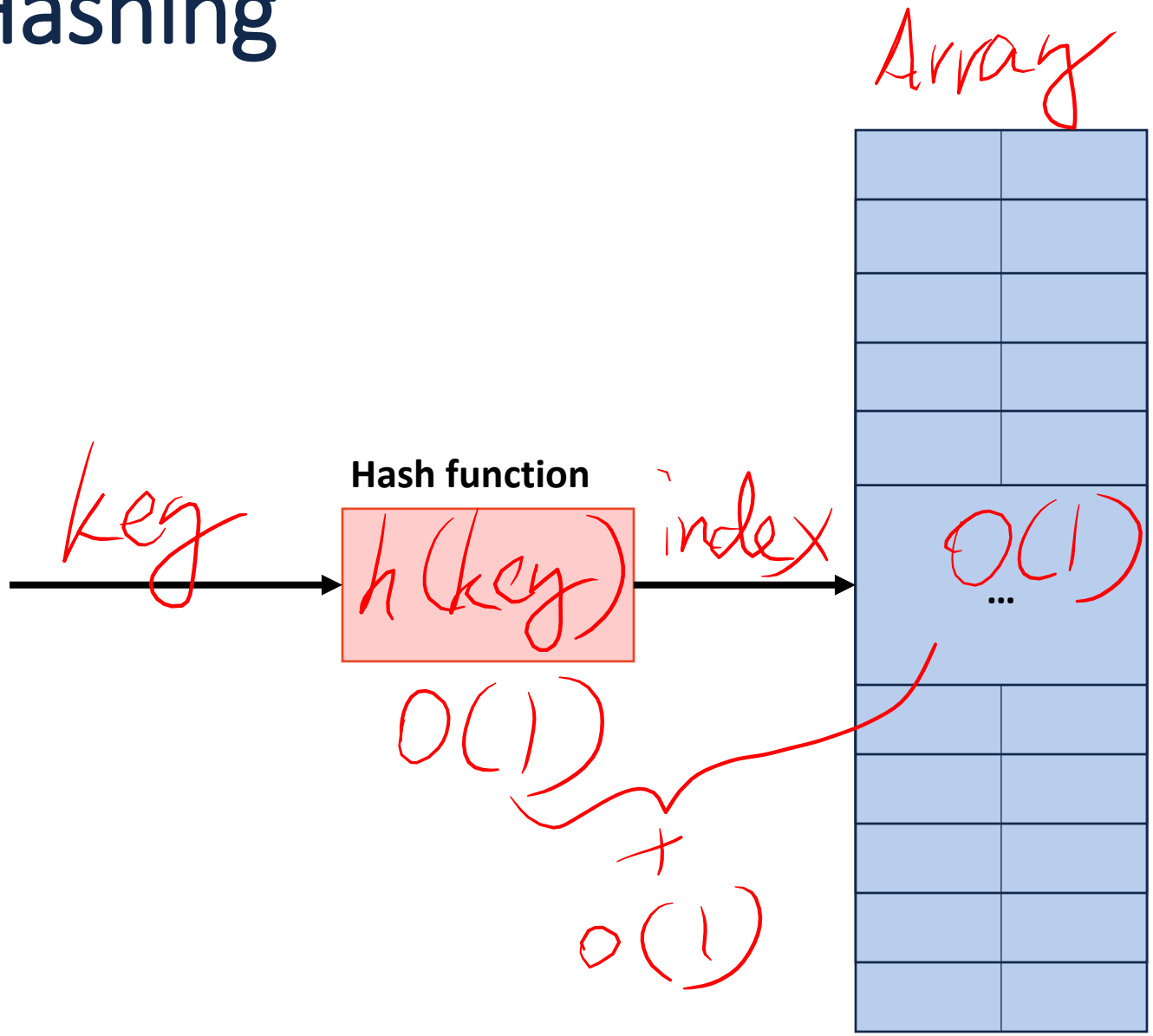

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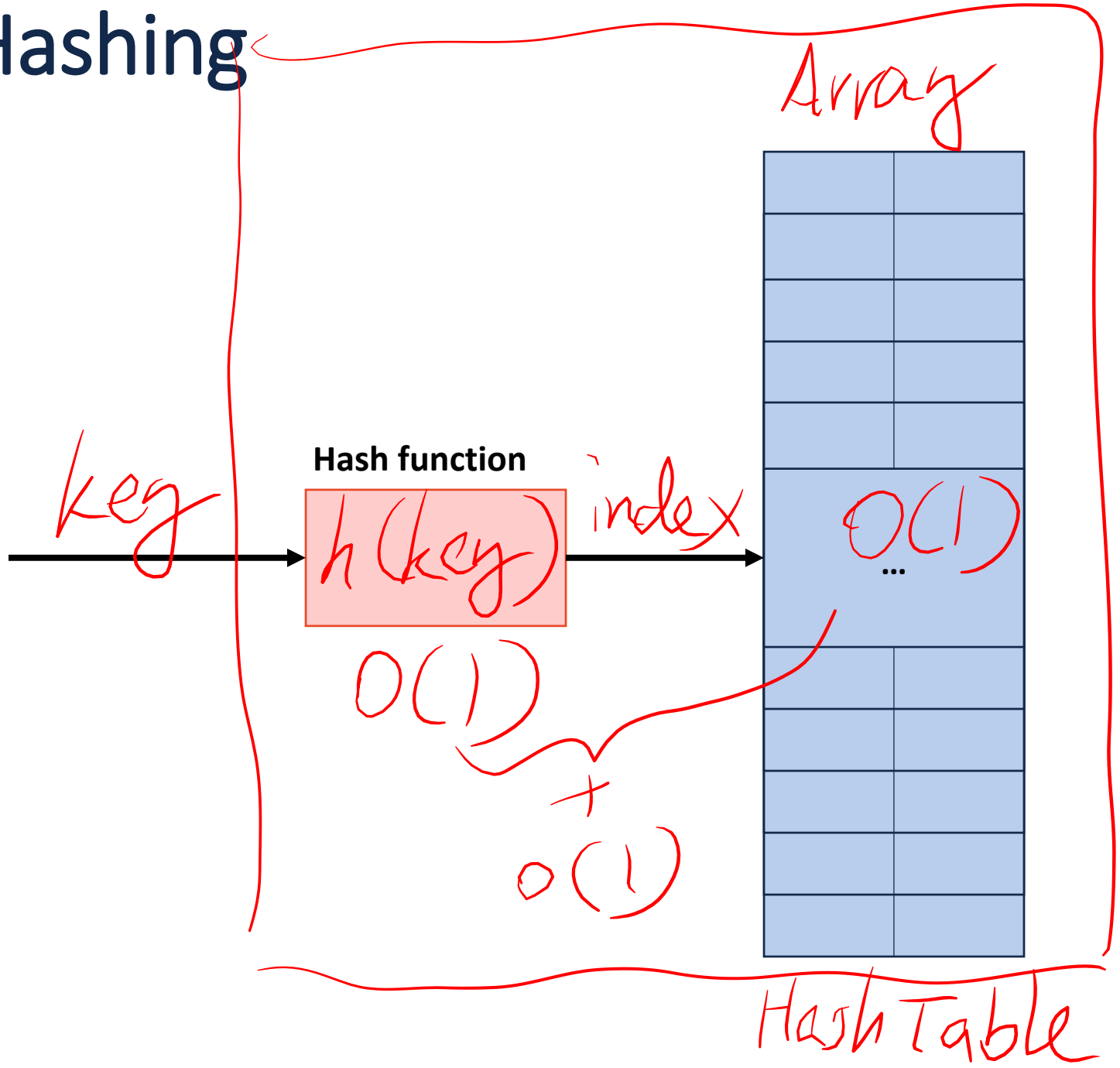

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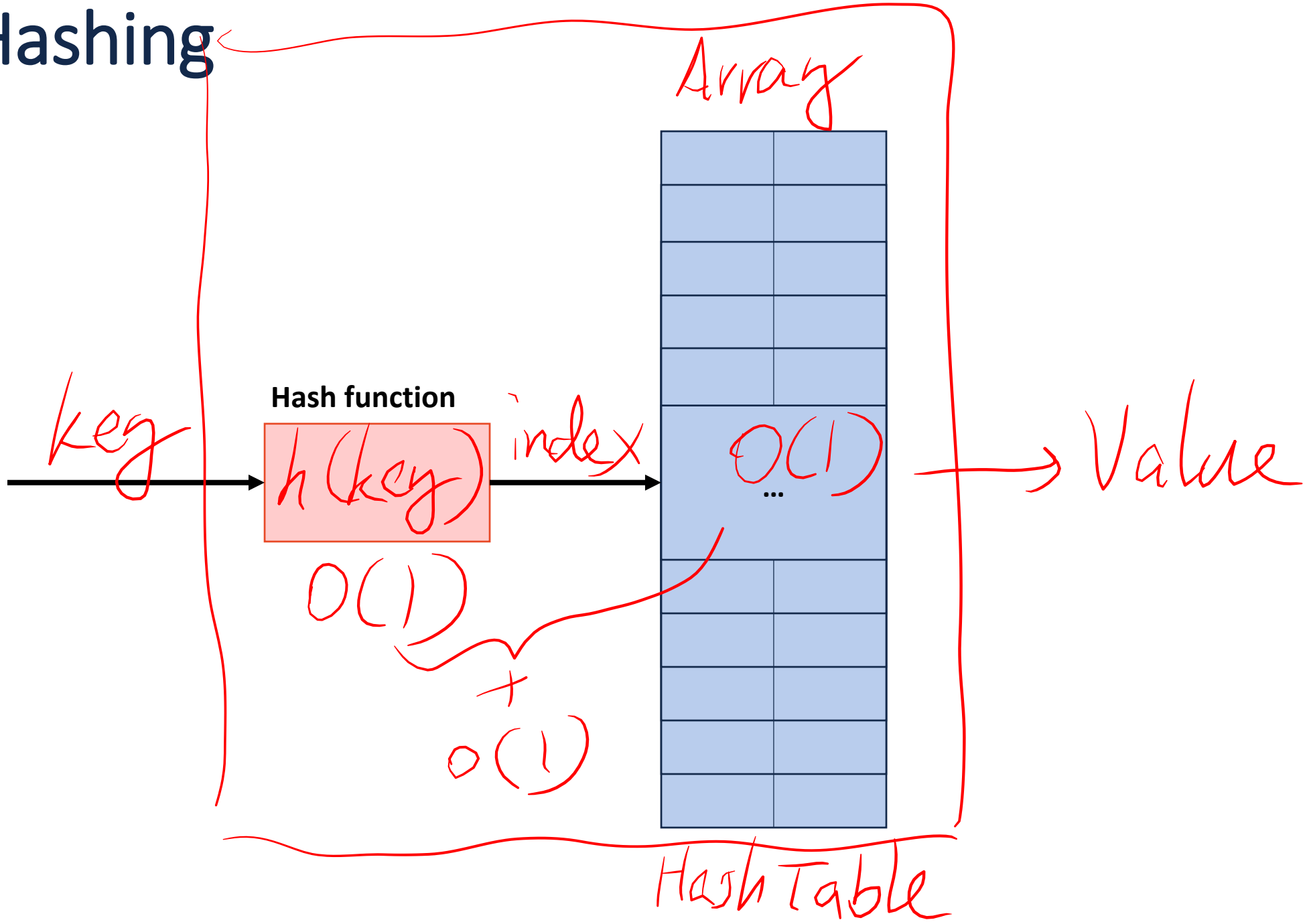

A Hash Table based Dictionary

Client Code:

```
1  Dictionary<KeyType, ValueType> d;
2  d[k] = v;
```

A **Hash Table** consists of three things:

1.

2.

3.

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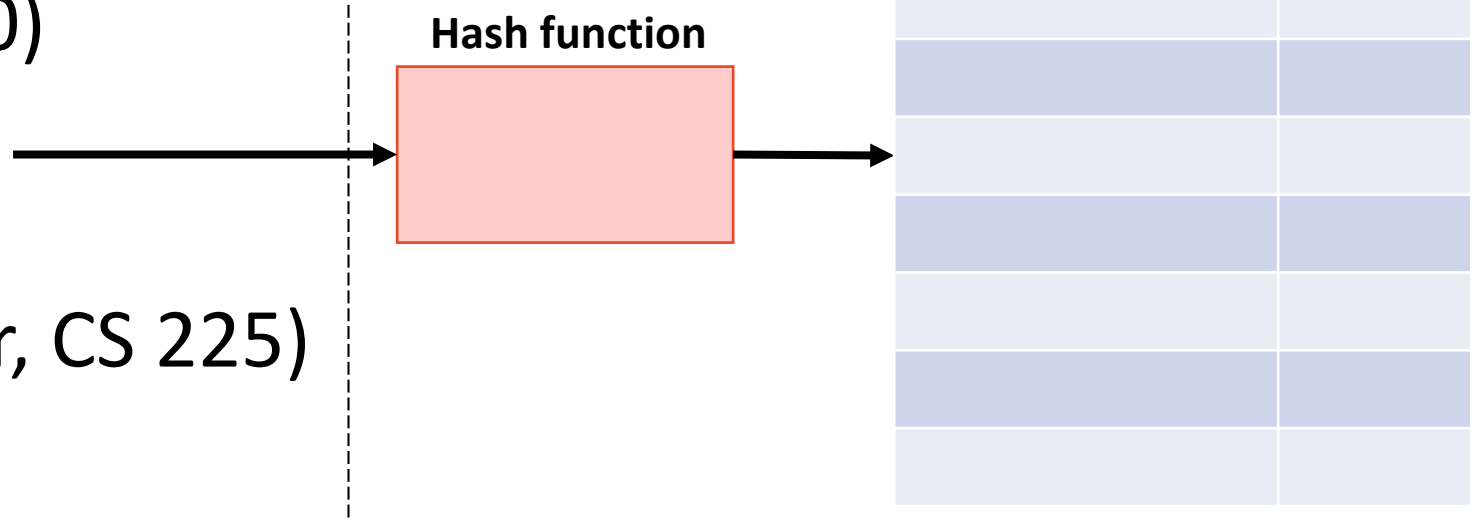
A Perfect Hash Function

(Angrave, CS 241)
(Beckman, CS 421)
(Cunningham, CS 210)
(Davis, CS 101)
(Evans, CS 126)
(Fagen-Ulmschneider, CS 225)
(Gunter, CS 422)
(Herman, CS 233)



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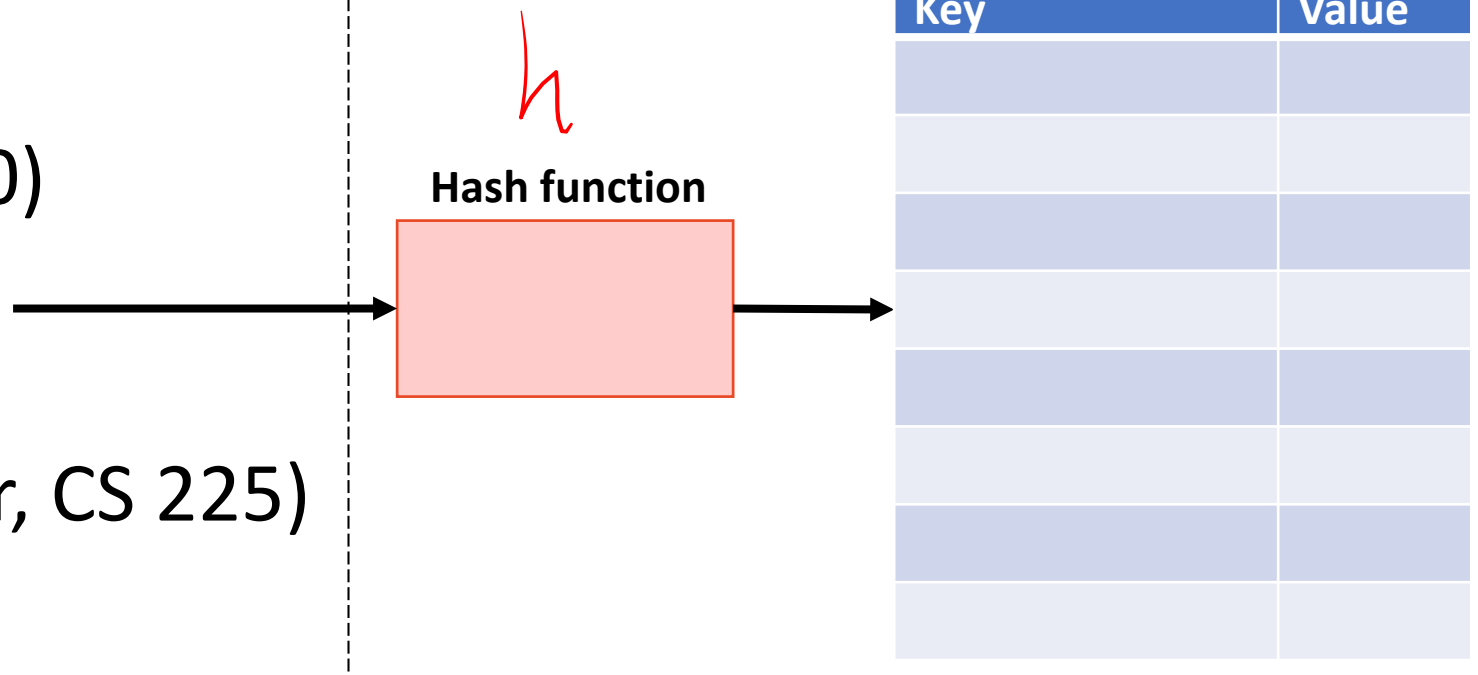

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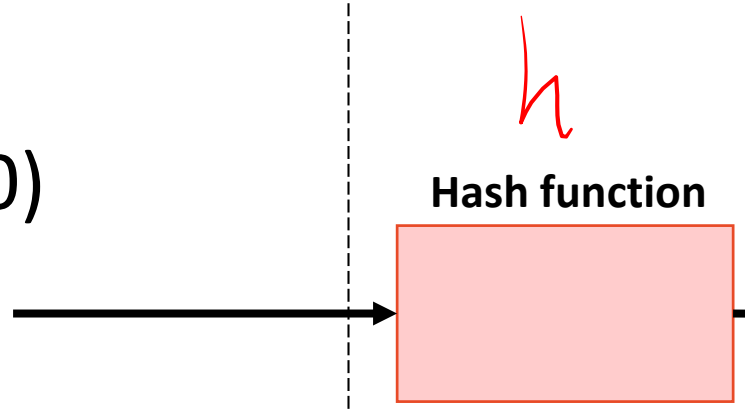

Key	Value
0	Angrave
1	Beckman
2	
3	
4	
5	
6	
7	

index

A Perfect Hash Function

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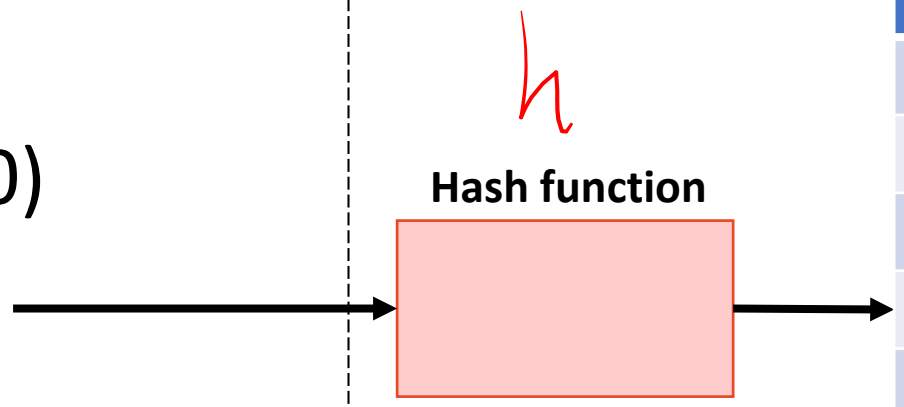

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A Perfect Hash Function

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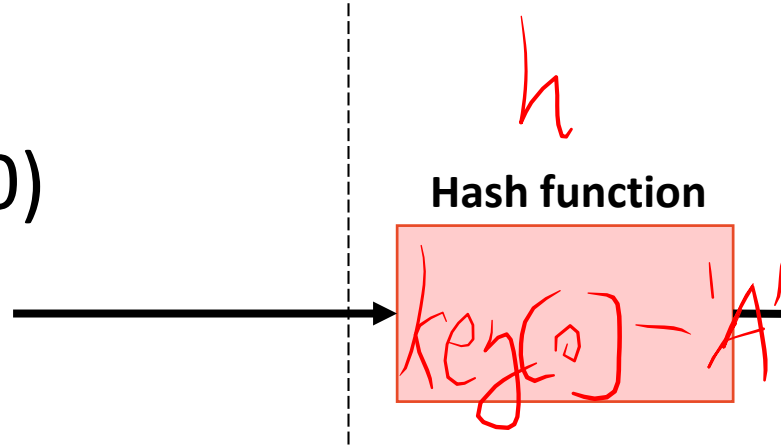

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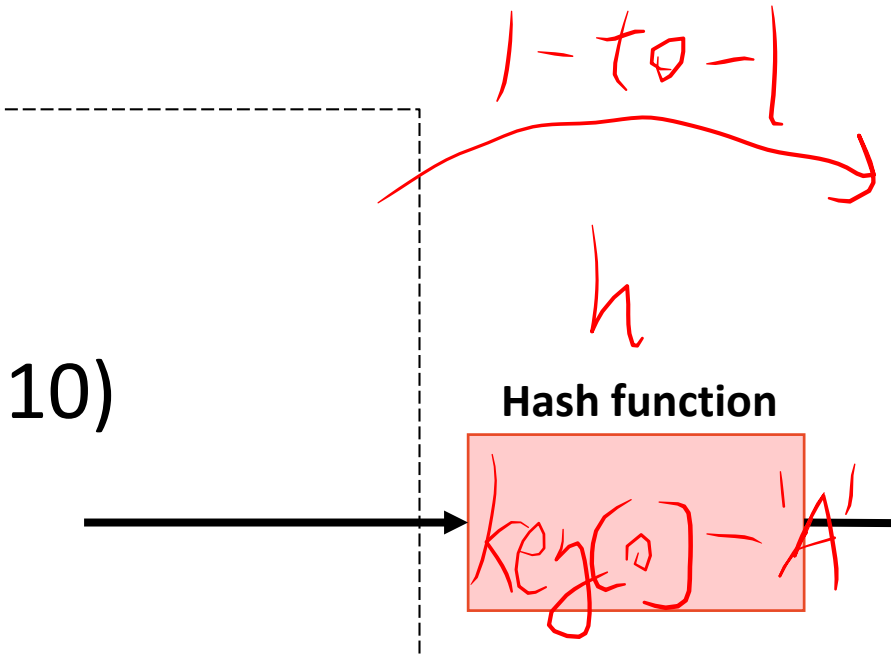

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5	
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7	

A Perfect Hash Function

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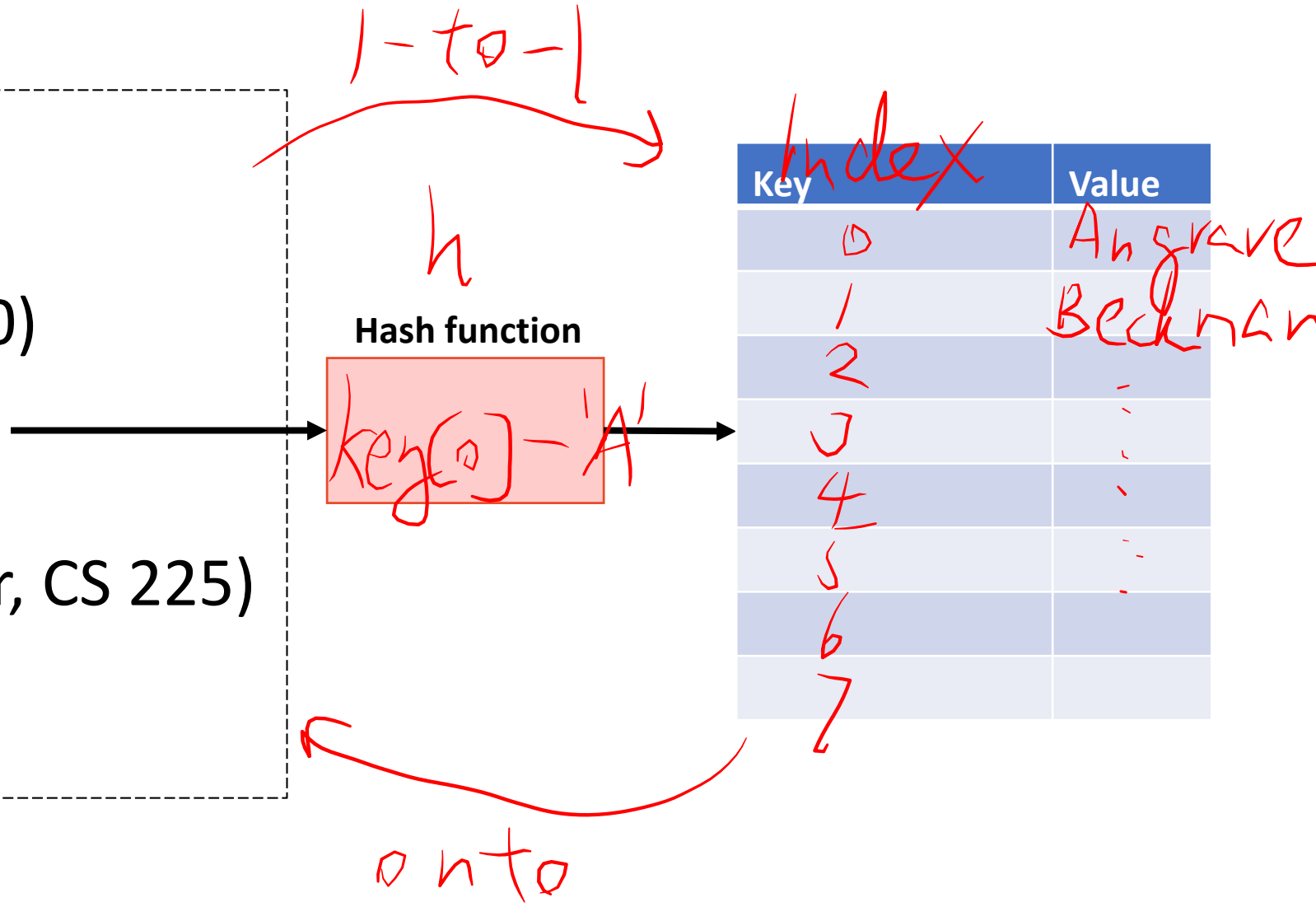

A Perfect Hash Function

bijective function

(Angrave, CS 241)
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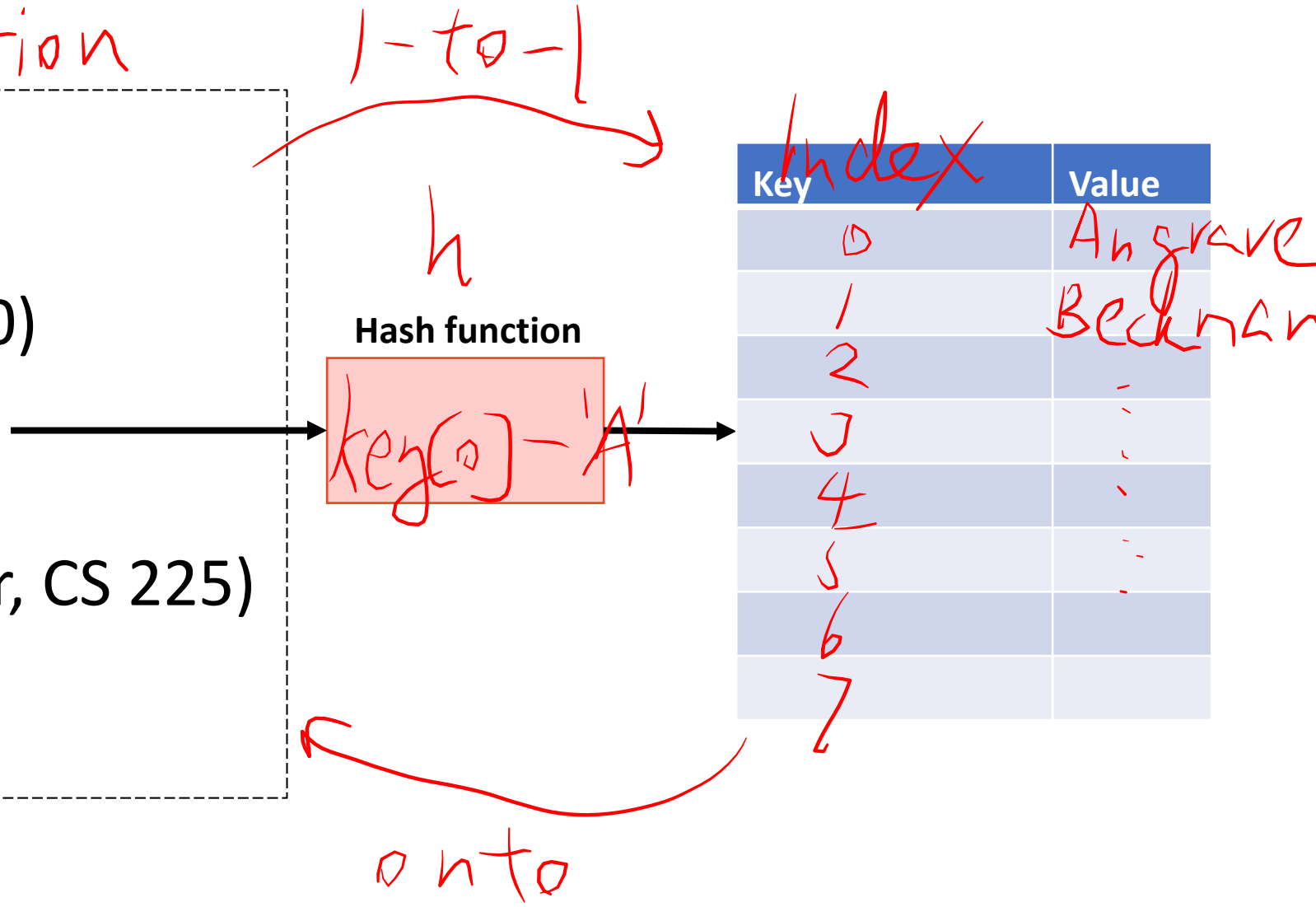

A Perfect Hash Function

bijective function

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(Herman, CS 233)

1-to-1

h

Hash function

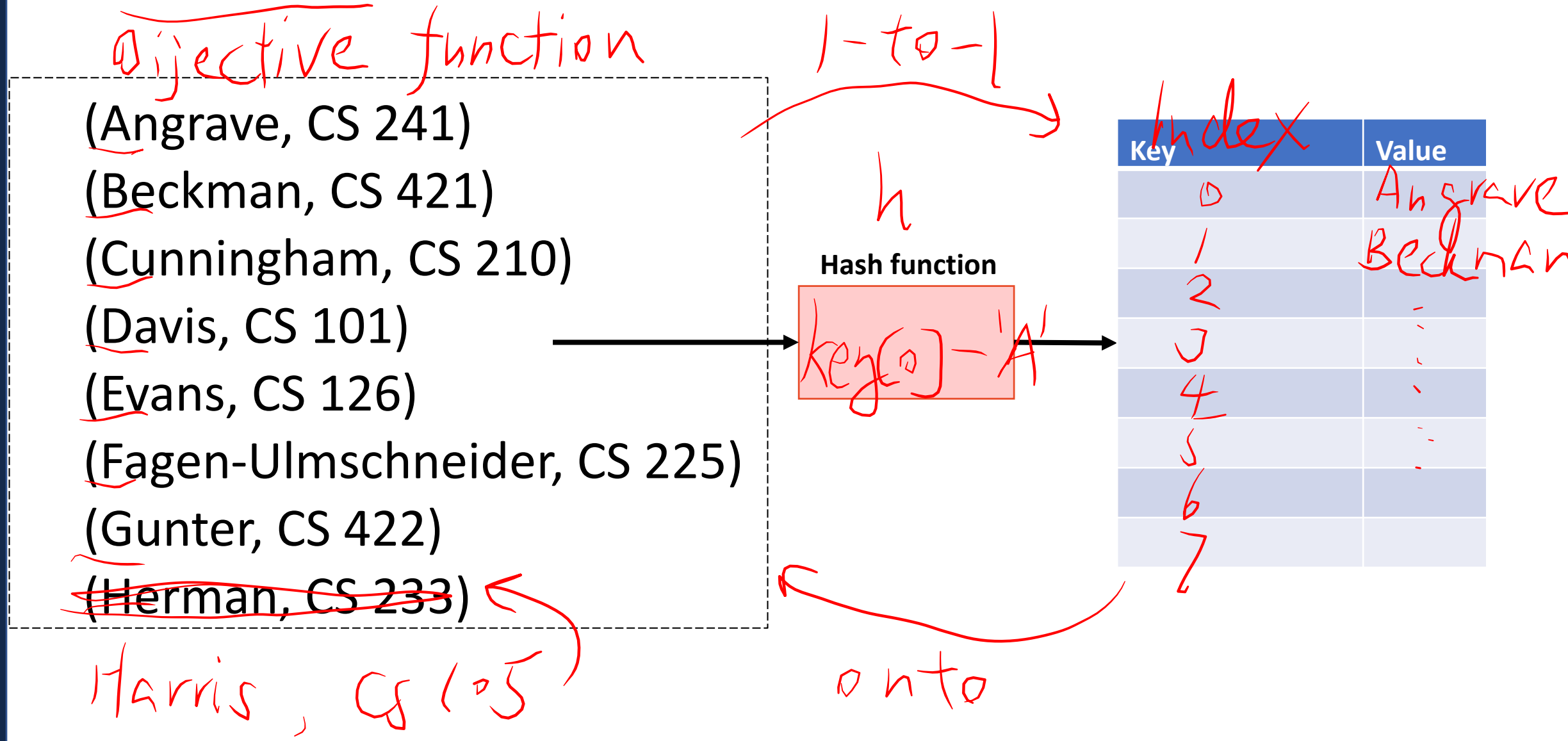
key[0] - 'A'

onto

Key (index)	Value
0	Angrave
1	Beckman
2	
3	
4	
5	
6	
7	

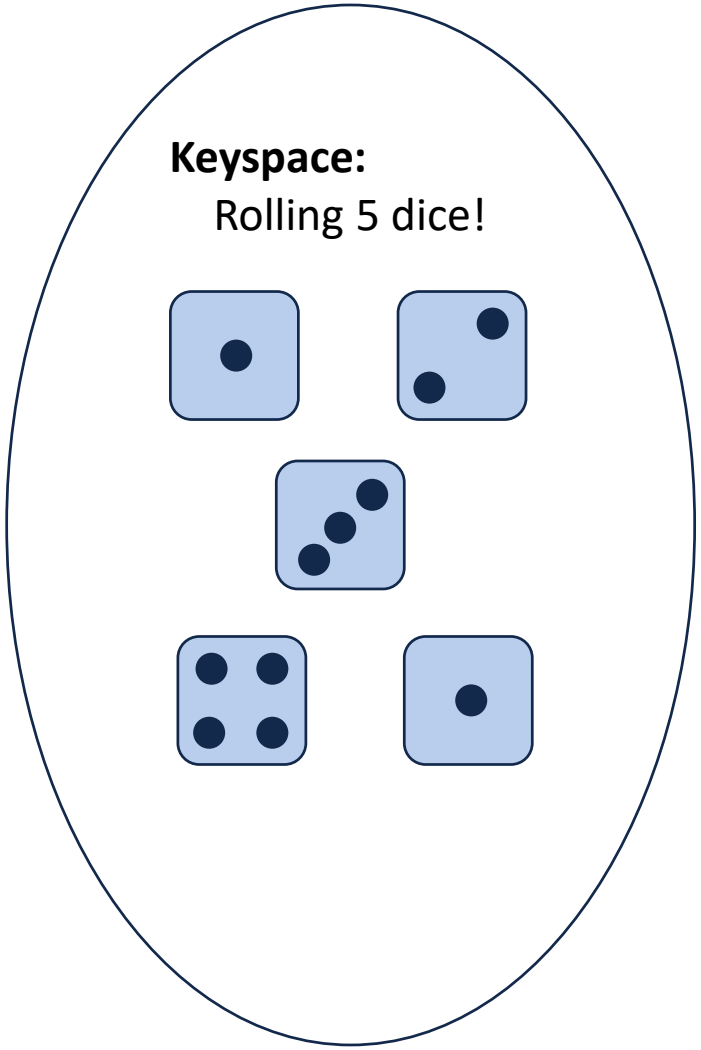
Harris, CS 105

A Perfect Hash Function



A Perfect Hash Function

https://www.random.org/dice/?num=5



Key	Value
0	
1	
2	
3	
4	
5	
6	
7	
8	
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11	
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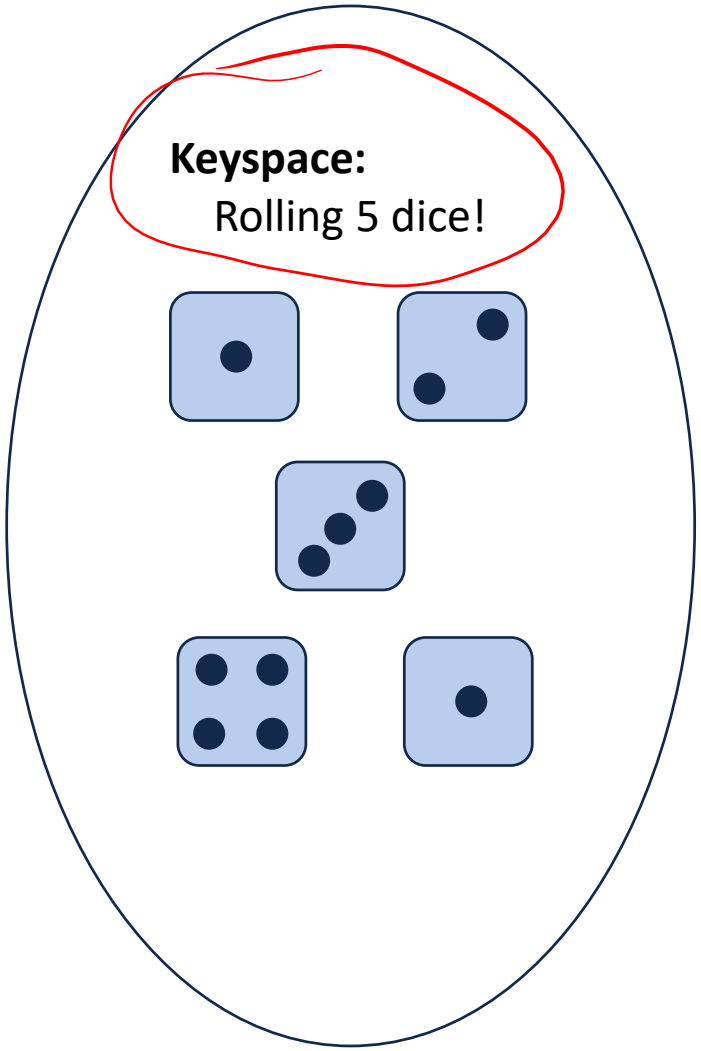
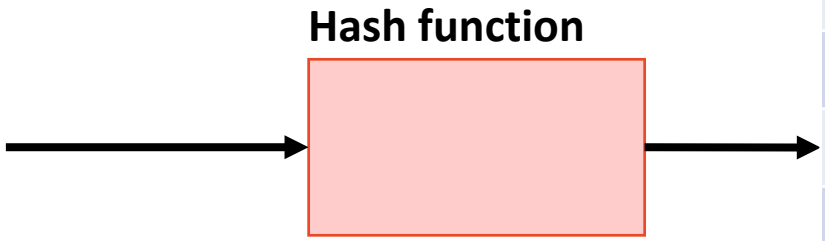
13	
14	
15	

A Perfect Hash Function





Key	Value
0	
1	
2	
3	
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5	
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7	
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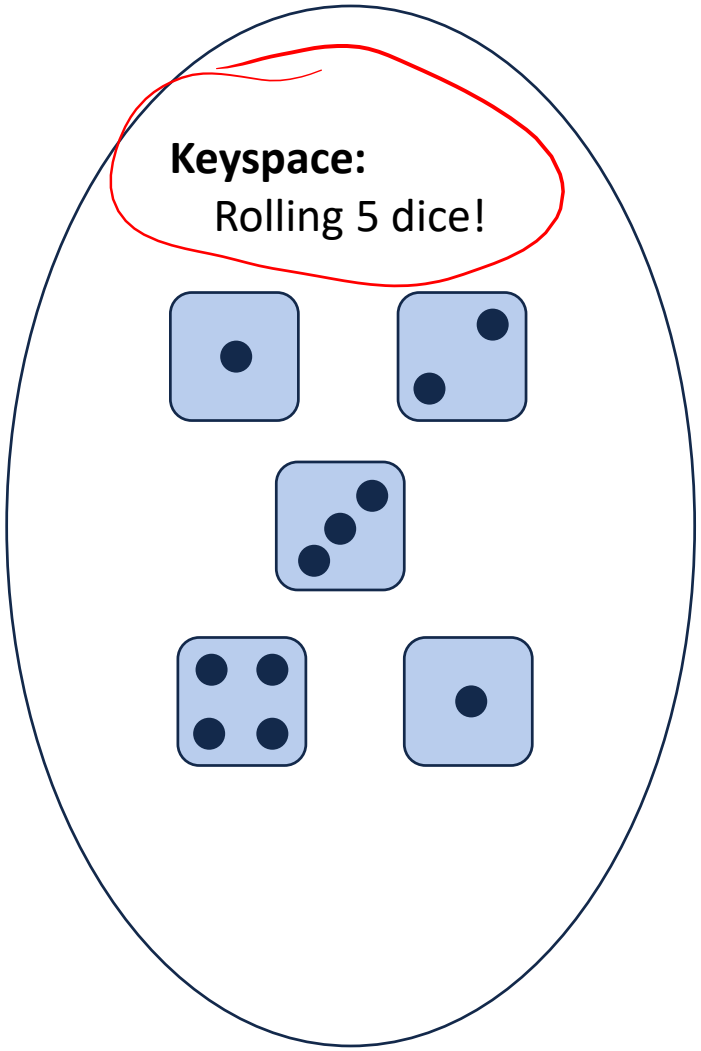
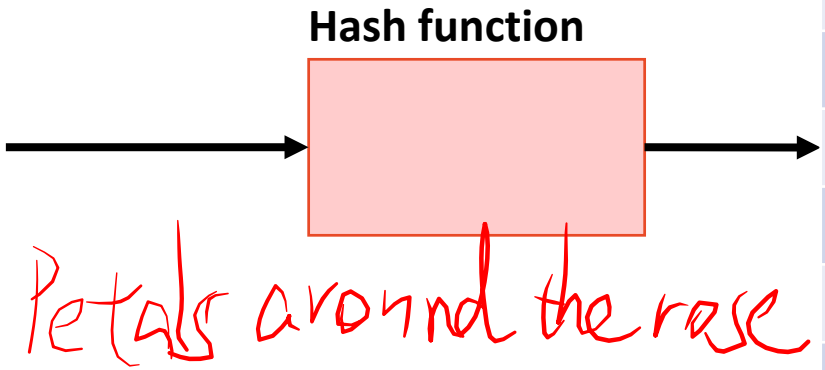
11	
12	
13	
14	
15	

A Perfect Hash Function





Key	Value
0	
1	
2	
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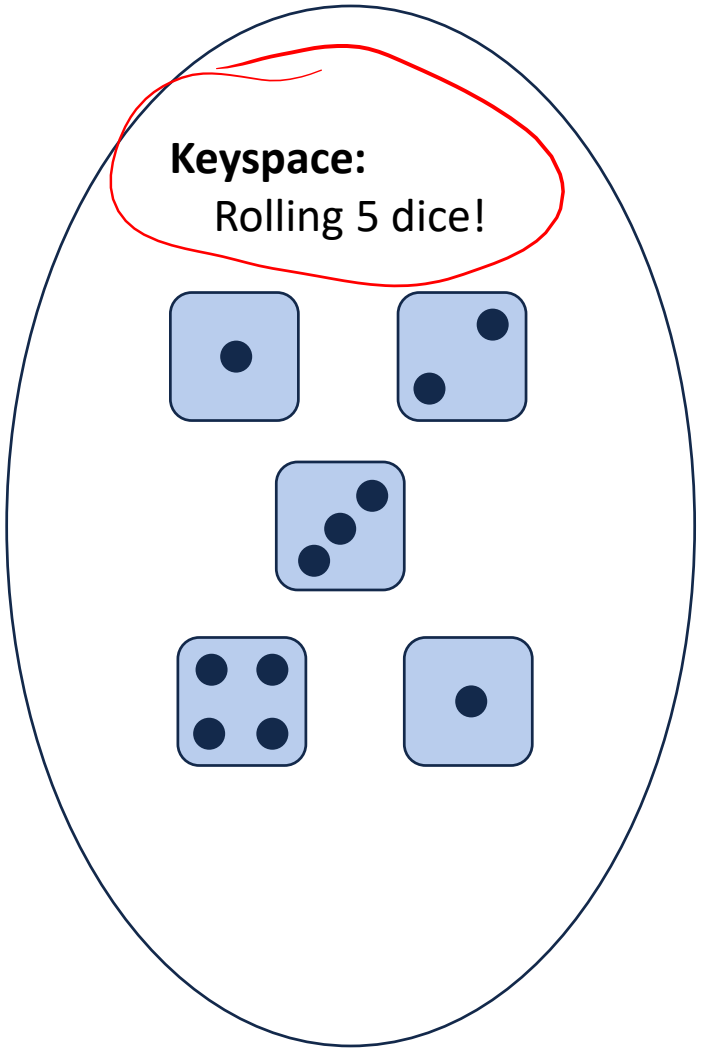
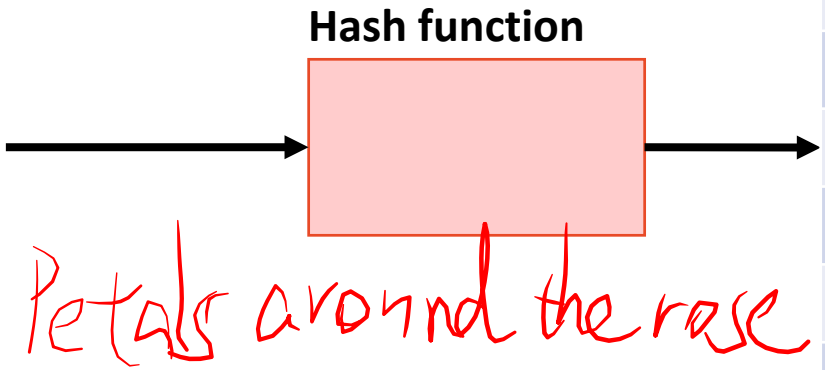
14	
15	

A Perfect Hash Function





Key	Value
0	
1	
2	
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11	
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13	
14	
15	

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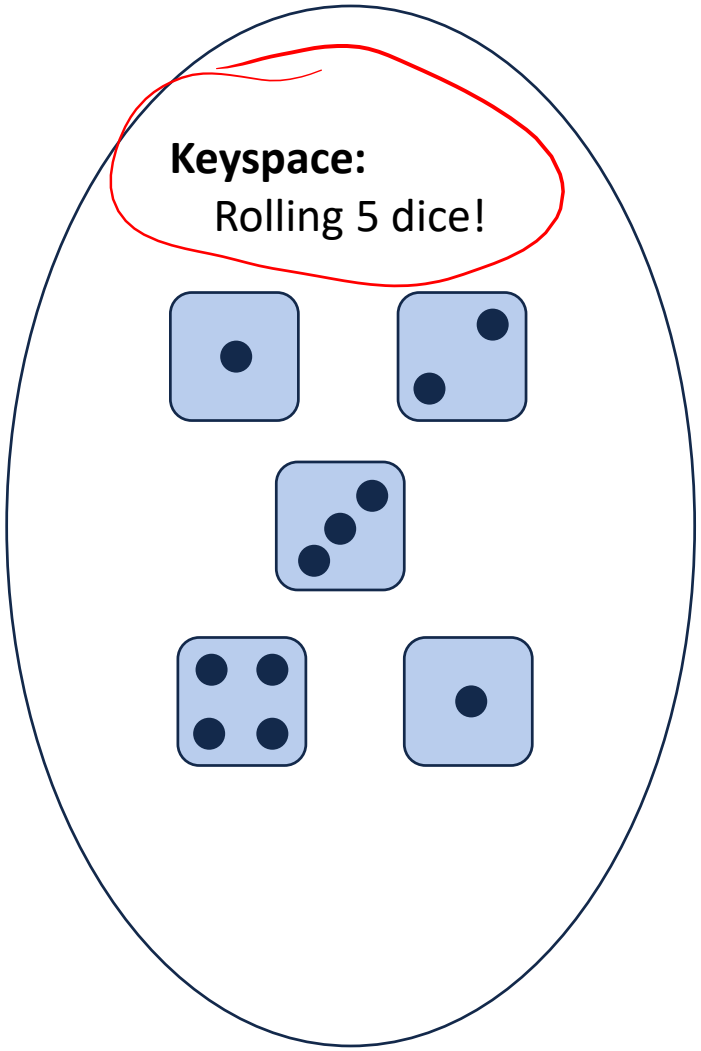

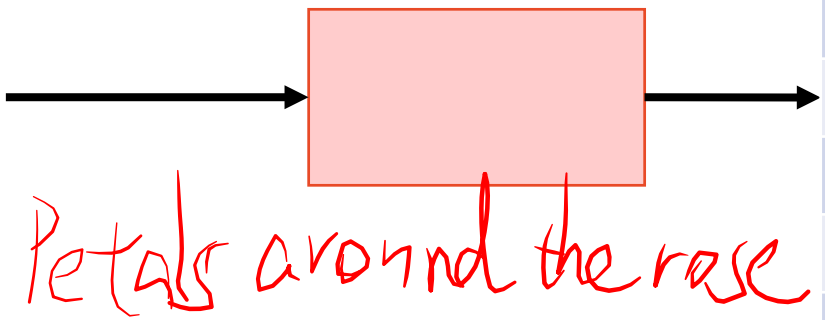

Petals around the rose

Key	Value
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	

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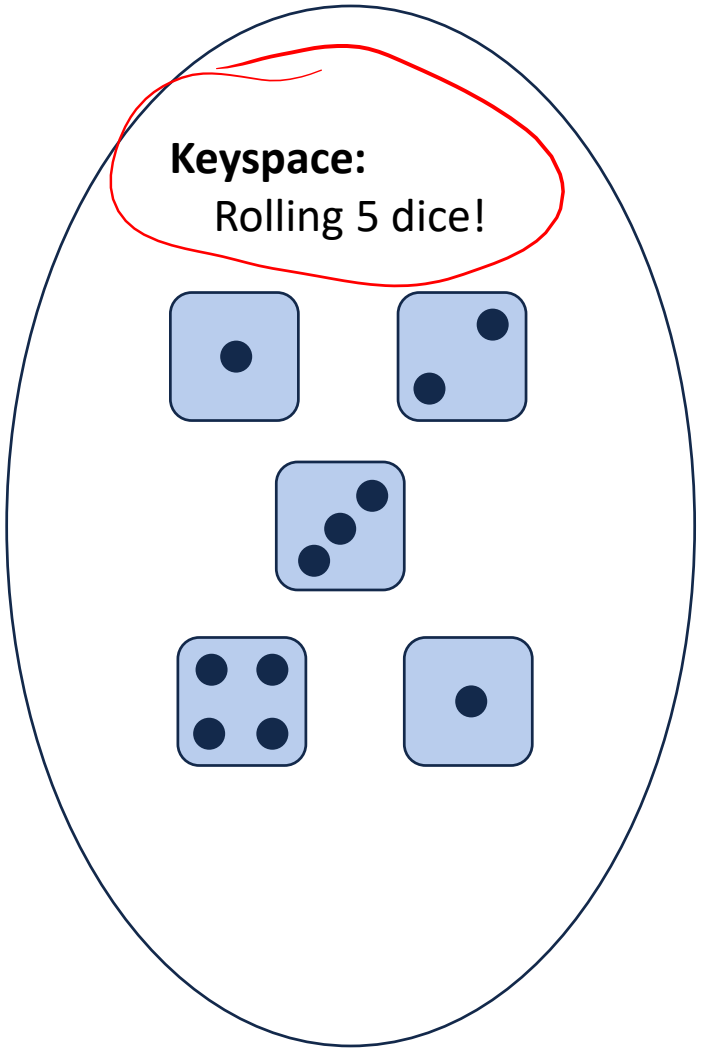
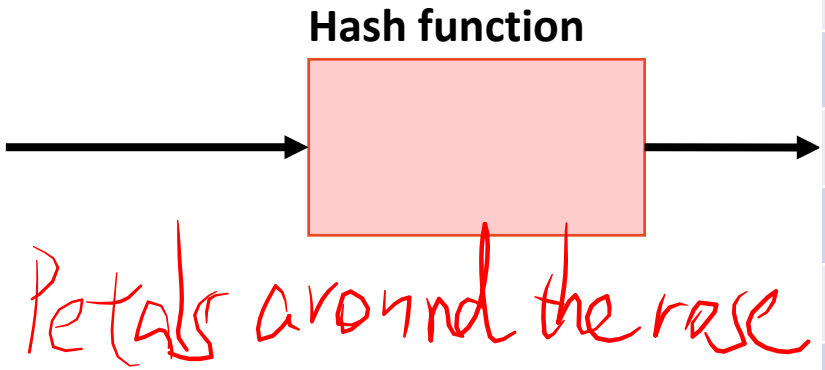
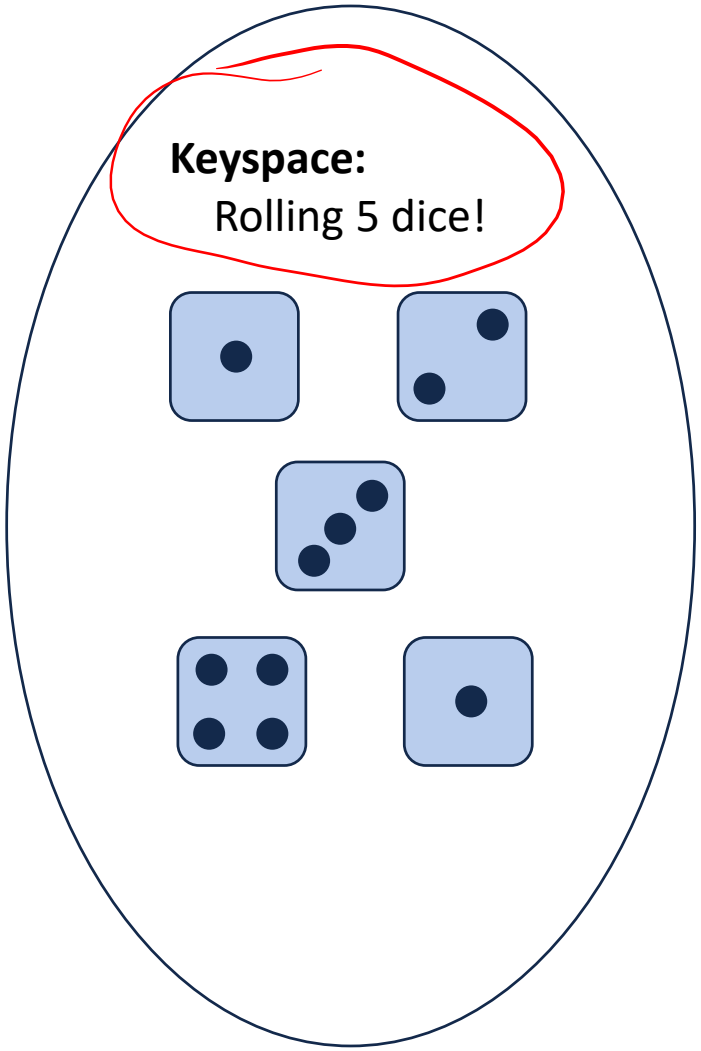
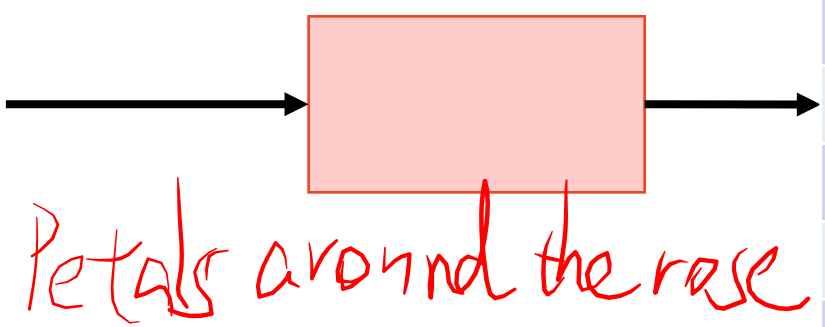




Key	Value
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	

A Perfect Hash Function

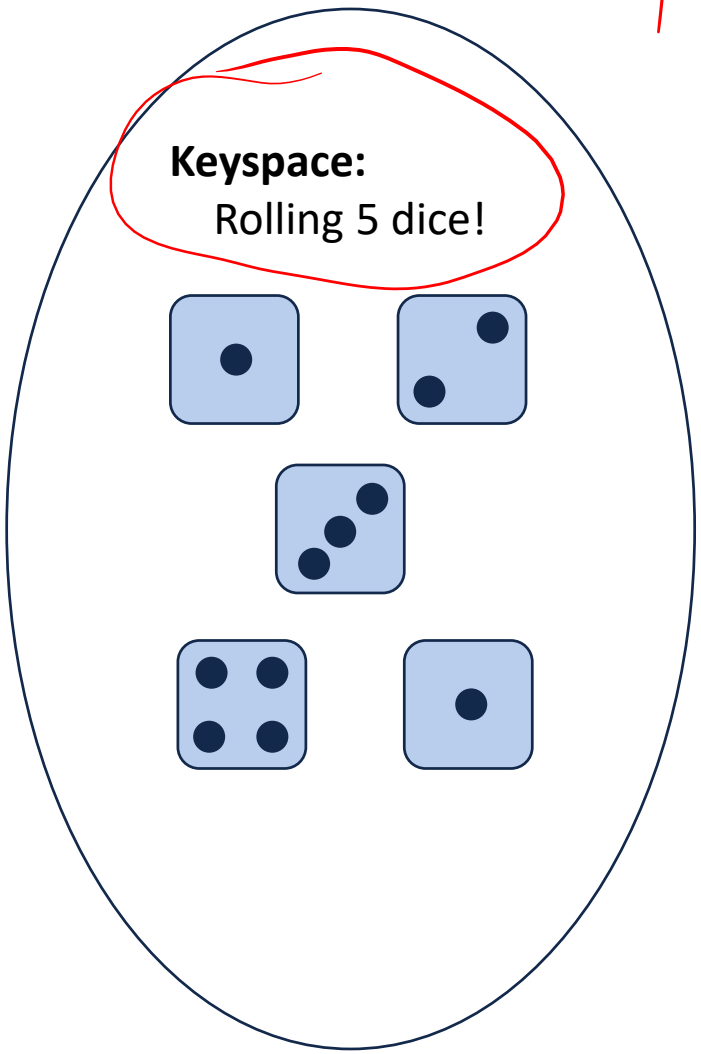




Key	Value
0	✓
1	✗
2	✓
3	✗
4	
5	✗
6	
7	✗
8	
9	✗
10	
11	✗
12	
13	✗
14	
15	✗

A Perfect Hash Function

1-to-1



Hash function

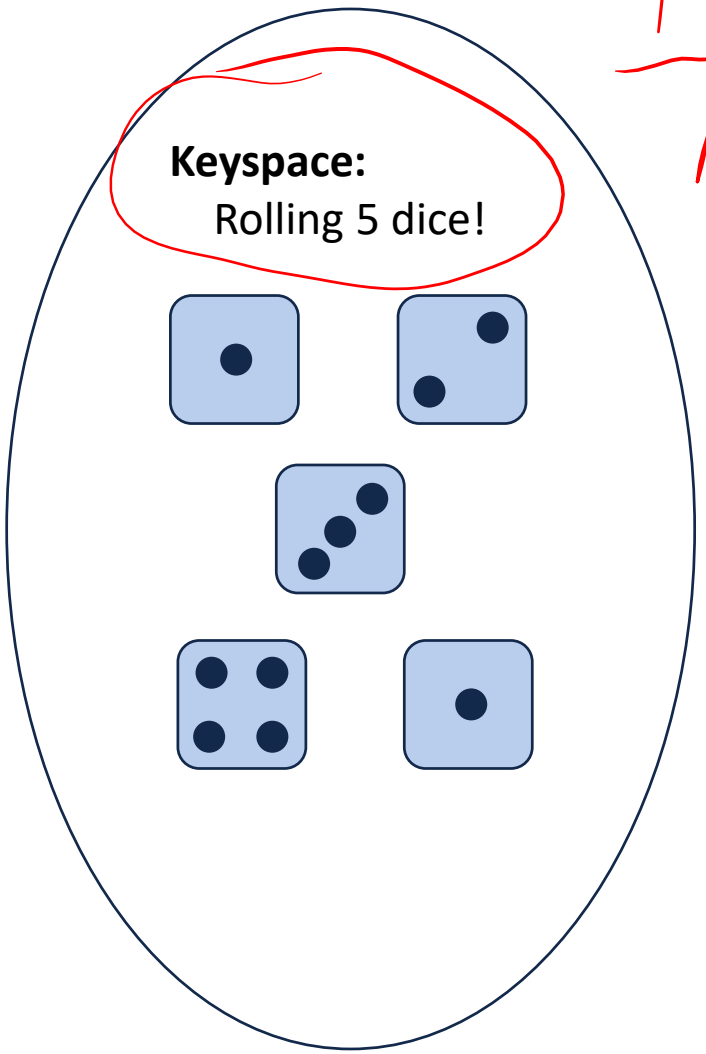
Petals around the rose

Key	Value
0	✓
1	✗
2	✓
3	✗
4	
5	✗
6	
7	✗
8	
9	✗
10	
11	✗
12	
13	✗
14	
15	✗

A Perfect Hash Function

1-to-1

No



Hash function

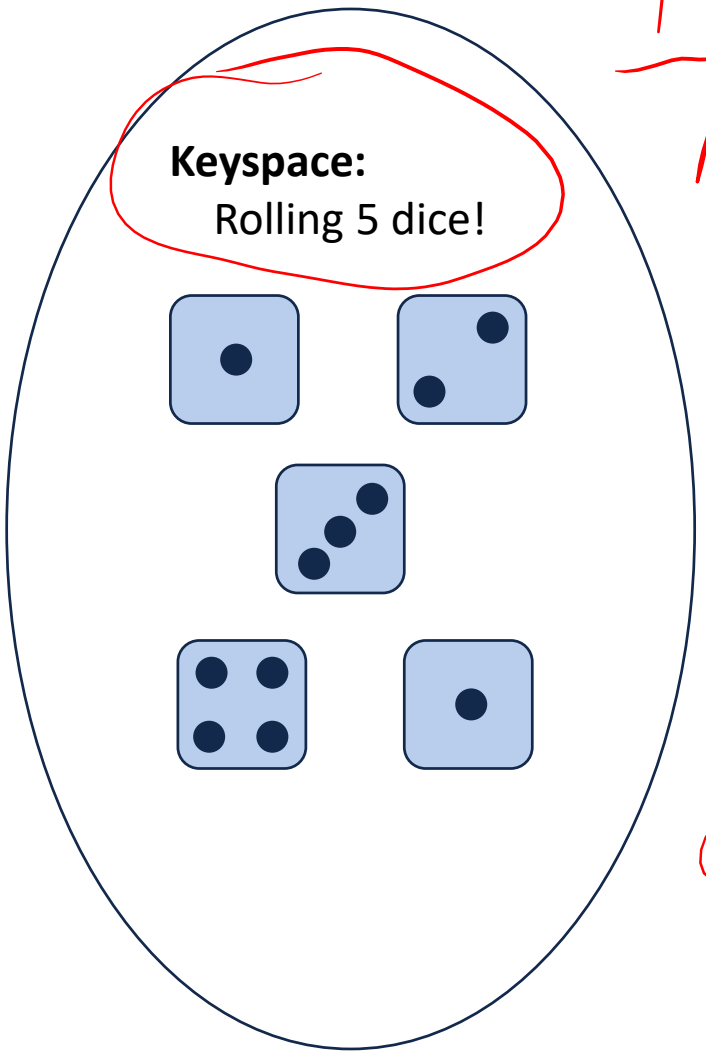
Petals around the rose

Key	Value
0	✓
1	X
2	✓
3	X
4	
5	X
6	
7	X
8	
9	X
10	
11	X
12	
13	X
14	
15	X

A Perfect Hash Function

1-to-1

No



Hash function

Petals around the rose

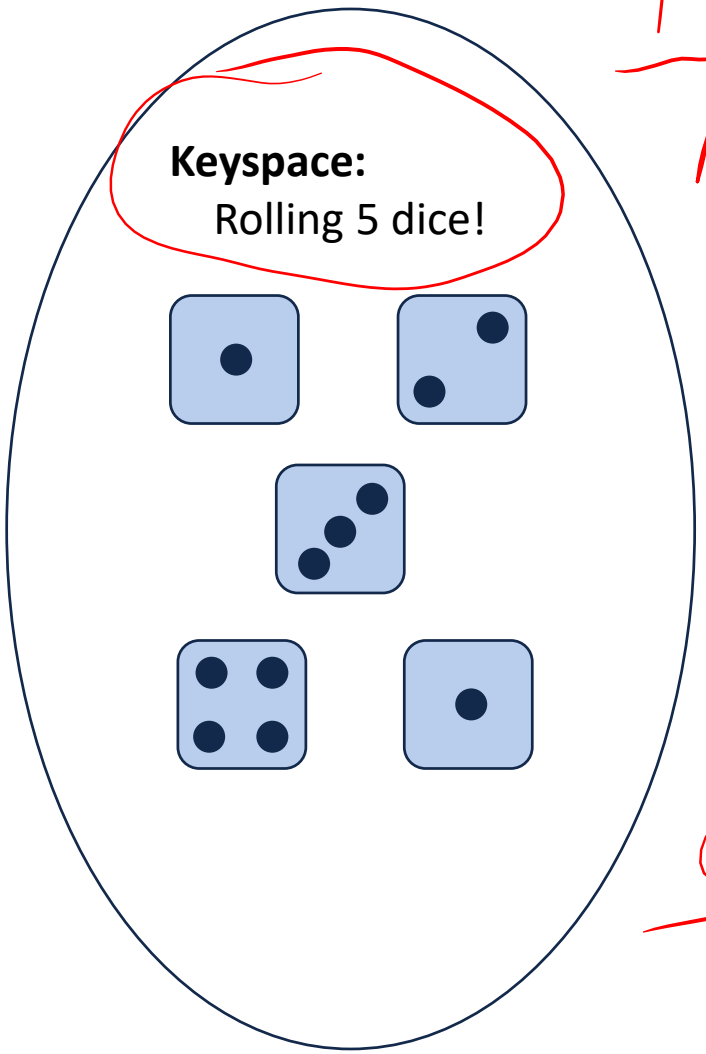
Onto

Key	Value
0	✓
1	X
2	✓
3	X
4	
5	X
6	
7	X
8	
9	X
10	
11	X
12	
13	X
14	
15	X

A Perfect Hash Function

1-to-1

No



Hash function

Petals around the rose

Onto

No : $h^{-1}(L)$ has no value

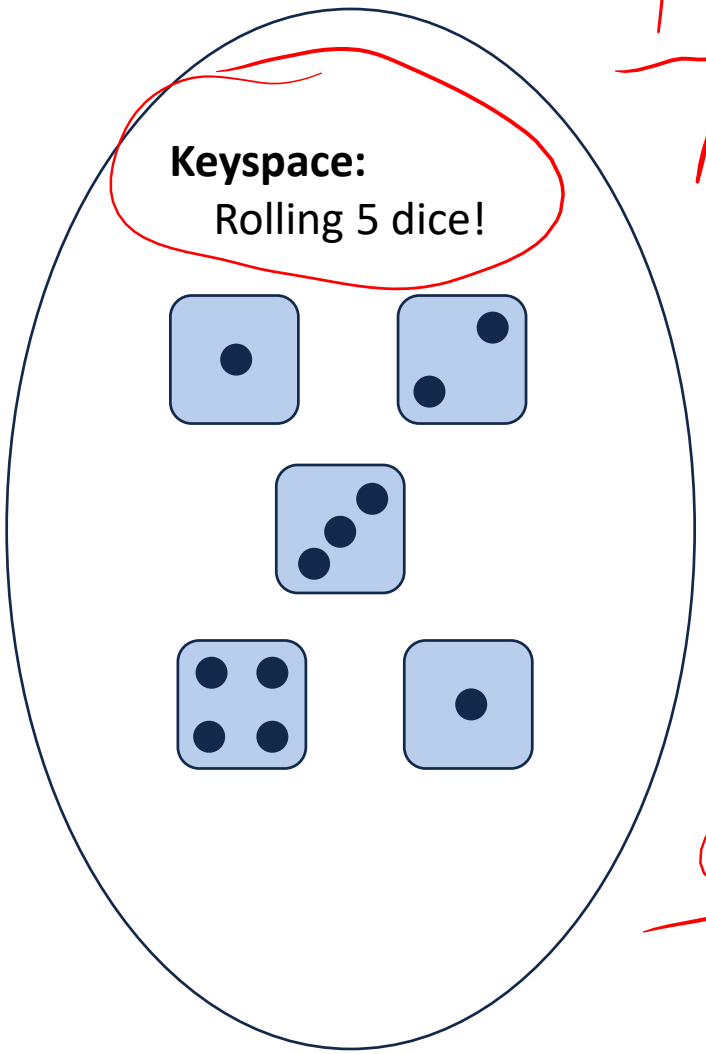
Key	Value
0	✓
1	X
2	✓
3	X
4	
5	X
6	
7	X
8	
9	X
10	
11	X
12	
13	X
14	
15	X

A Perfect Hash Function

P(20) =

1-to-1

No



Hash function

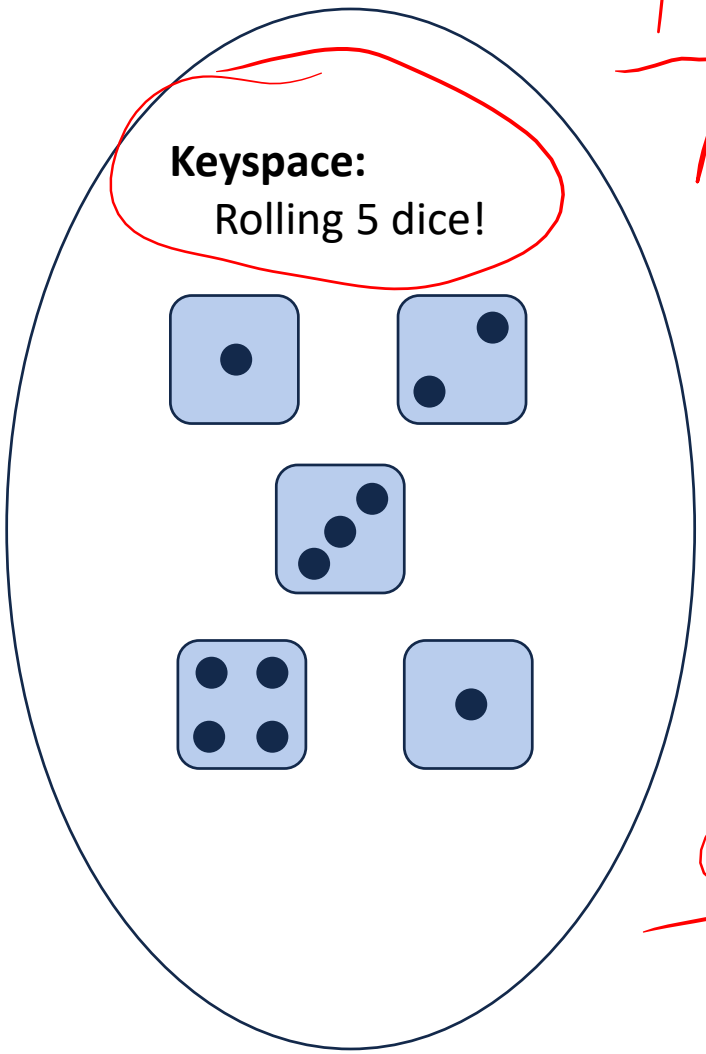
Petals around the rose

Onto

No: $h^{-1}(L)$ has no value

Key	Value
0	✓
1	X
2	✓
3	X
4	
5	X
6	
7	X
8	
9	X
10	
11	X
12	
13	X
14	
15	X

A Perfect Hash Function



1-to-1

No

Hash function

Petals around the rose

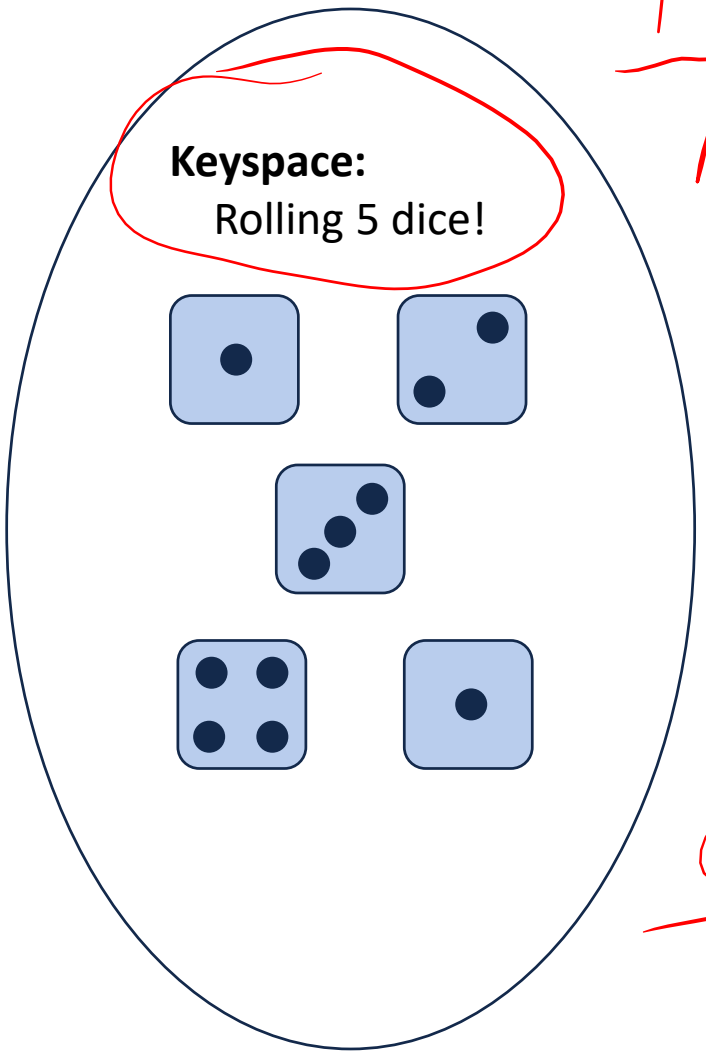
Onto

No: $h^{-1}(L)$ has no value

$P(20) = \frac{1}{6^5}$

Key	Value
0	✓
1	✗
2	✓
3	✗
4	
5	✗
6	
7	✗
8	
9	✗
10	
11	✗
12	
13	✗
14	
15	✗

A Perfect Hash Function



1-to-1

No

$P(20) = 1/6^5$

$P(0) > 1/6^5$

Hash function

Petals around the rose

Key	Value
0	✓
1	X
2	✓
3	X
4	
5	X
6	
7	X
8	
9	X
10	
11	X
12	
13	X
14	
15	X

Onto

No : $h^{-1}(L)$ has no value

Hash Function

Our **hash function** consists of two parts:

- A **hash**:

- A **compression**:

Choosing a good hash function is tricky…

- Don’t create your own (yet*)
- Very smart people have created very bad hash functions

Hash Function

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Hash Function

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- A **hash**: h_H key $\rightarrow$ i (int)

- A **compression**:

$$h_C: i \rightarrow [0, N-1]$$

Choosing a good hash function is tricky…

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Hash Function

Our **hash function** consists of two parts:

- A **hash**: h_H key $\rightarrow$ i (int)

- A **compression**: h_C: i $\rightarrow$ [0, N-1] $\%N$

Choosing a good hash function is tricky…

- Don’t create your own (yet*)
- Very smart people have created very bad hash functions

Hash Function

Characteristics of a good hash function:

1. Computation Time:

2. Deterministic:

3. Satisfy the SUHA:

Hash Function

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Runs in constant time, O(1)

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No random element

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Characteristics of a good hash function:

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Runs in constant time, O(1)

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No random element

if $k_1 == k_2 \rightarrow h(k_1) == h(k_2)$

3. Satisfy the SUHA:

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if $k_1 == k_2 \rightarrow h(k_1) == h(k_2)$

3. Satisfy the SUHA:

Simple Uniform hashing assumption

Hash Function

Characteristics of a good hash function:

1. Computation Time:

   Runs in constant time, O(1)

2. Deterministic:

   No random element

   if $k_1 == k_2 \rightarrow h(k_1) == h(k_2)$

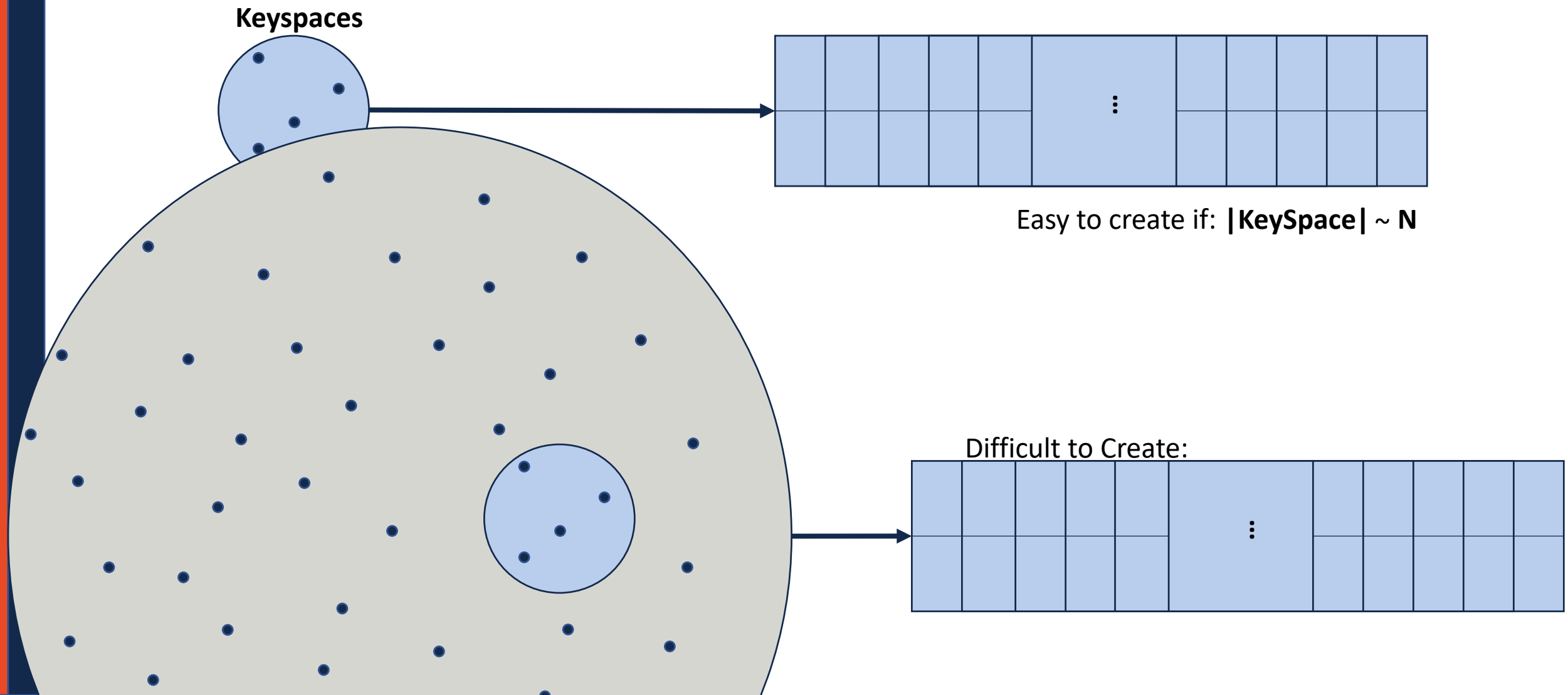
3. Satisfy the SUHA:

   Simple Uniform hashing assumption

   for keys k, j where k != j

   $P(h(k) == h(j)) = 1/N$

General Purpose Hash Function



Hash Function

Given: Easy to create a hash function of strings of length 8

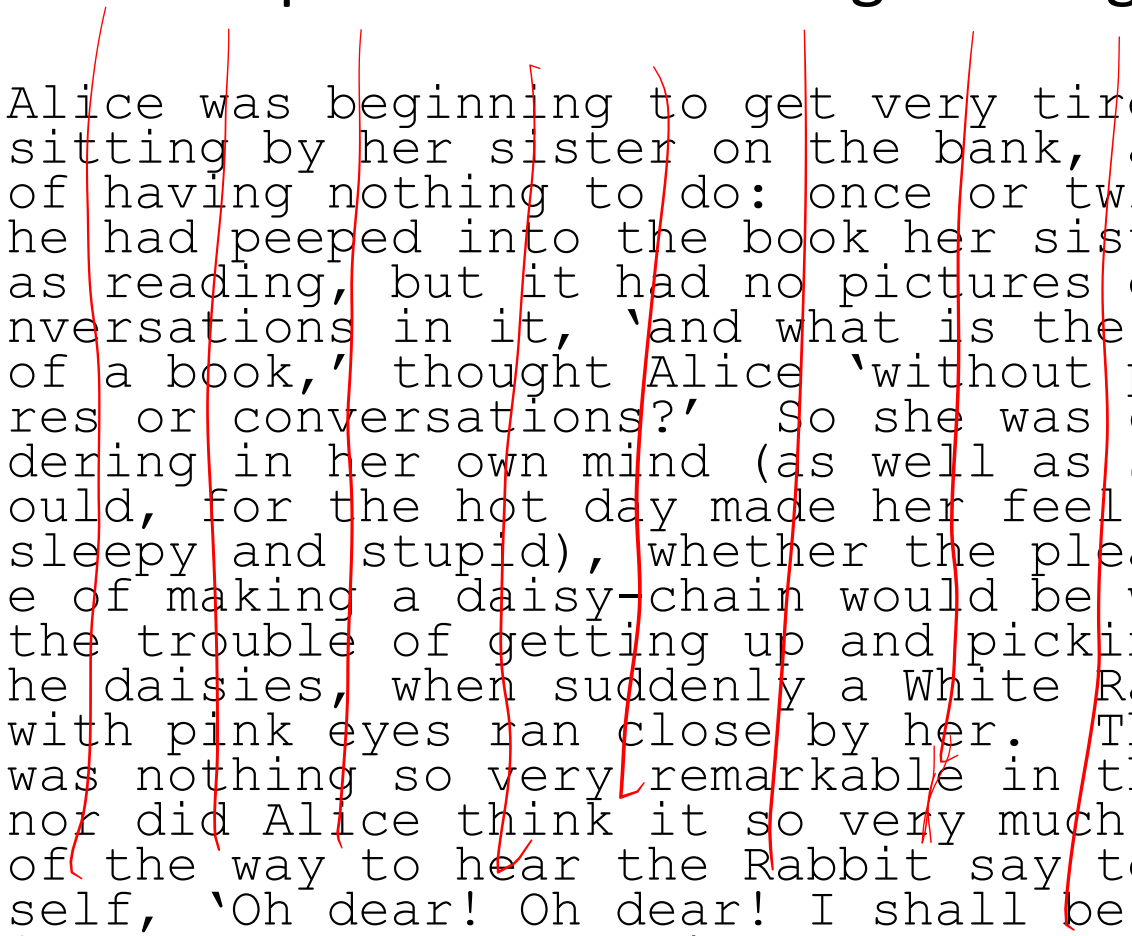
Hash Function

Idea: Map 40 character things to length 8:

```
Alice was beginning to get very tired of
sitting by her sister on the bank, and
of having nothing to do: once or twice s
he had peeped into the book her sister w
as reading, but it had no pictures or co
nversations in it, ‘and what is the use
of a book,’ thought Alice ‘without pictu
res or conversations?’  So she was consi
dering in her own mind (as well as she c
ould, for the hot day made her feel very
sleepy and stupid), whether the pleasur
e of making a daisy-chain would be worth
the trouble of getting up and picking t
he daisies, when suddenly a White Rabbit
with pink eyes ran close by her.  There
was nothing so very remarkable in that;
nor did Alice think it so very much out
of the way to hear the Rabbit say to it
self, ‘Oh dear! Oh dear! I shall be late
!’ (when she thought it over afterwards,
it occurred to her that she ought to ha
```

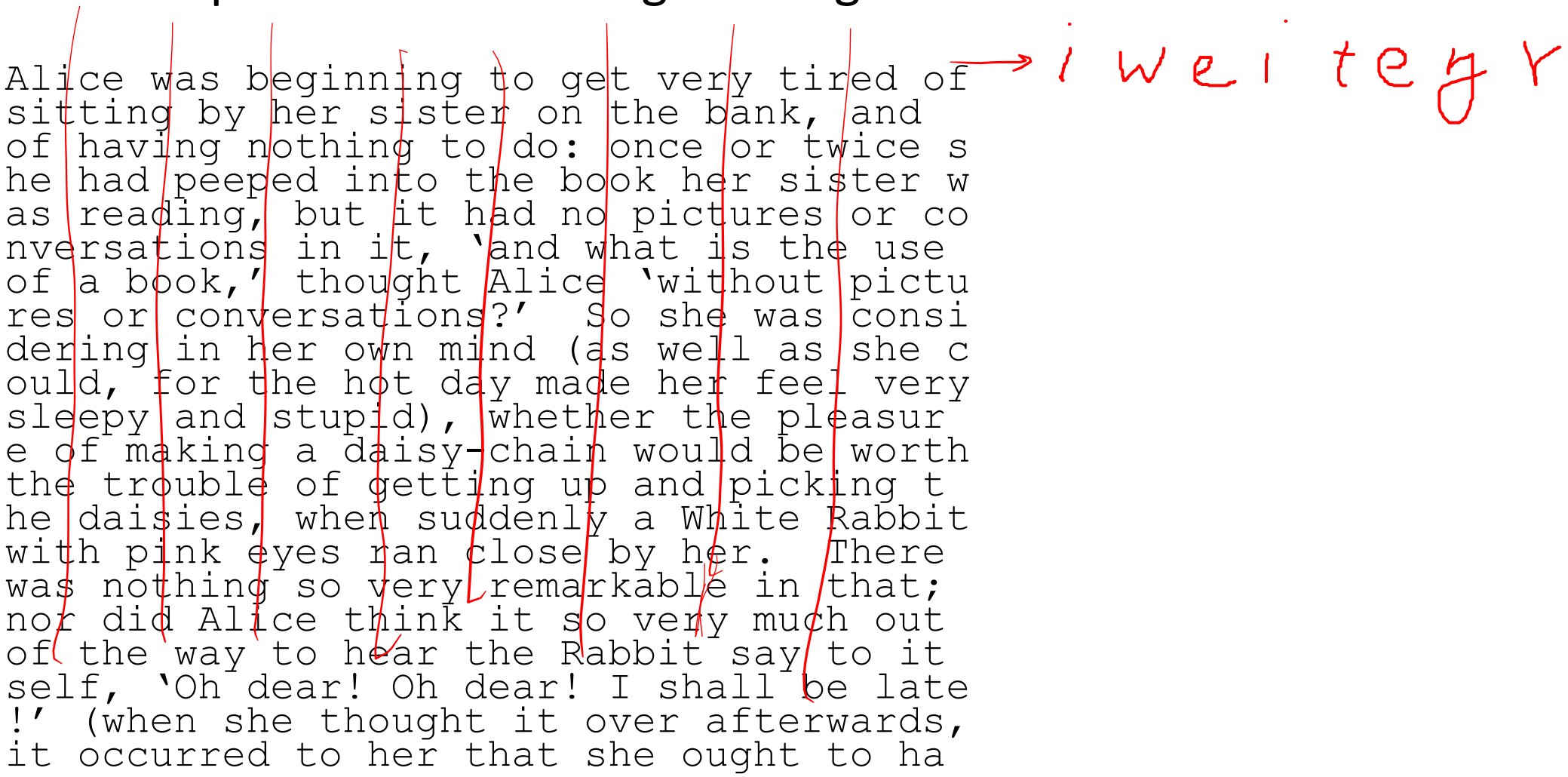
Hash Function

Idea: Map 40 character things to length 8:



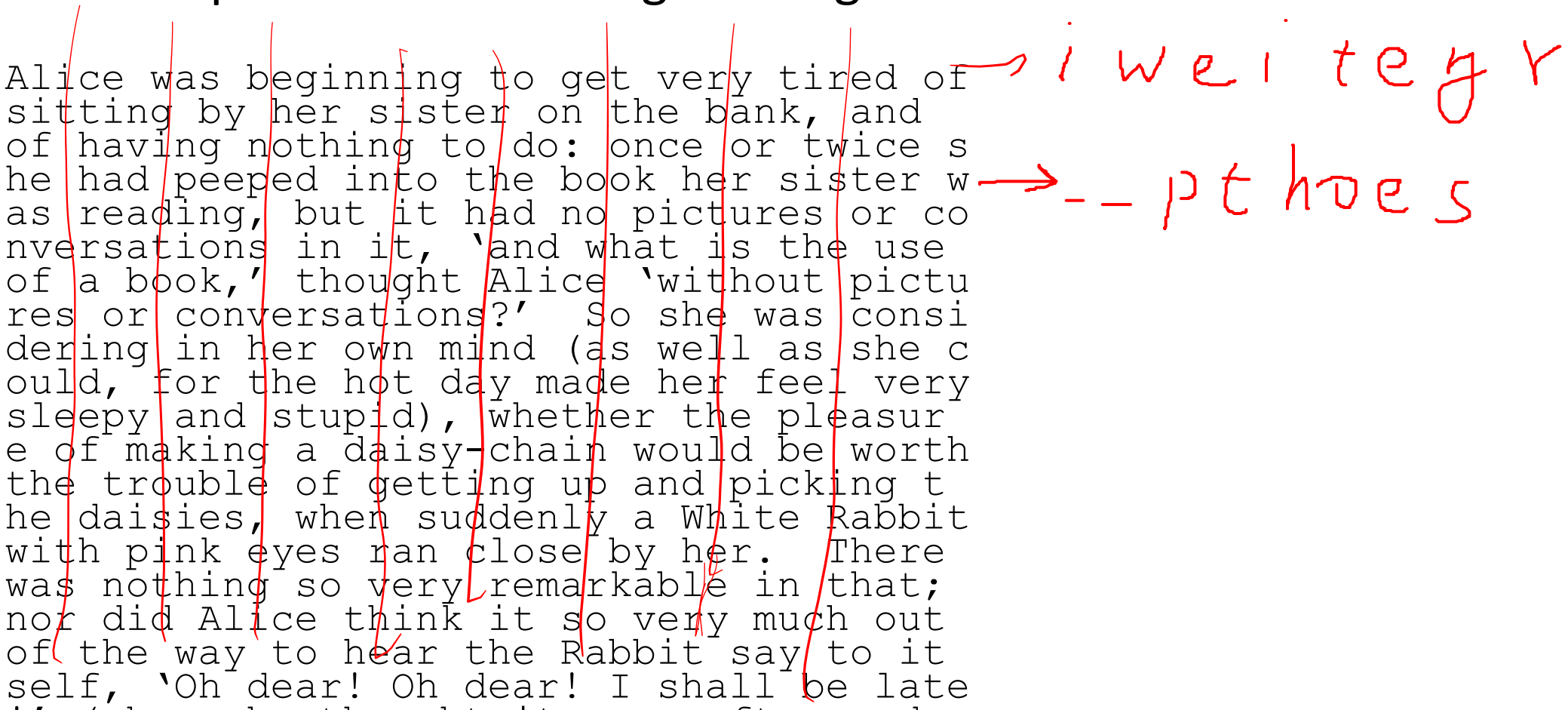
Hash Function

Idea: Map 40 character things to length 8:



Hash Function

Idea: Map 40 character things to length 8:



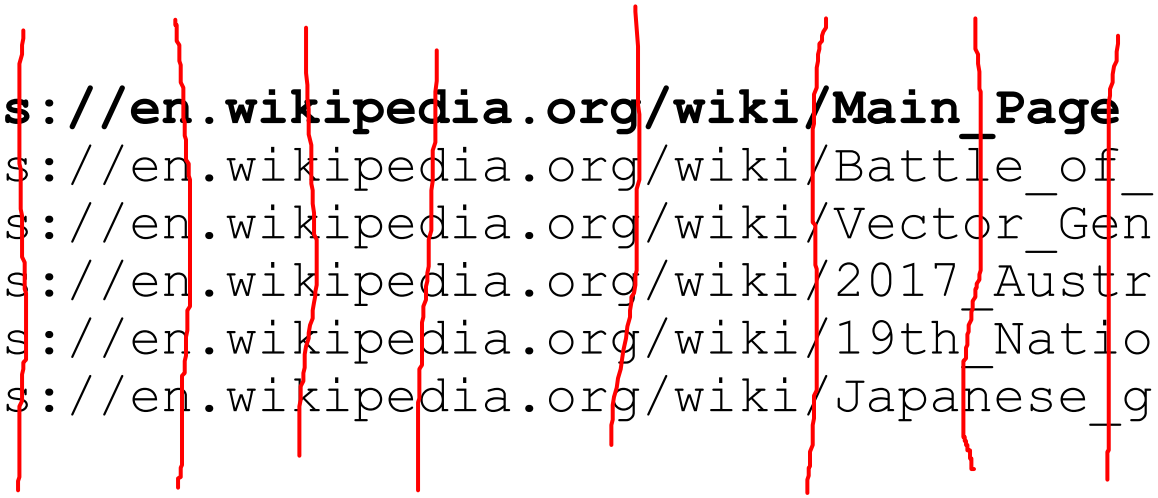
Hash Function

Idea: Map 40 character things to length 8:

```
https://en.wikipedia.org/wiki/Main_Page
https://en.wikipedia.org/wiki/Battle_of_
https://en.wikipedia.org/wiki/Vector_Gen
https://en.wikipedia.org/wiki/2017_Austr
https://en.wikipedia.org/wiki/19th_Natio
https://en.wikipedia.org/wiki/Japanese_g
```

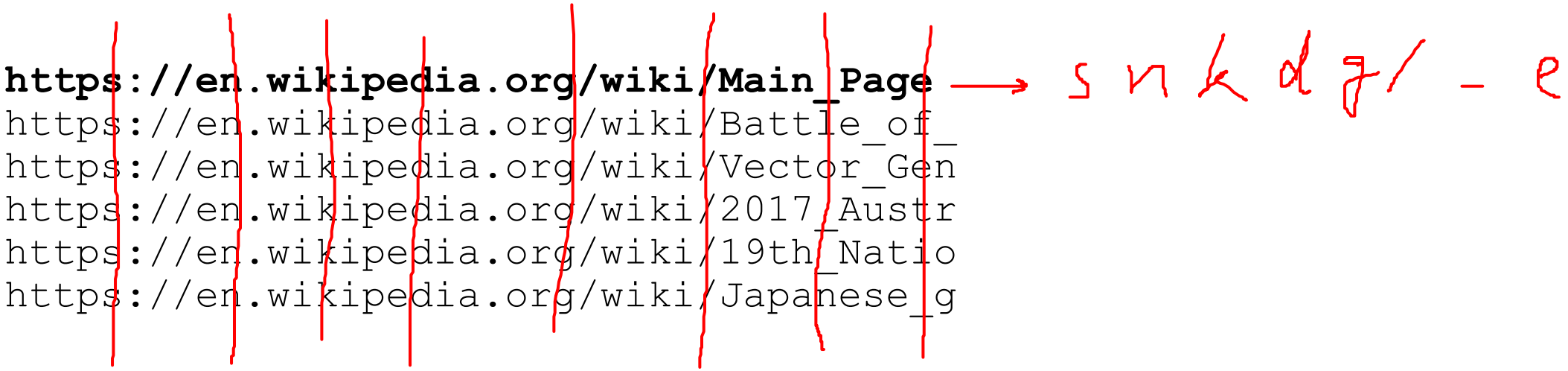
Hash Function

Idea: Map 40 character things to length 8:



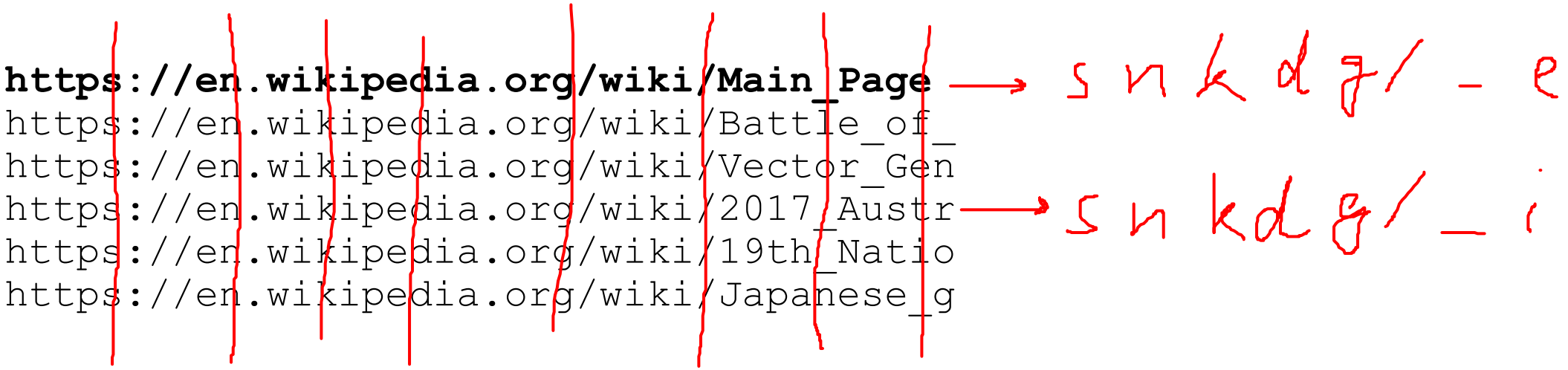
Hash Function

Idea: Map 40 character things to length 8:



Hash Function

Idea: Map 40 character things to length 8:



Hash Function

In CS 225, we focus on **general purpose** hash functions.

Other hash functions exist with different properties
(eg: cryptographic hash functions)

CS 225

Data Structures

Volodymyr Kindratenko

Slides courtesy of Wade Fagen-Ulmschneider

A Hash Table based Dictionary

Client Code:

```
1  Dictionary<KeyType, ValueType> d;
2  d[k] = v;
```

A **Hash Table** consists of three things:

1.

2.

3.

A Hash Table based Dictionary

Client Code:

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1  Dictionary<KeyType, ValueType> d;
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1. hash function h(k)
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A **Hash Table** consists of three things:

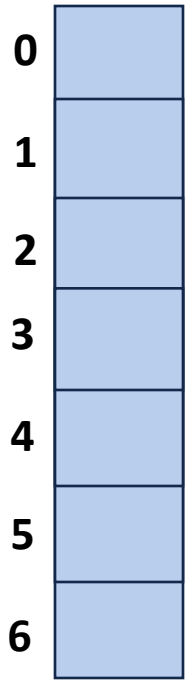
1. hash function h(k)
2. array
3. collision resolution strategies

(Example of open hashing)

Collision Handling: Separate Chaining

$S = \{ 16, 8, 4, 13, 29, 11, 22 \}$ $|S| = n$

$h(k) = k \% 7$ $|Array| = N$



	Worst Case	SUHA
Insert		
Remove/Find		

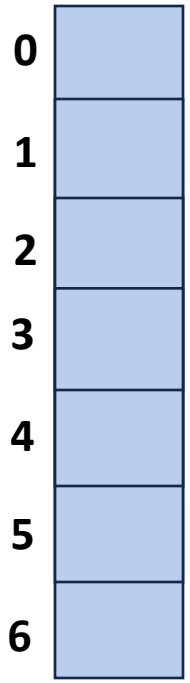
(Example of open hashing)

Collision Handling: Separate Chaining

Linked list of every node

S = { 16, 8, 4, 13, 29, 11, 22 } **|S| = n**

h(k) = k % 7 **|Array| = N**



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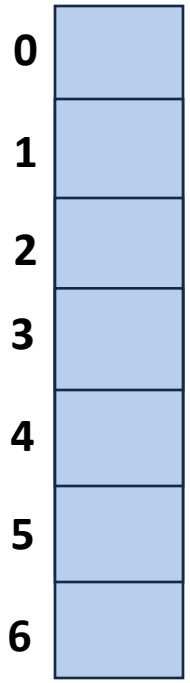

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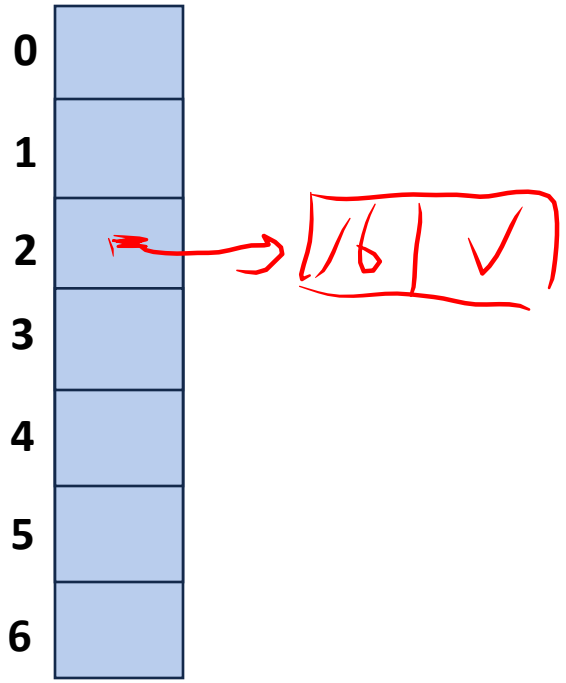

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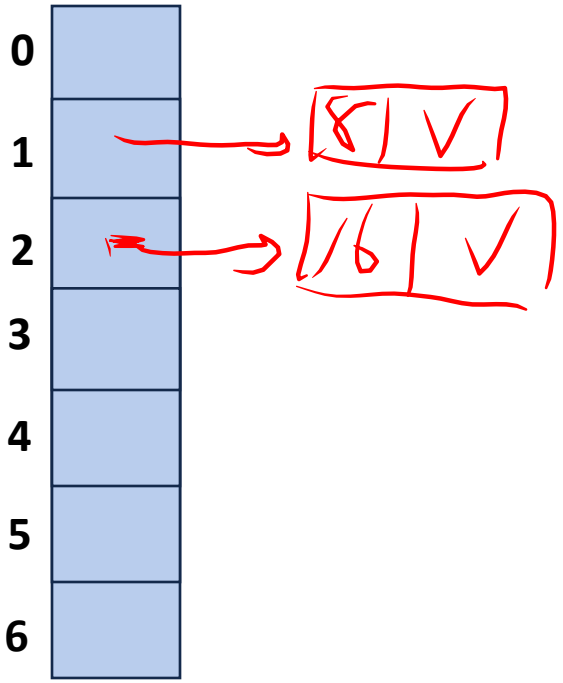

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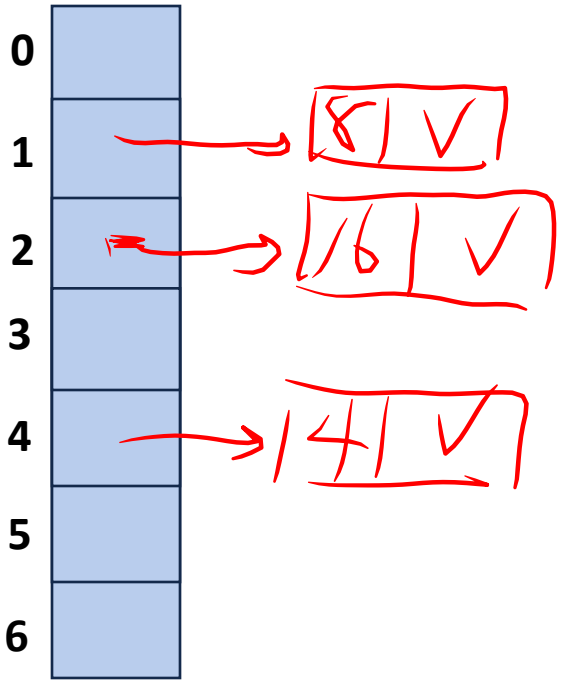

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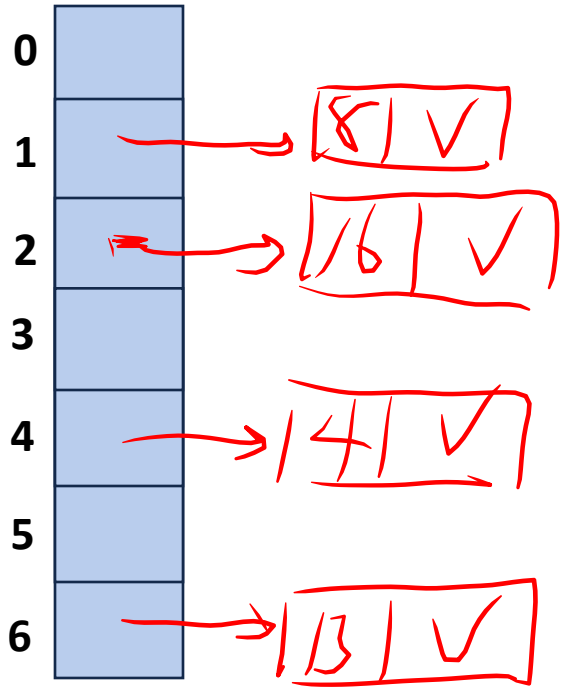

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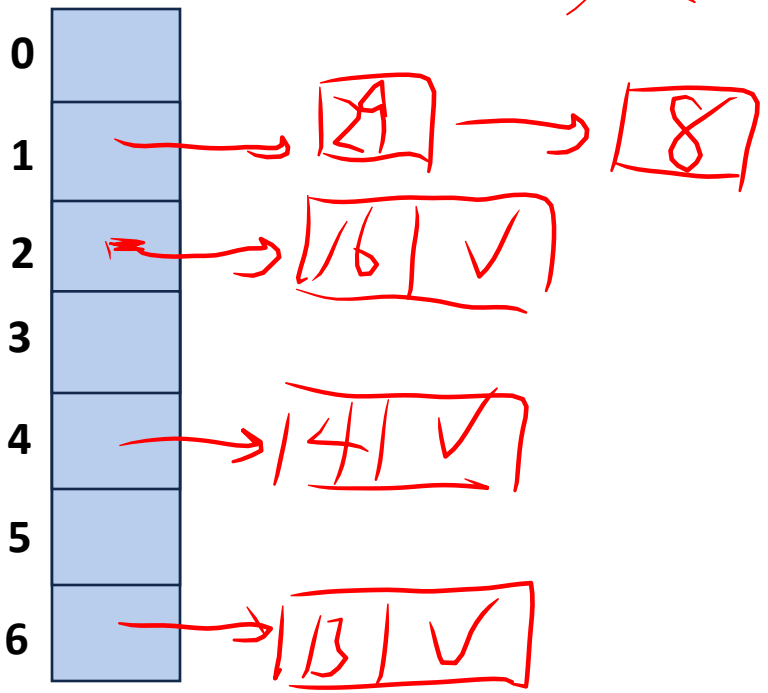

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(Example of open hashing)

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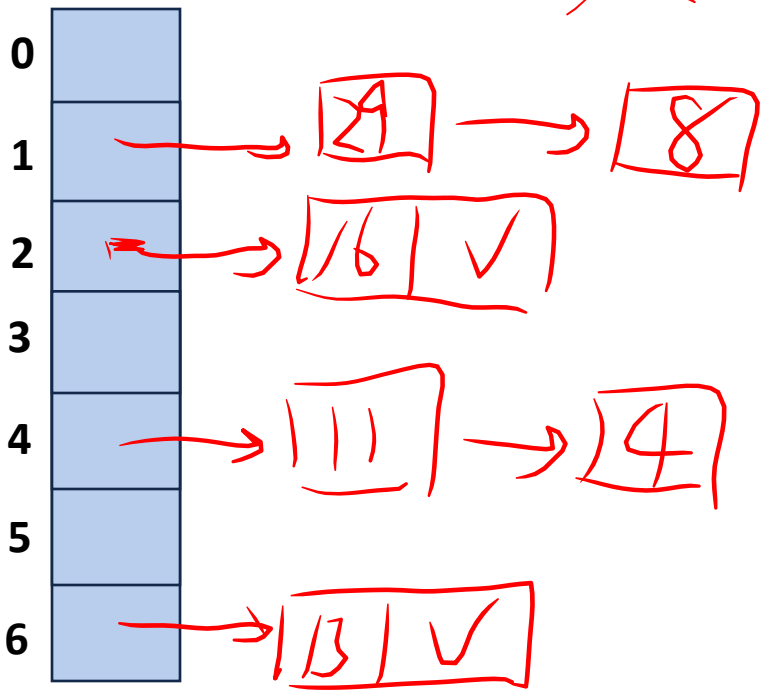

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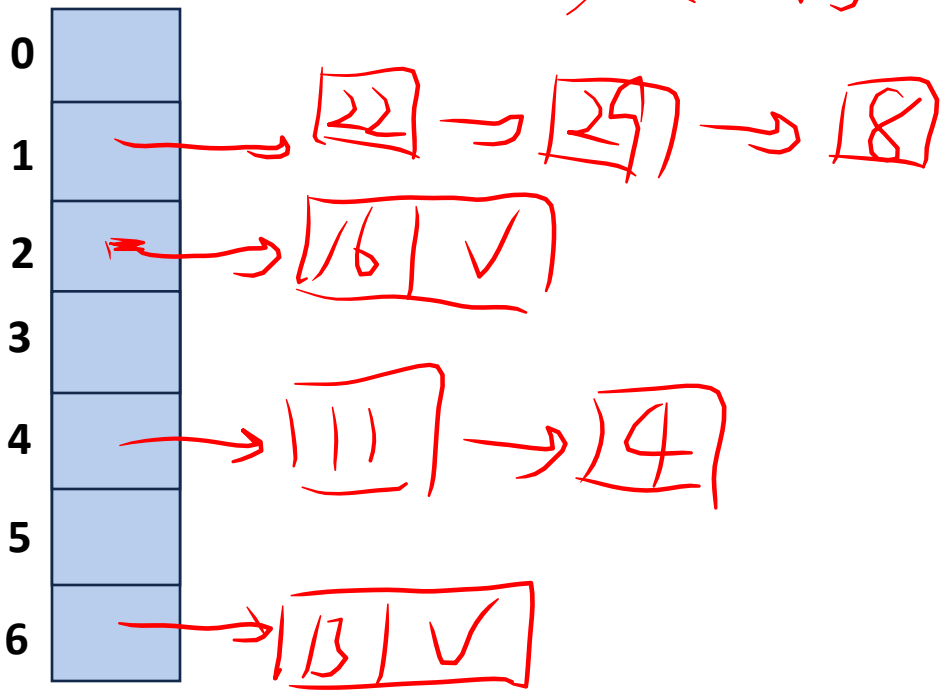

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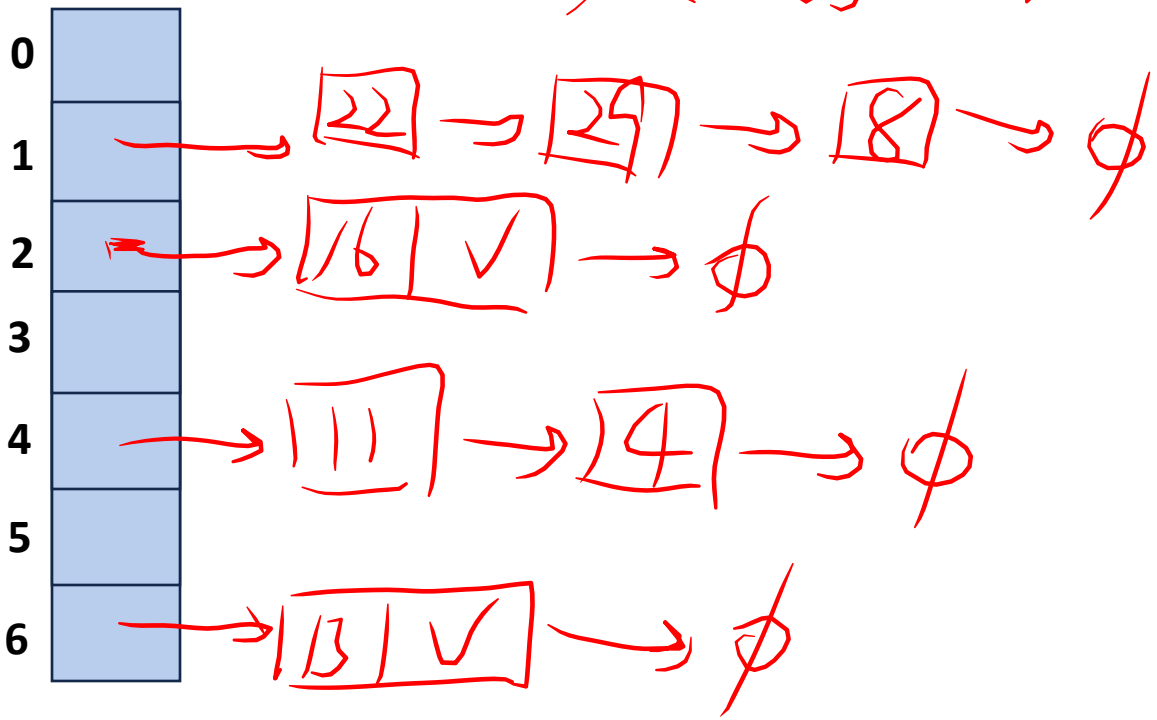

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Remove/Find		

(Example of open hashing)

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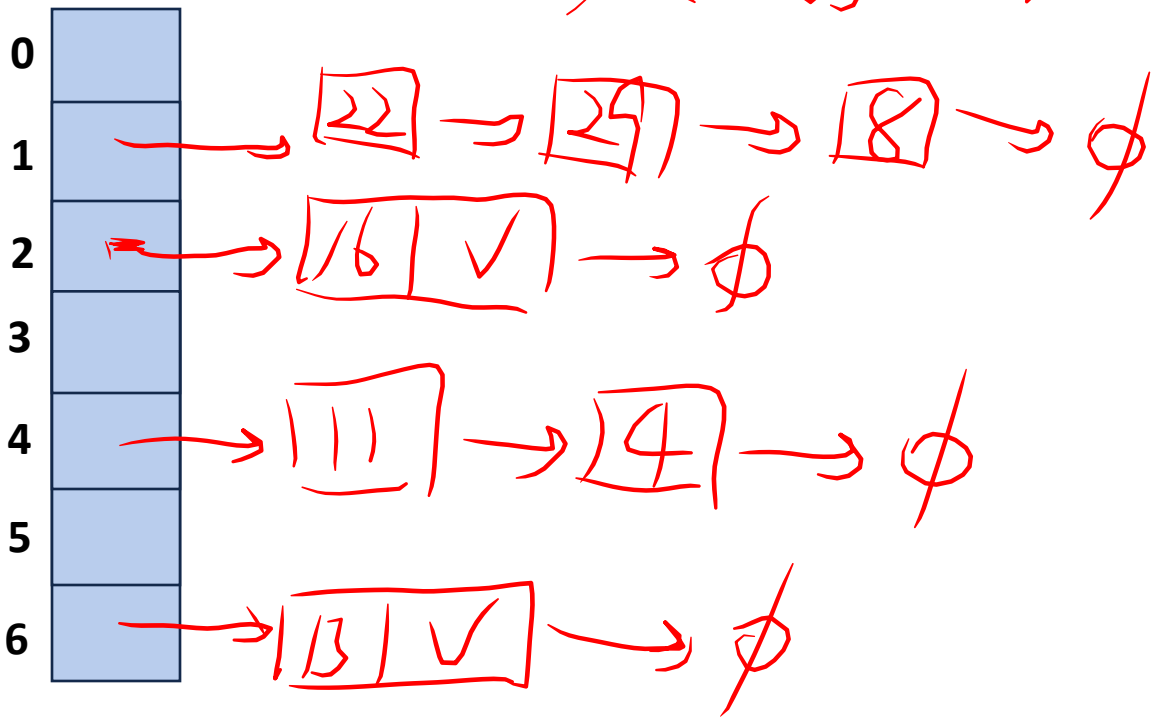

	Worst Case	SUHA
Insert	O(1)	
Remove/Find		

(Example of open hashing)

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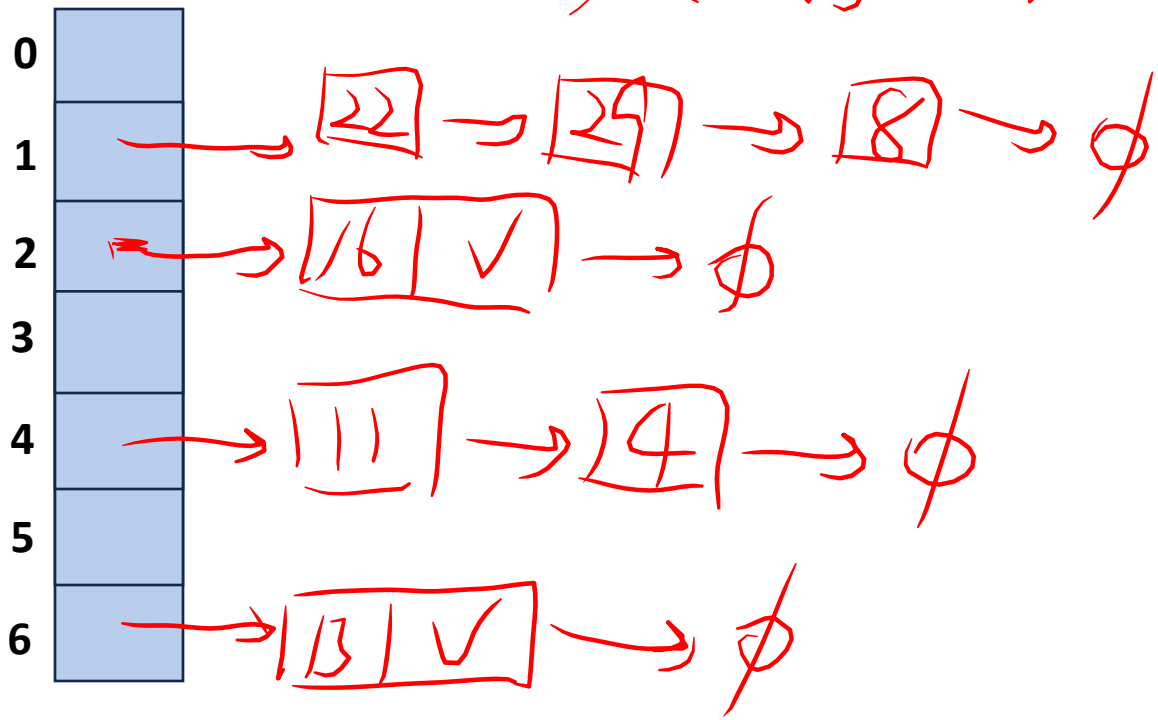

	Worst Case	SUHA
Insert	O(1)	
Remove/Find	O(n)	

(Example of open hashing)

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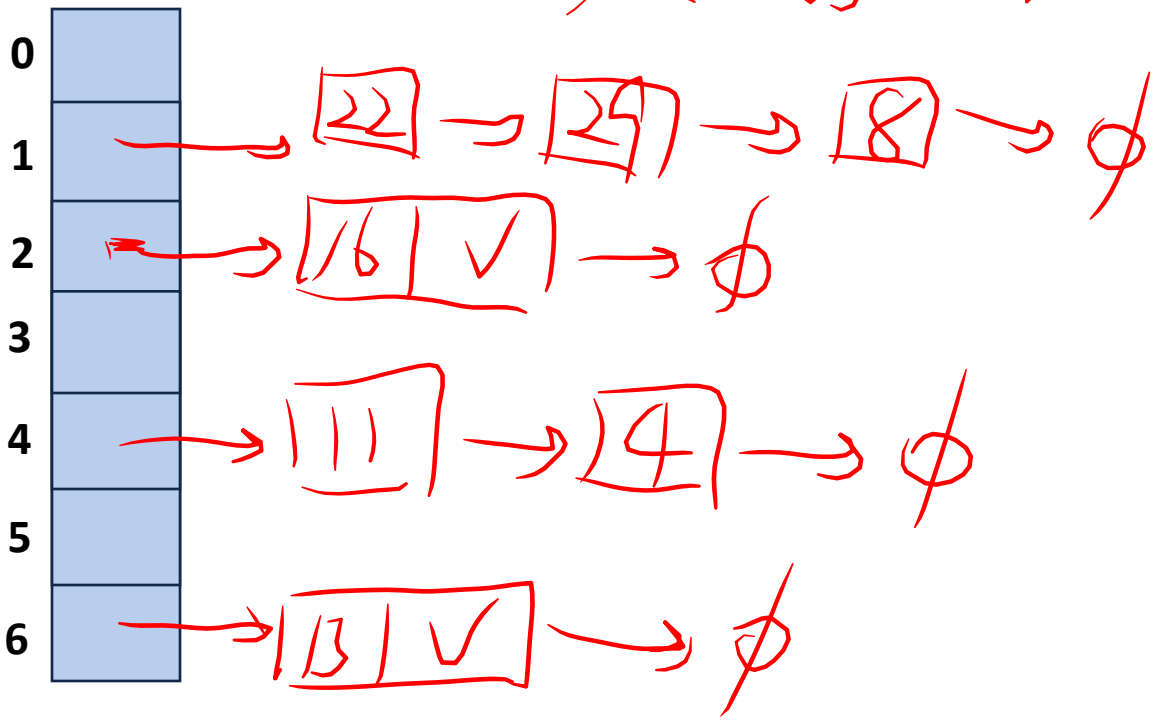

	Worst Case	SUHA
Insert	O(1)	
Remove/Find	O(n)	O(n/N)

(Example of open hashing)

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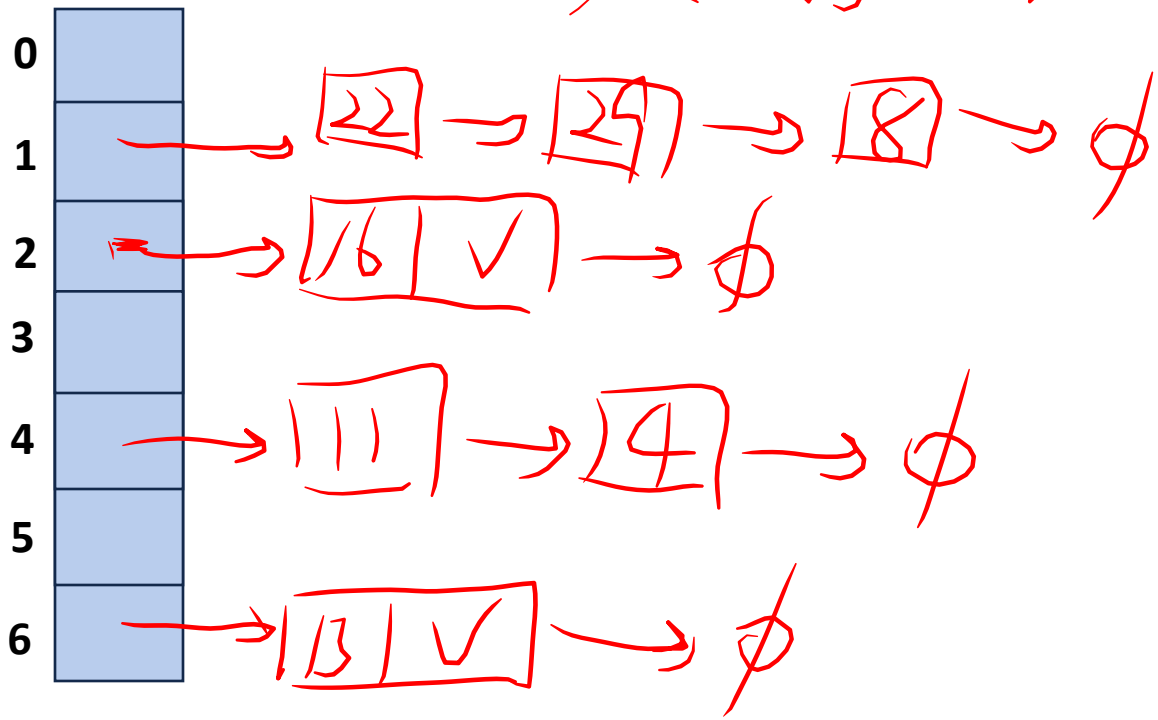

	Worst Case	SUHA
Insert	O(1)	
Remove/Find	O(n) ≡ O(n/N)	

(Example of open hashing)

Collision Handling: Separate Chaining

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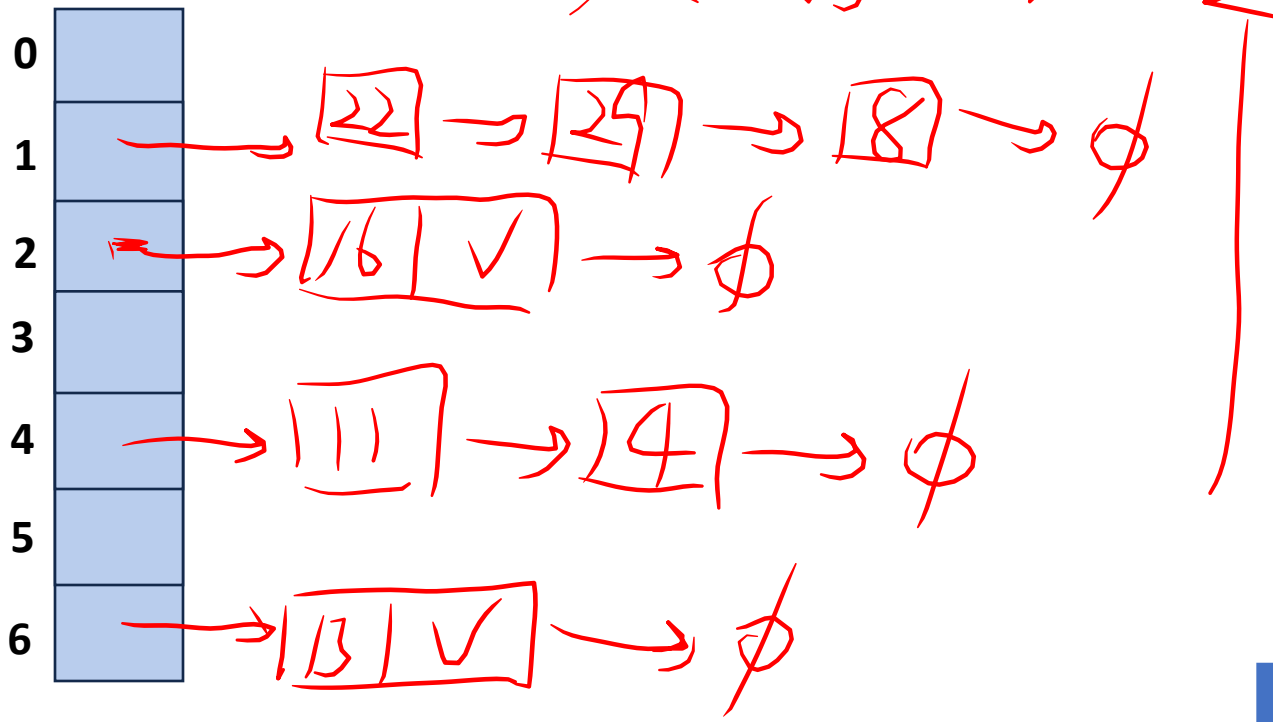
load factor



	Worst Case	SUHA
Insert	O(1)	
Remove/Find	O(n) ≡ O(n/N)	

(Example of open hashing)

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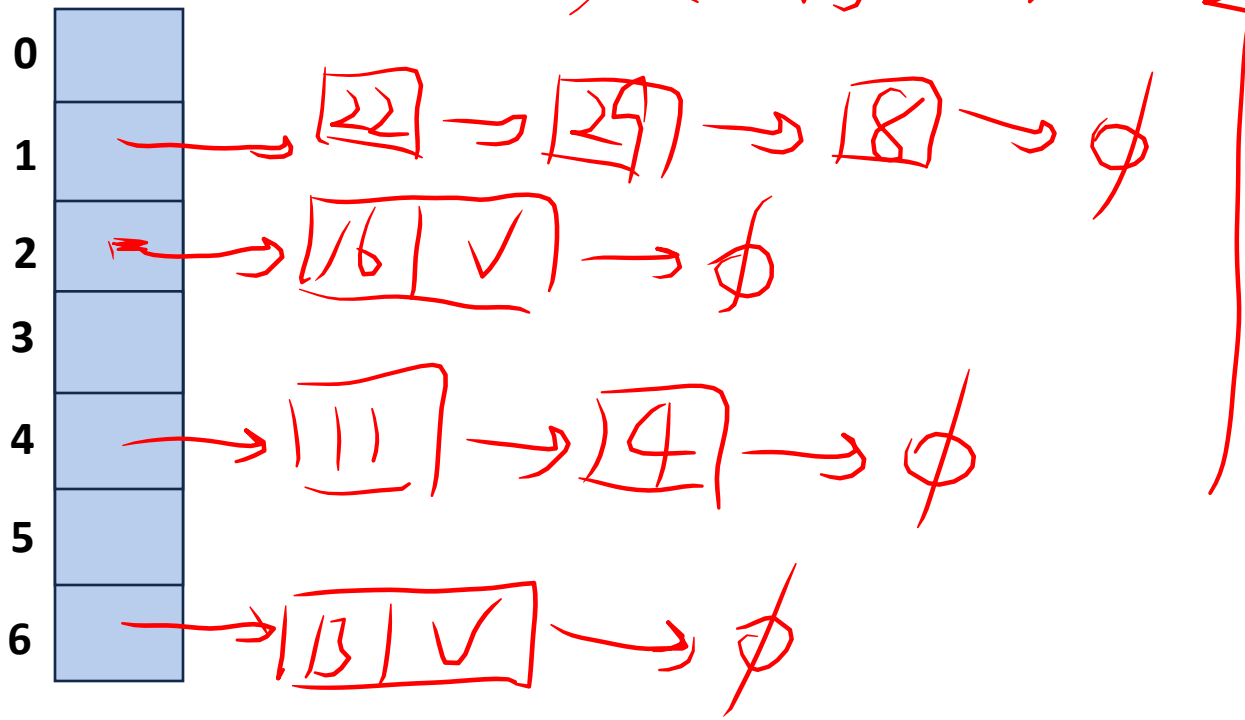

load factor

$\alpha = n/N$

	Worst Case	SUHA
Insert	O(1)	
Remove/Find	O(n) ≡ O(n/N)	

(Example of open hashing)

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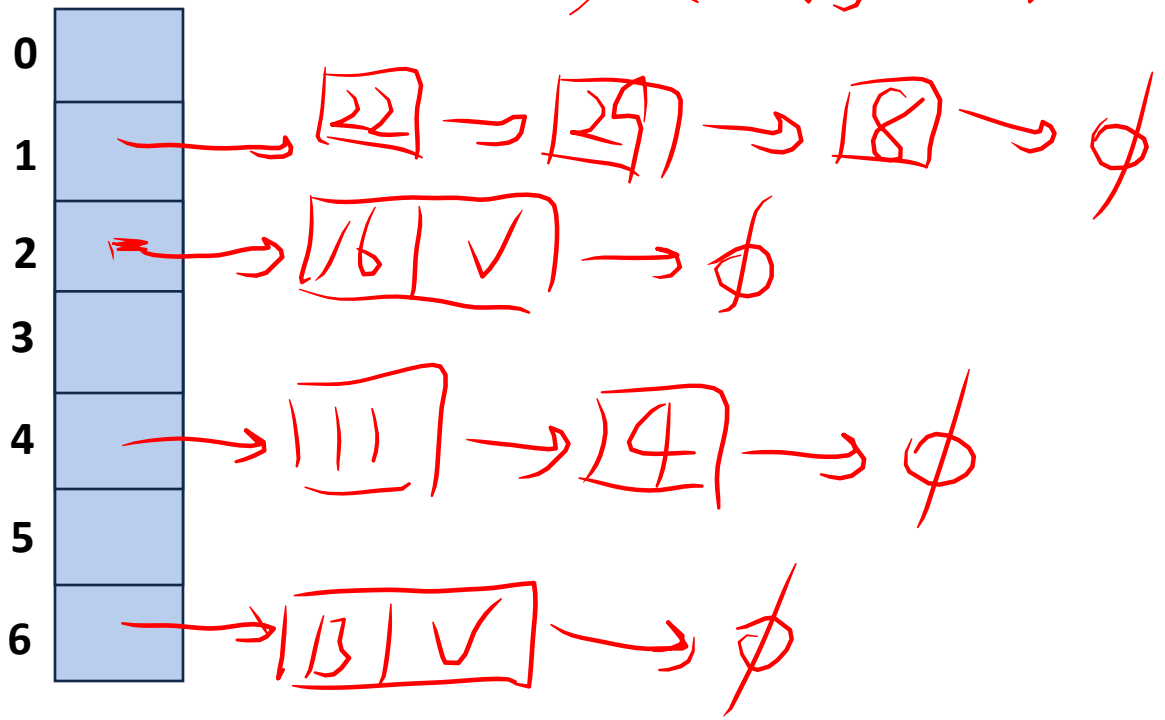


load factor

$\alpha = n/N$

$\alpha \Rightarrow$ unbounded

	Worst Case	SUHA
Insert	O(1)	
Remove/Find	O(n) ≡ O(n/N)	

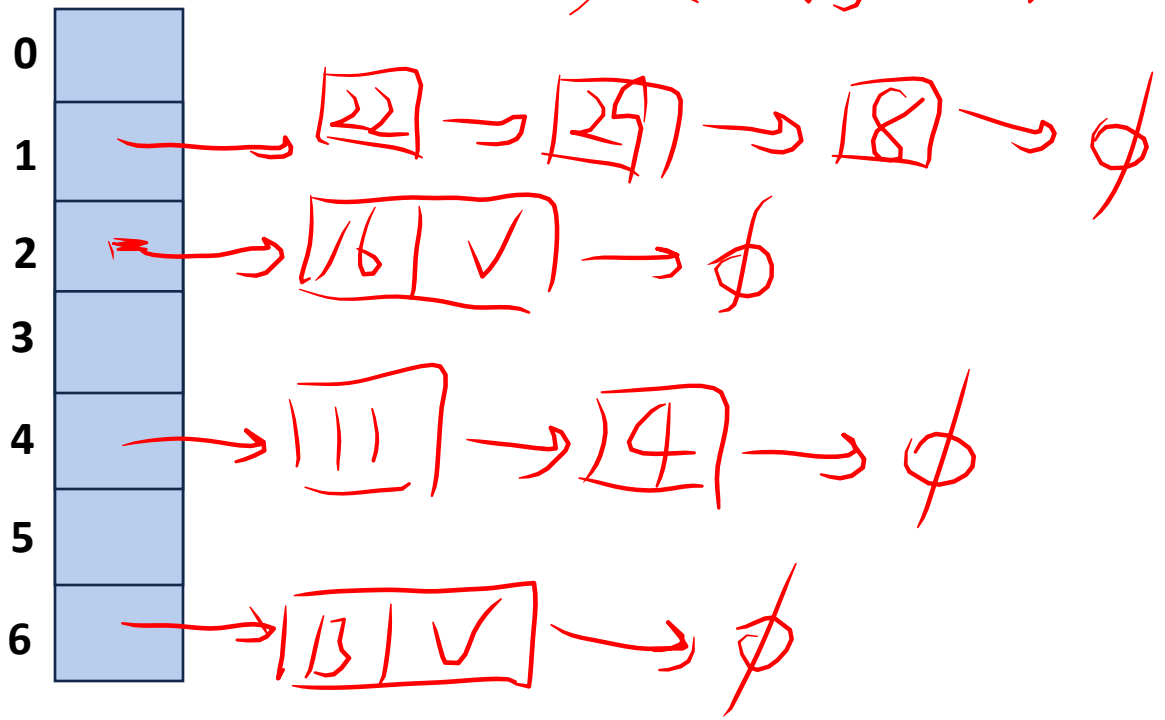
(Example of open hashing)

Collision Handling: Separate Chaining

Linked list of every node

S = { 16, 8, 4, 13, 29, 11, 22 } **|S| = n**

h(k) = k % 7 **|Array| = N**



load factor

$\alpha = n/N$

$\alpha \Rightarrow$ unbounded

open hashing

	Worst Case	SUHA
Insert	$O(1)$	
Remove/Find	$O(n) \equiv O(n/N)$	

(Example of closed hashing)

Collision Handling: Probe-based Hashing

S = { 16, 8, 4, 13, 29, 11, 22 } **|S| = n**

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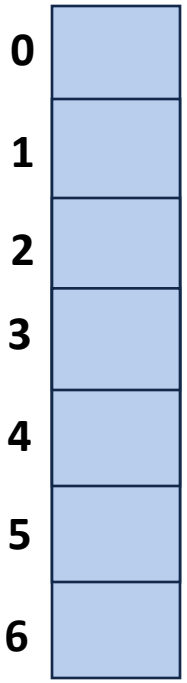

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(Example of closed hashing)

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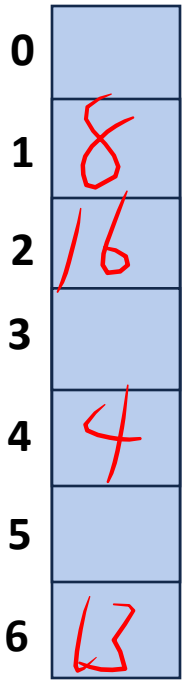

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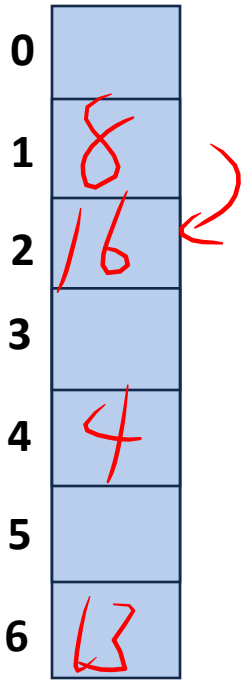

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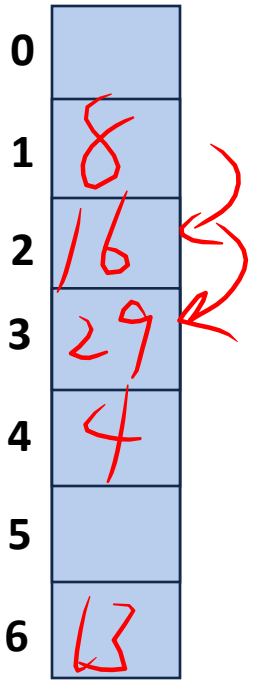

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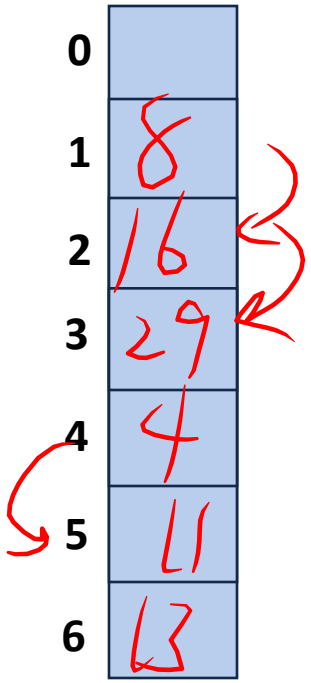

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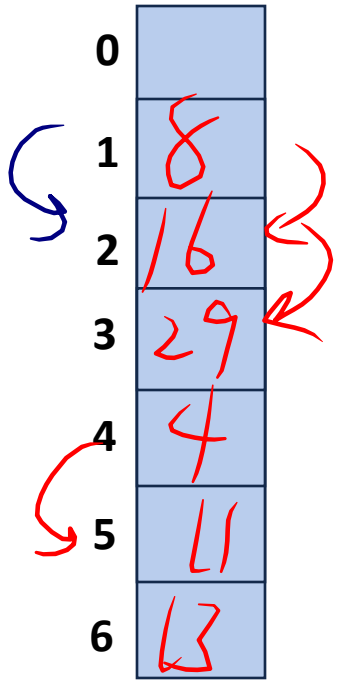

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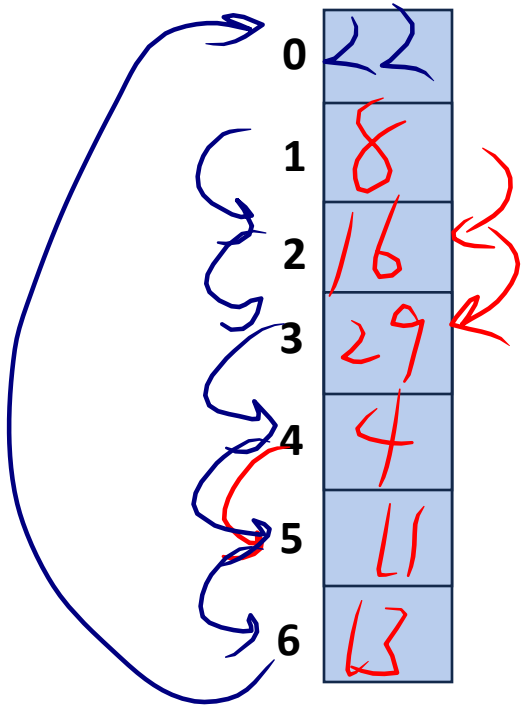

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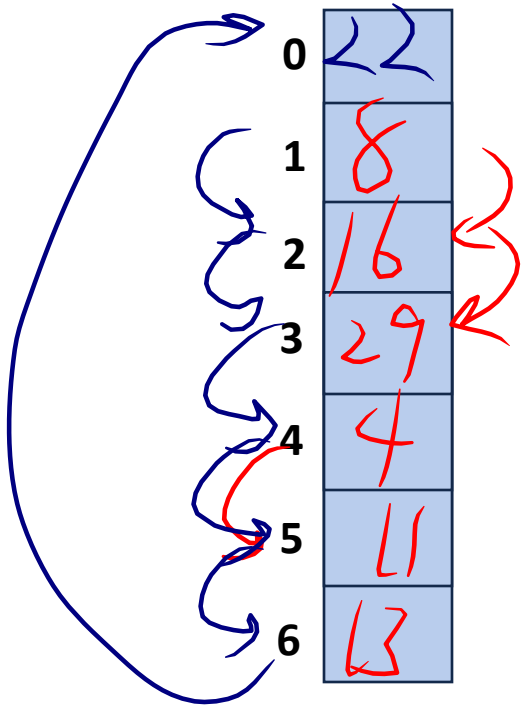

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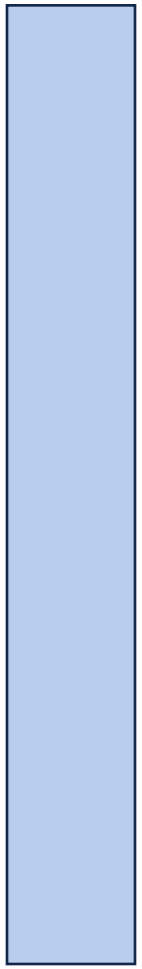
$2 \leq 1.0 \quad \frac{n}{N} = \frac{7}{7}$

A Problem w/ Linear Probing

Primary clustering:

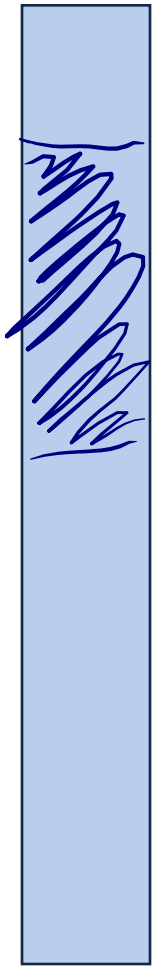


Description:

Remedy:

A Problem w/ Linear Probing

Primary clustering:



Description:

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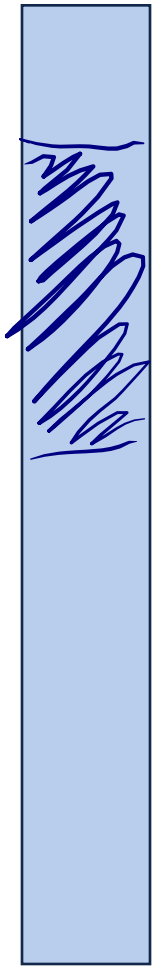
A Problem w/ Linear Probing

Primary clustering:



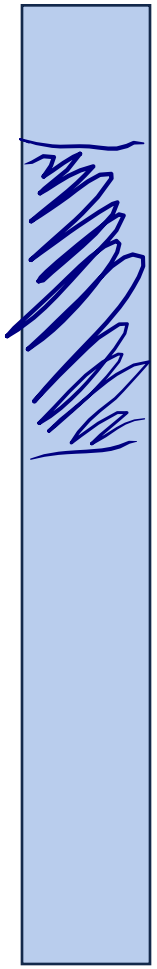
Description:

hash locations tend to cluster in primary clusters even under low load 2

Remedy:

A Problem w/ Linear Probing

Primary clustering:



Description:

hash locations tend to cluster in primary clusters even under low load 2

Remedy:

double hashing

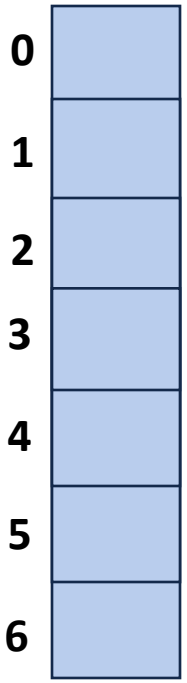
(Example of closed hashing)

Collision Handling: Double hashing

$S = \{ 16, 8, 4, 13, 29, 11, 22 \}$ $|S| = n$

$h_1(k) = k \% 7$ $|Array| = N$

$h_2(k) = 5 - k \% 5$



Try $h(k) = (k + 0*h_2(k)) \% 7$, if full…

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$$h(k, i) = (h_1(k) + i*h_2(k)) \% 7$$

(Example of closed hashing)

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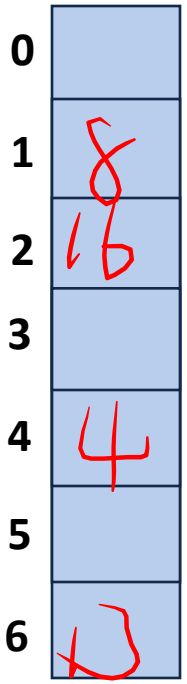

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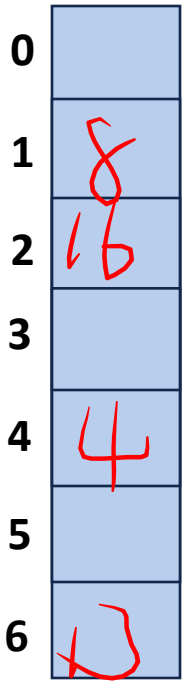

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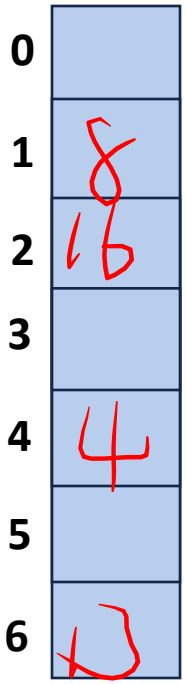

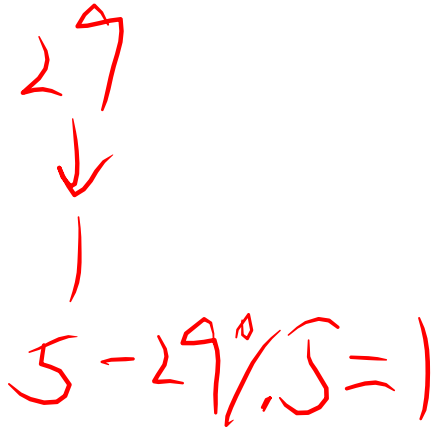

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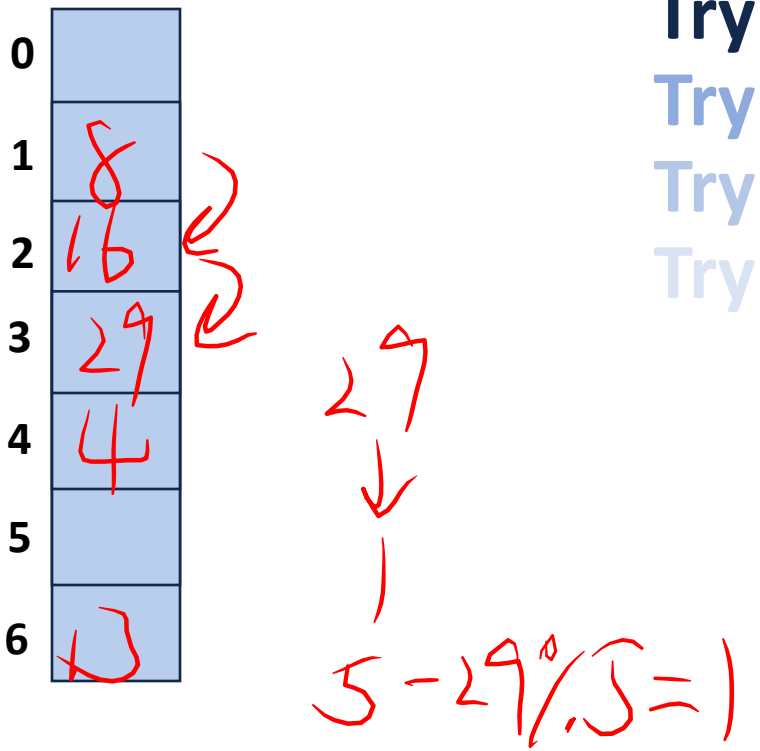

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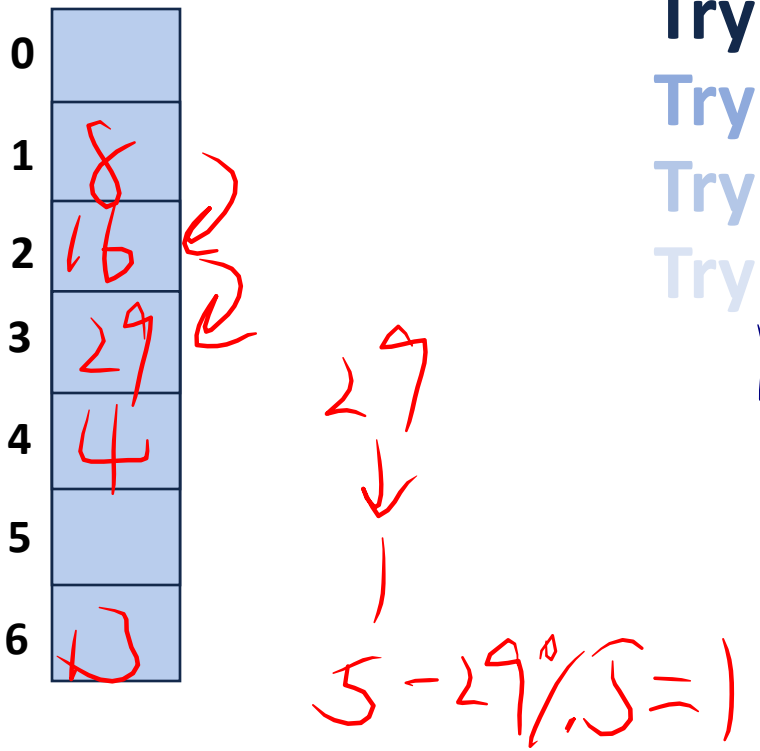

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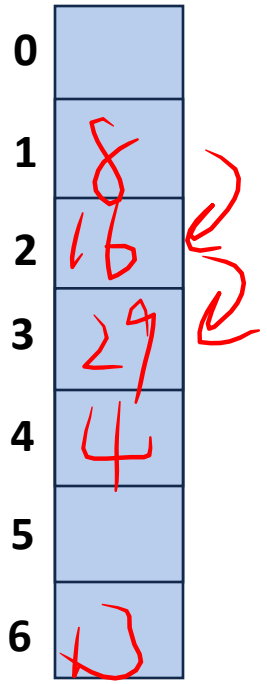

$h_2(11) = 4$

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Try …

29 → 1 → $5 - 29\%5 = 1$

11 → 4

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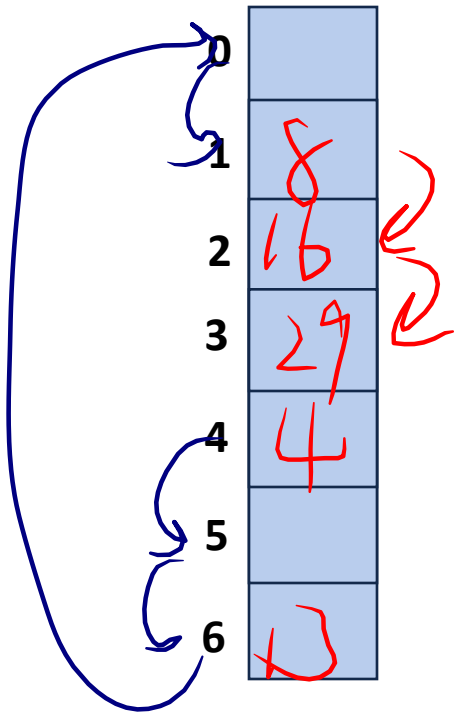

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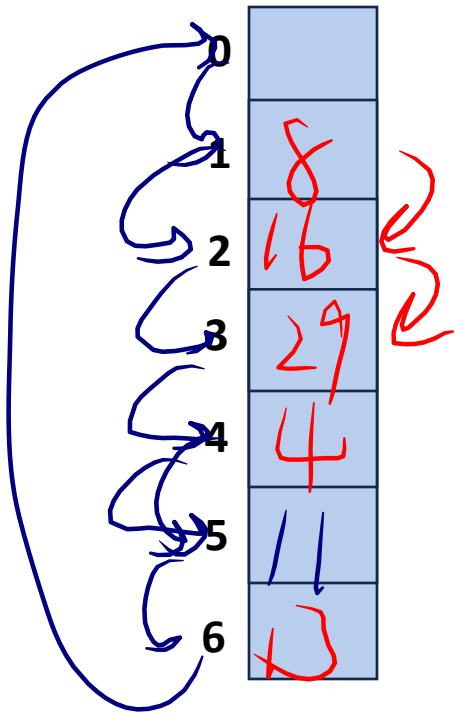

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Running Times

The expected number of probes for find(key) under SUHA

Linear Probing:

- Successful: **½(1 + 1/(1-α))**
- Unsuccessful: **½(1 + 1/(1-α))²**

Double Hashing:

- Successful: **1/α * ln(1/(1-α))**
- Unsuccessful: **1/(1-α)**

Separate Chaining:

- Successful: **1 + α/2**
- Unsuccessful: **1 + α**

(Don't memorize these equations, no need.)

Instead, observe:

- As α increases:

- If α is constant:

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Running time does not change

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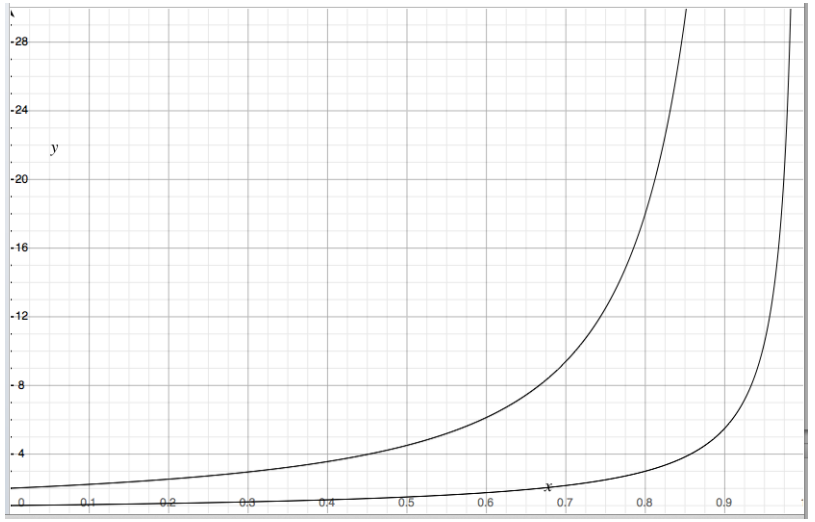

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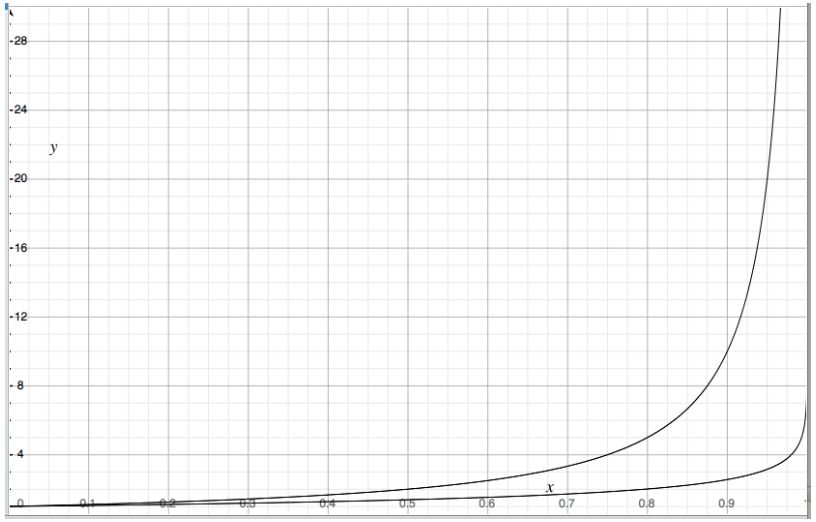

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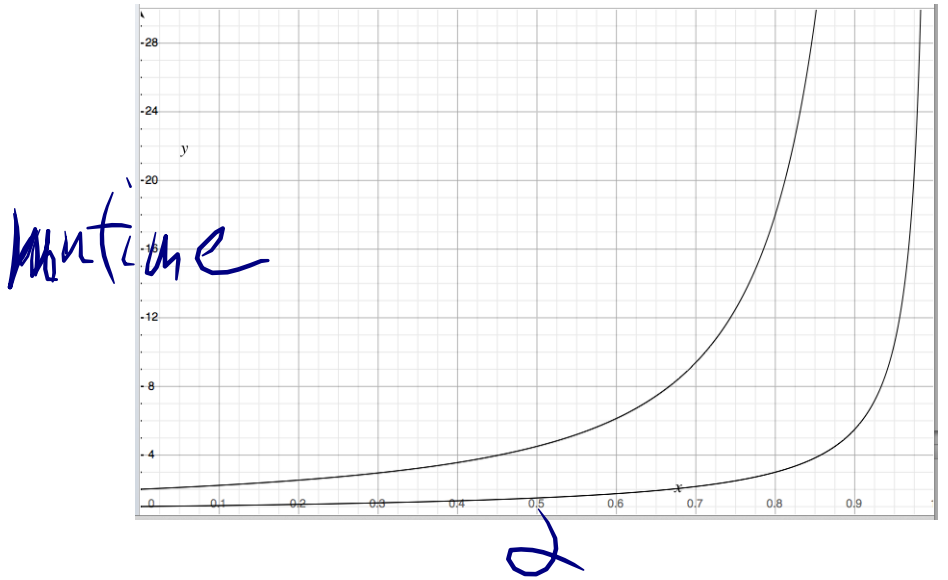

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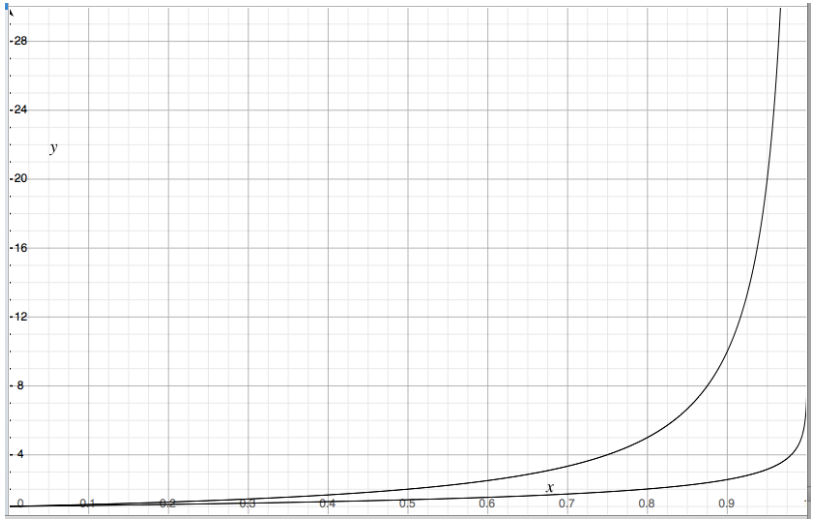

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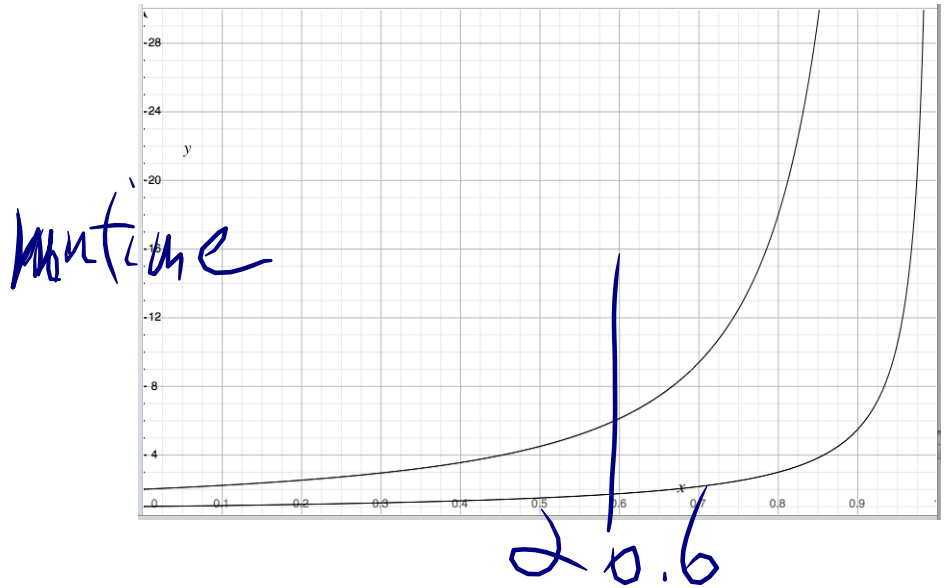

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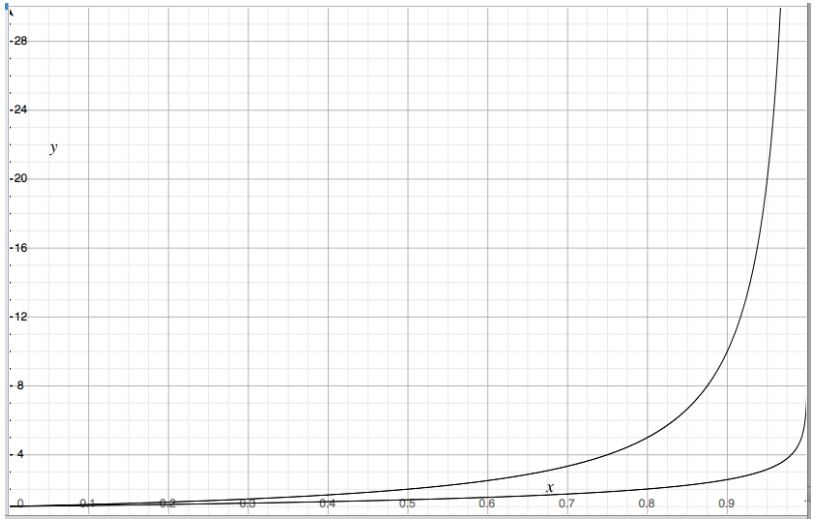
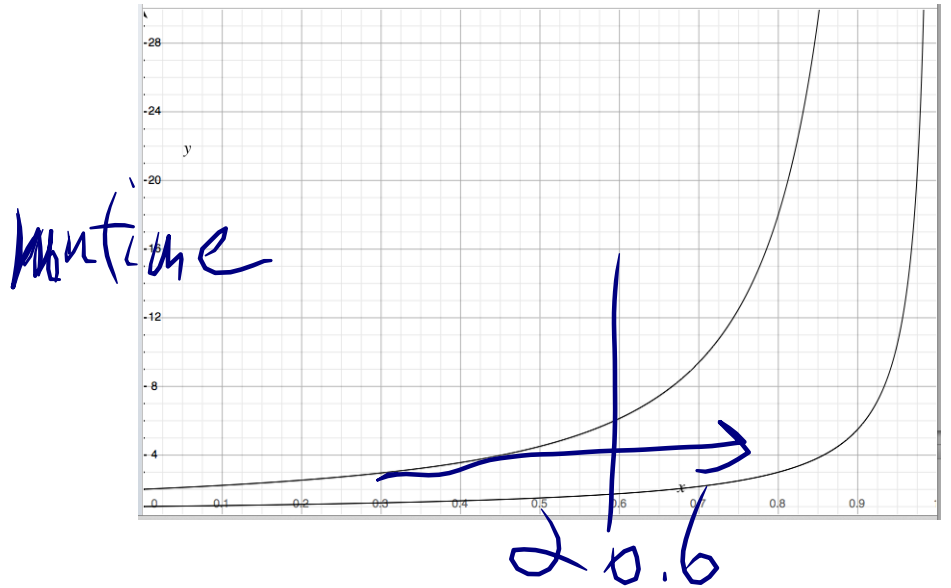


Running Times

The expected number of probes for find(key) under SUHA

Linear Probing:

- Successful: **½(1 + 1/(1-α))**
- Unsuccessful: **½(1 + 1/(1-α))²**



Double Hashing:

- Successful: **1/α * ln(1/(1-α))**
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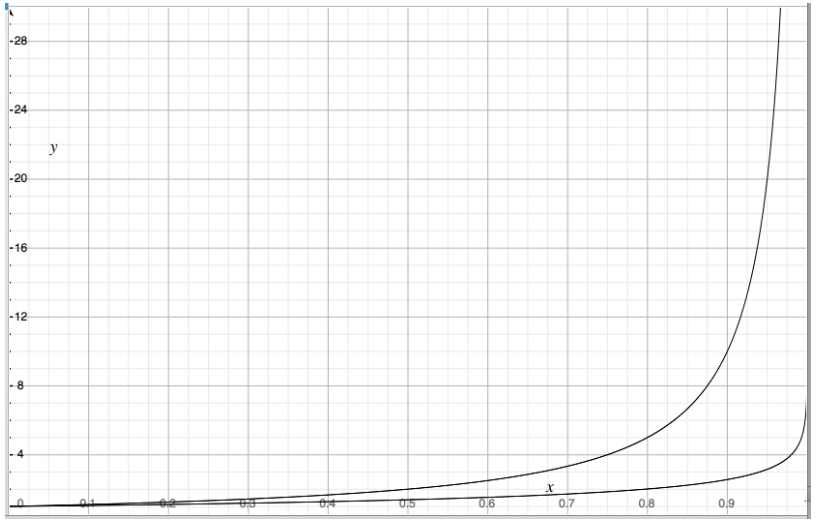

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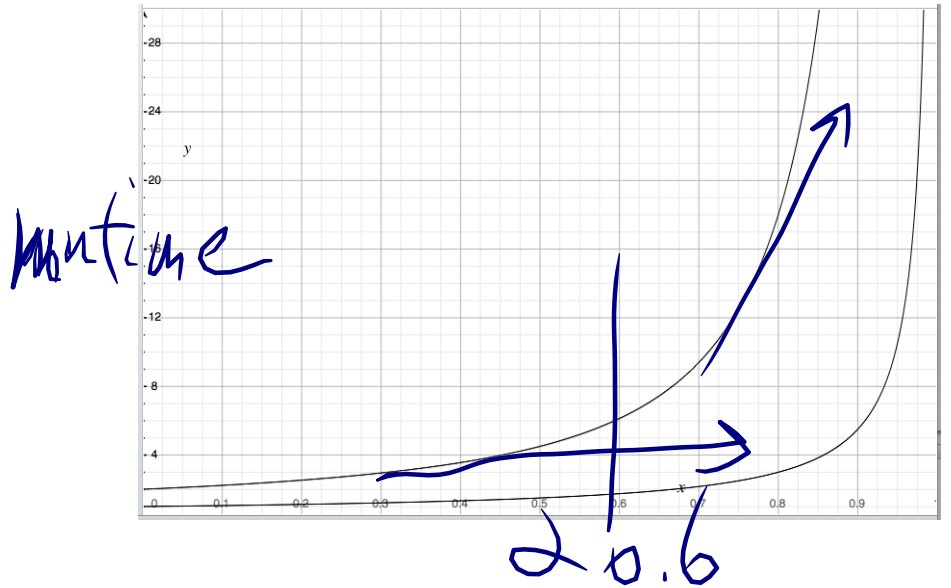

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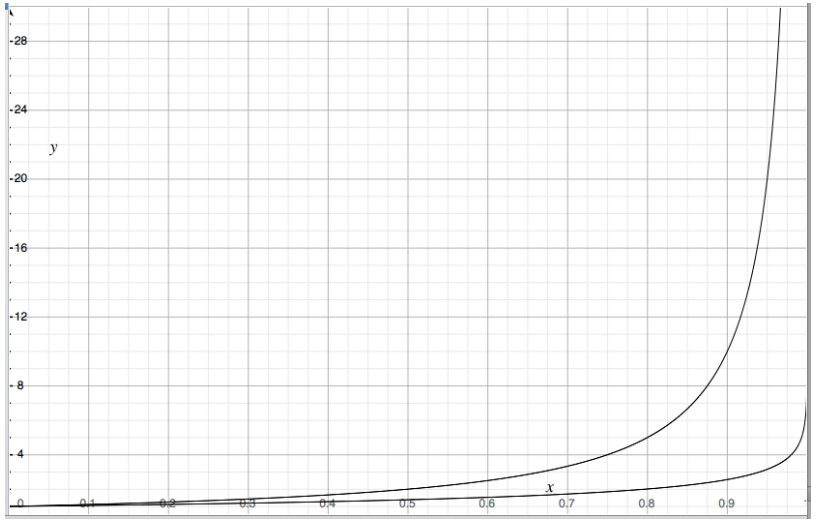

CS 225

Data Structures

Volodymyr Kindratenko

Slides courtesy of Wade Fagen-Ulmschneider

Running Times

The expected number of probes for find(key) under SUHA

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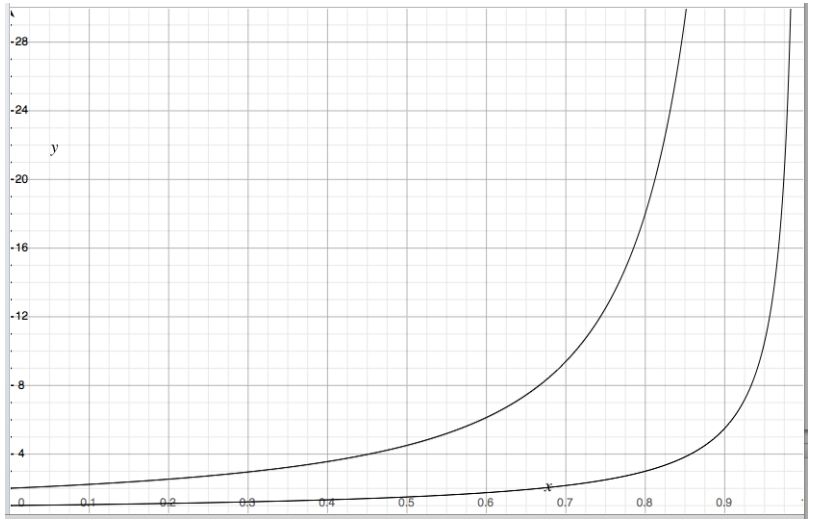

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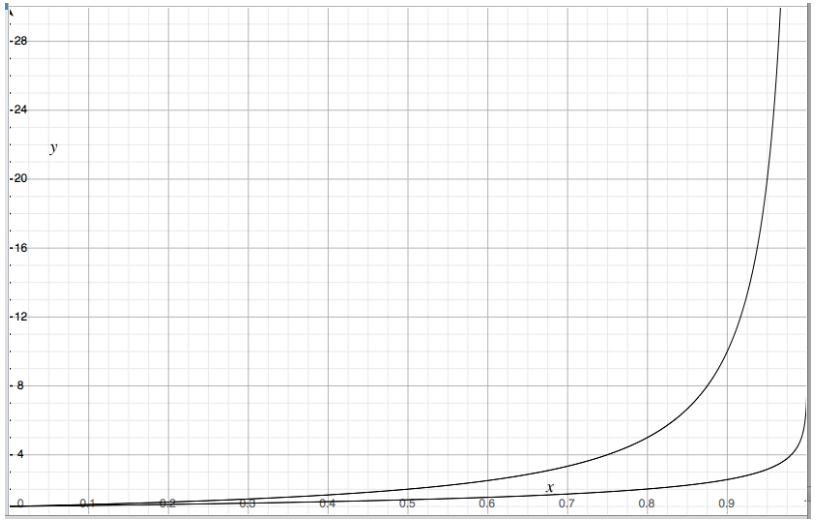

Running Times

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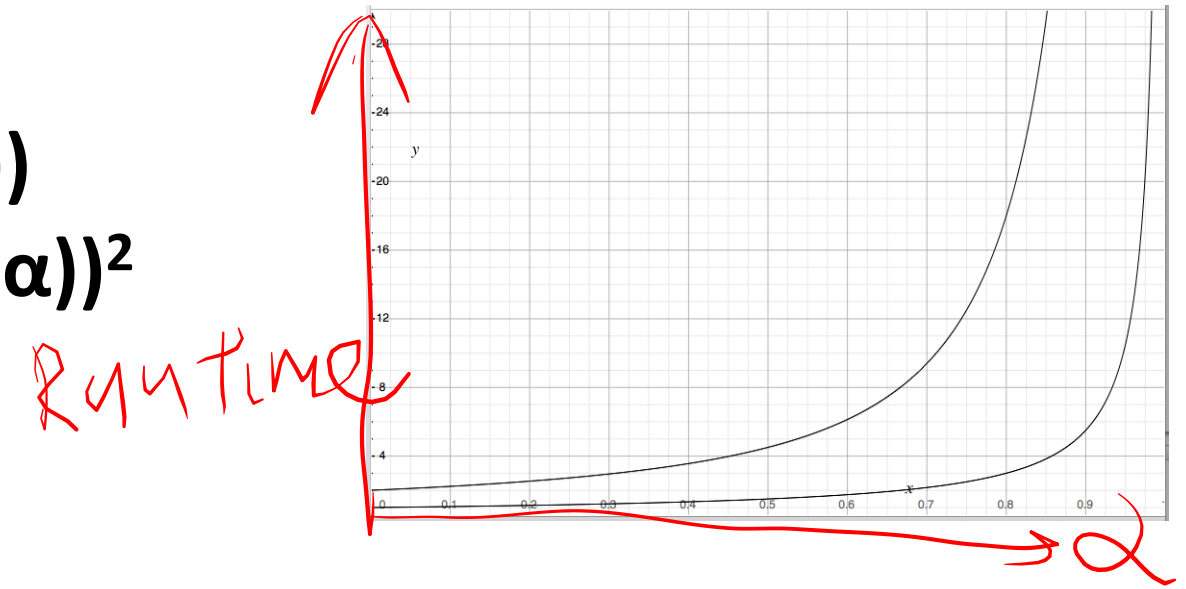

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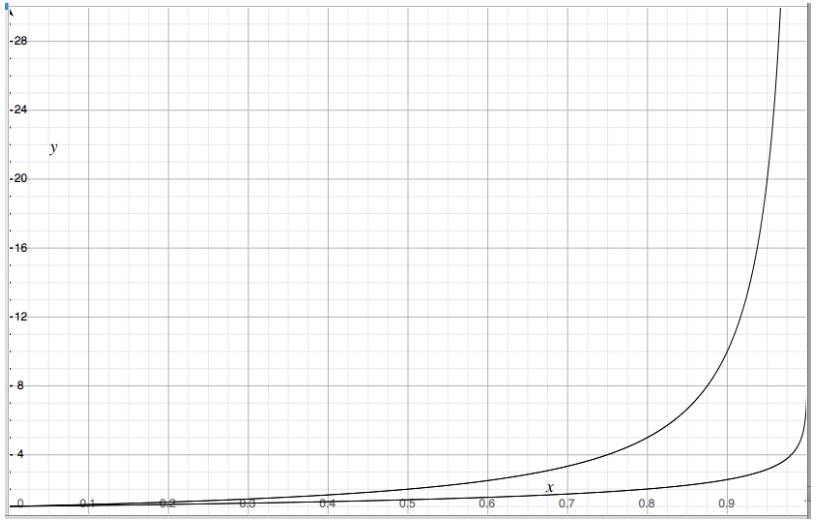

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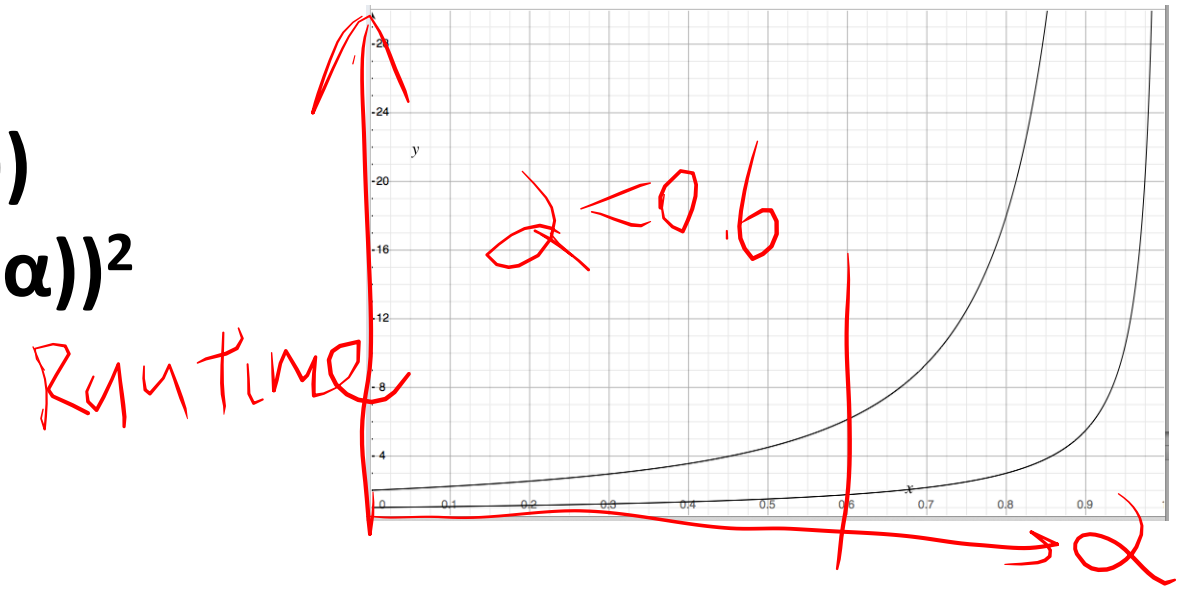

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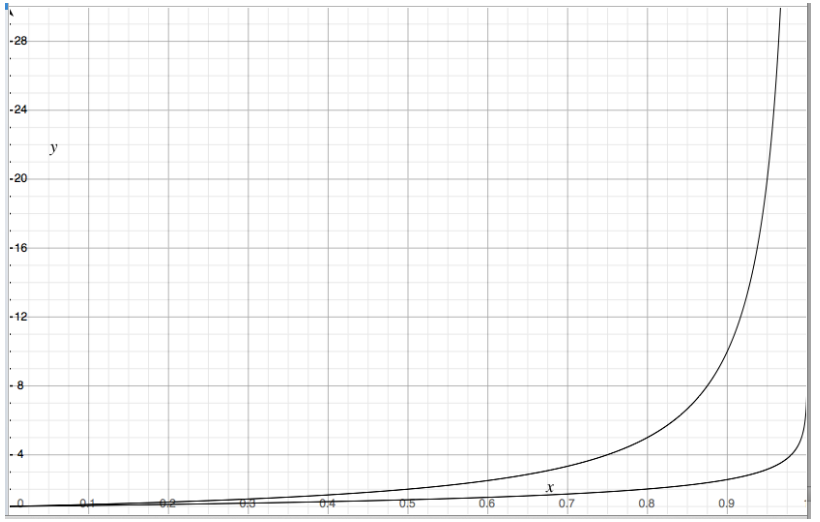

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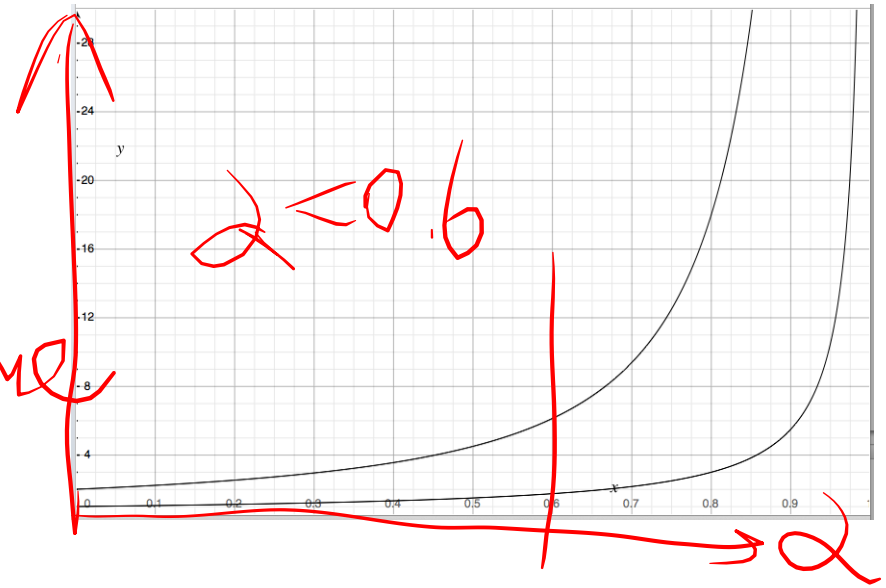

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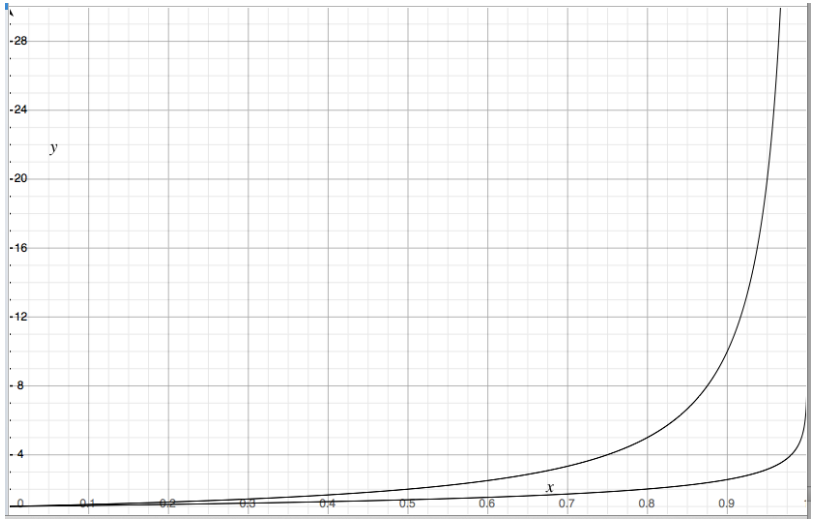

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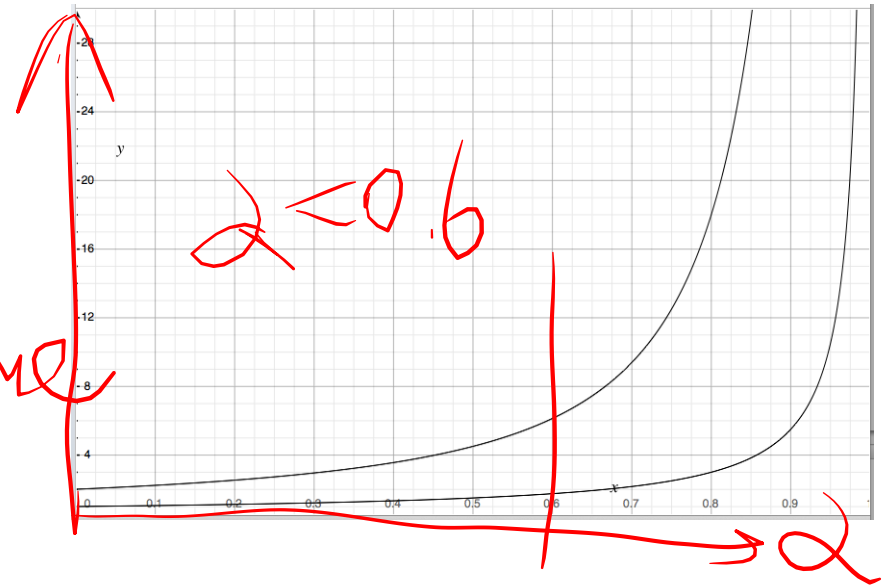

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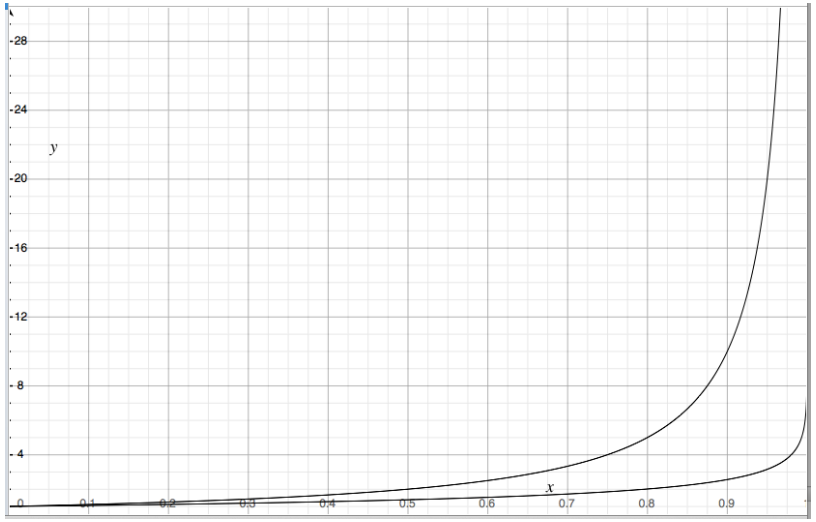
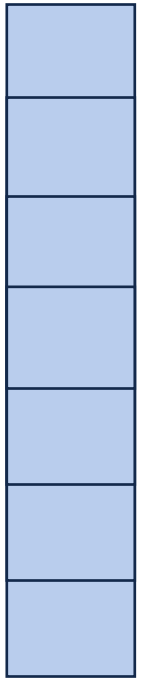
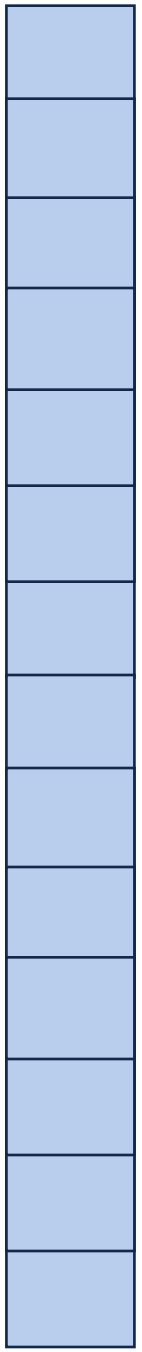


ReHashing

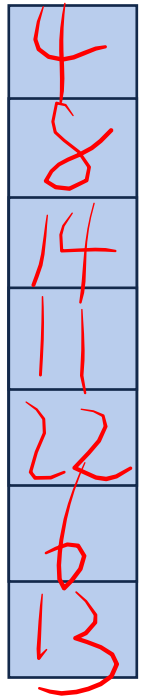
What if the array fills?

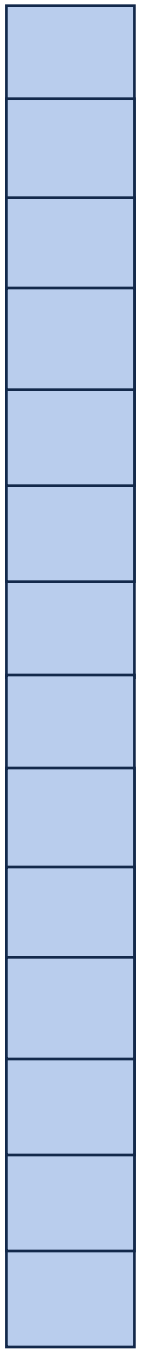




ReHashing

What if the array fills?

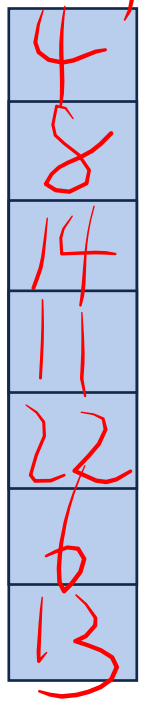


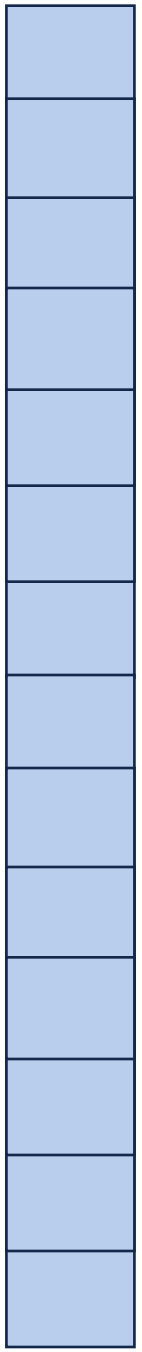


ReHashing

~~What if the array fills?~~

What should happen as $\alpha \to 0.6$?





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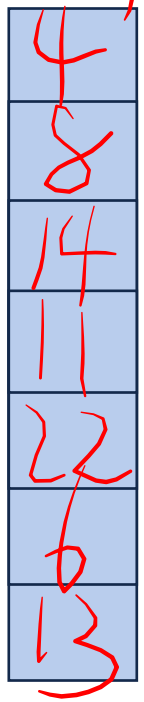

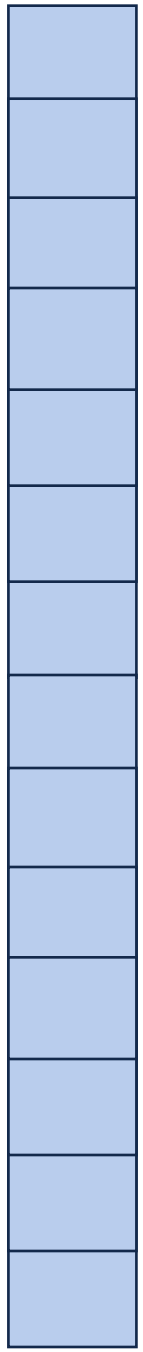

Double size

Copy elements

ReHashing

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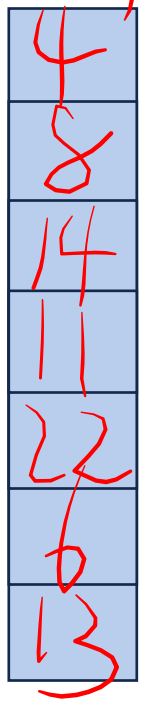
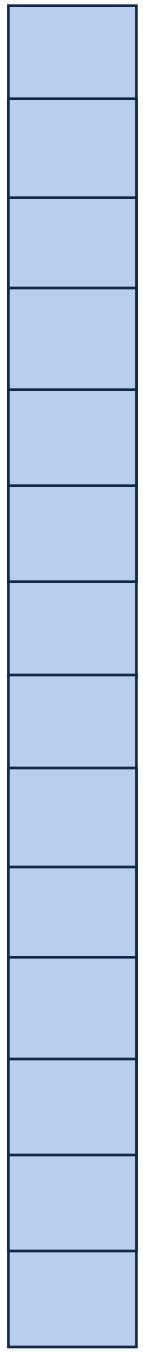
What should happen as $\alpha \to 0.6$?



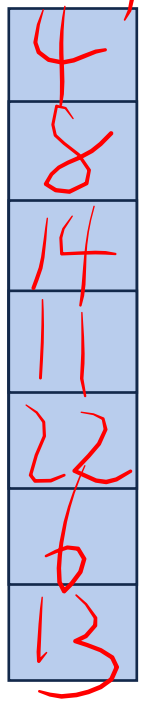


Double size
update hash f
Copy elements

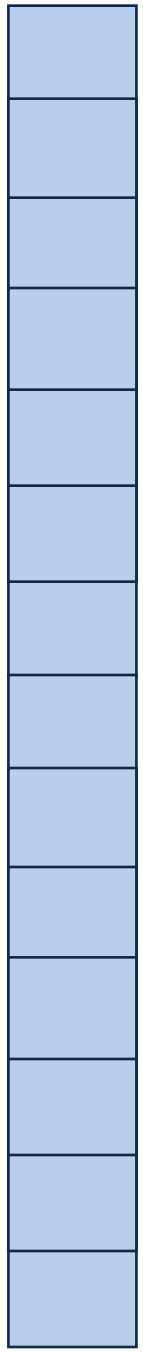
ReHashing

~~What if the array fills?~~

What should happen as $\alpha \to 0.6$?



$h(k) = k \% 7$



Double size
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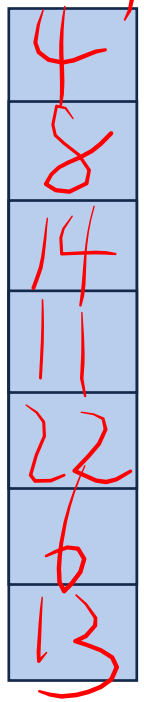

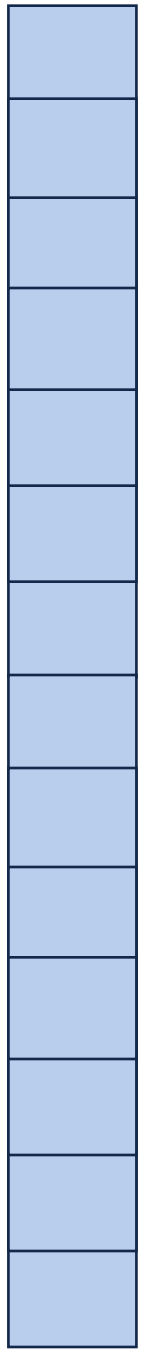

Double size
update hash f
Copy elements

$h(k) = k\%7$

$k\%11$

ReHashing

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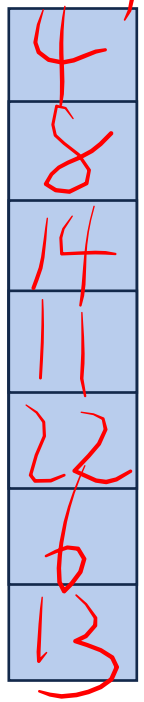
What should happen as $\alpha \to 0.6$?



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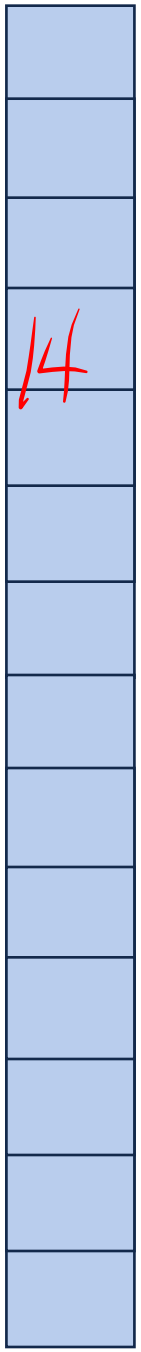

$k\%11$

Double size
update hash f
Copy elements

ReHashing

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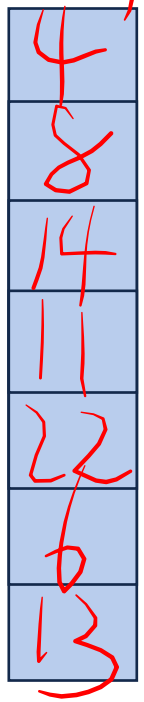
What should happen as $\alpha \to 0.6$?



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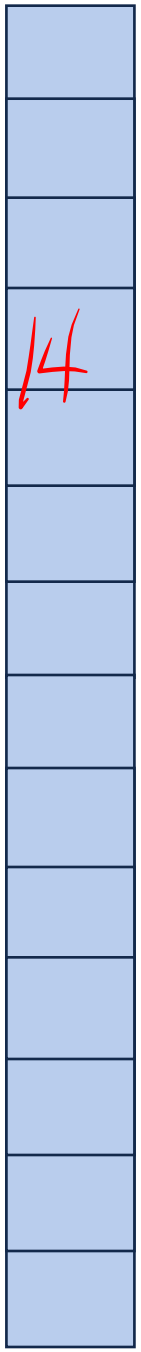


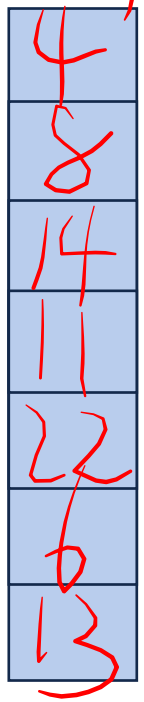
$k \% 11$

Double size
: update hash f
Copy elements
- Rehash elements

ReHashing

~~What if the array fills?~~

What should happen as $\alpha \to 0.6$?



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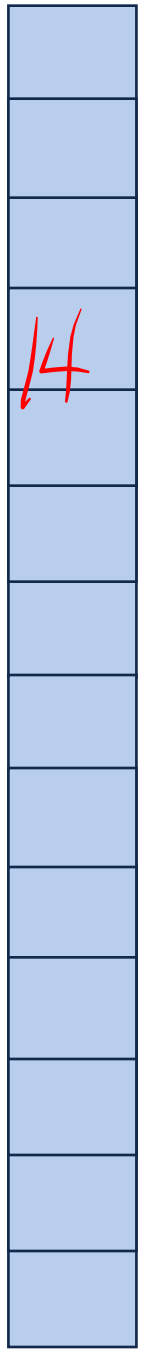


Double size
: update hash f

Copy elements

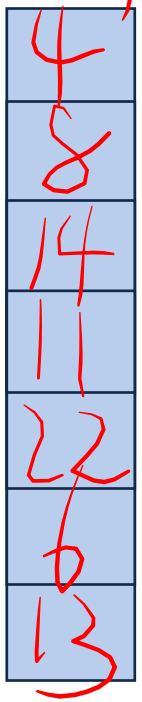
- Rehash elements
- Make to new location

$k \% 11$

ReHashing

~~What if the array fills?~~

What should happen as $2 \to 0.6$?



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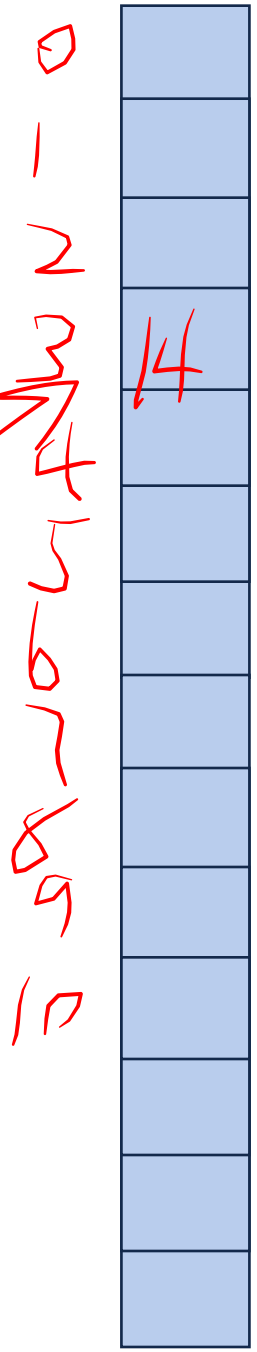


Double size
: update hash f

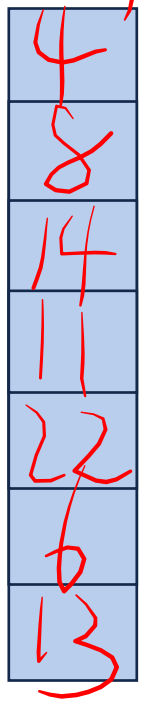
Copy elements
- Rehash elements
- Make to new location

$k\%11$

ReHashing

~~What if the array fills?~~

What should happen as 2 → 0.6?



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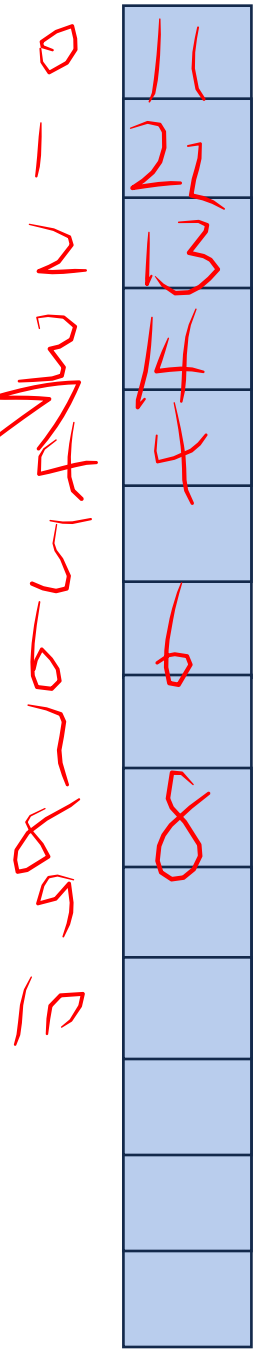

Double size
: update hash f

Copy elements
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- Make to new location

$k\%11$

ReHashing

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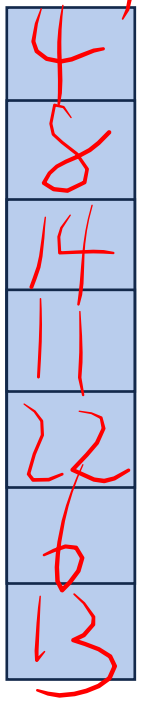
What should happen as a → 0.6?



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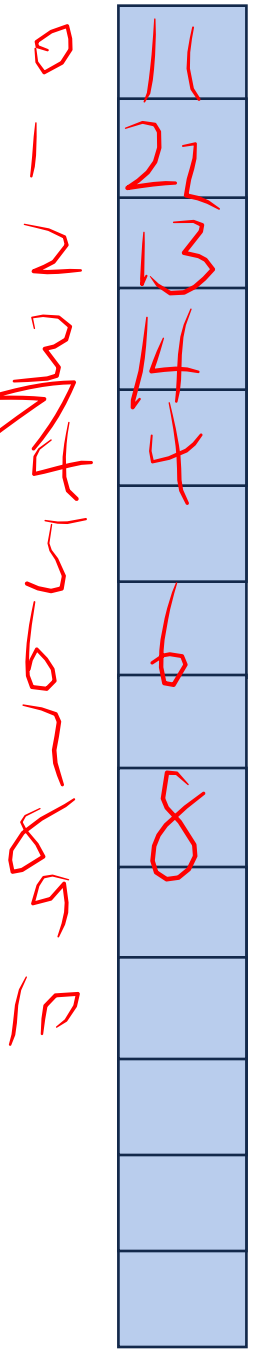

Double size
: update hash f

Copy elements
- Rehash elements
- Make to new location O(1)

$k\%11$

O(1)

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: Worst Case:		
Insert	Amortized: Worst Case:		
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: O(1) Worst Case:		
Insert	Amortized: Worst Case:		
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: O(1) Worst Case: O(n)		
Insert	Amortized: Worst Case:		
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: Worst Case:		
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case:		
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$		
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space			

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space			$O(n)$

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space		$O(n)$	$O(n)$

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space		$O(n)$	$O(n)$ 1 xptr/elem

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space		$O(n)$ 2 x ptrs / elem	$O(n)$ 1 x ptr / elem

Running Times

	Hash Table	AVL	Linked List
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Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space	$N = \frac{n}{0.6} \approx 1.5n$	$O(n)$ 2 x ptrs/elem	$O(n)$ 1 x ptr/elem

Running Times

	Hash Table	AVL	Linked List
Find	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(n)$
Insert	Amortized: $O(1)$ Worst Case: $O(n)$	$O(\lg(n))$	$O(1)$
Storage Space	$N = \frac{n}{0.6} \approx 1.5n$ $O(n)$	$O(n)$ 2 xptrs/elem	$O(n)$ 1 xptr/elem

Which collision resolution strategy is better?

- Big Records:

- Structure Speed:

What structure do hash tables replace?

What constraint exists on hashing that doesn't exist with BSTs?

Why talk about BSTs at all?

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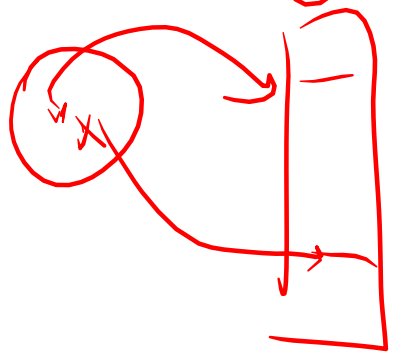

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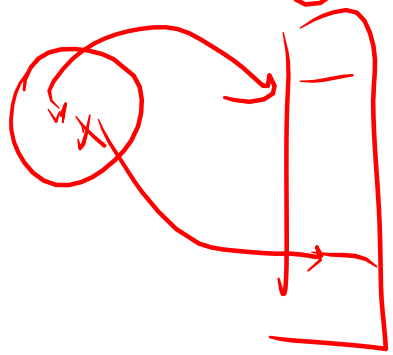


★ Nearest neighbor

Ranges [10, 100]

std data structures

std::map

std data structures

std::map

operator[]

insert

evase

std data structures

std::map

::operator[]
::insert
::erase

::lower_bound(key) ➔ Iterator to first element ≤ key
::upper_bound(key) ➔ Iterator to first element > key

std data structures

→ Impl w/ a balanced BST

std::map

::operator[]

::insert

::erase

::lower_bound(key) ➔ Iterator to first element ≤ key

::upper_bound(key) ➔ Iterator to first element > key

std data structures

std::unordered_map

::operator[]

::insert

::erase

~~– ::lower_bound(key) → Iterator to first element ≤ key~~

~~– ::upper_bound(key) → Iterator to first element > key~~

std data structures

std::unordered_map

::operator[]

::insert

::erase

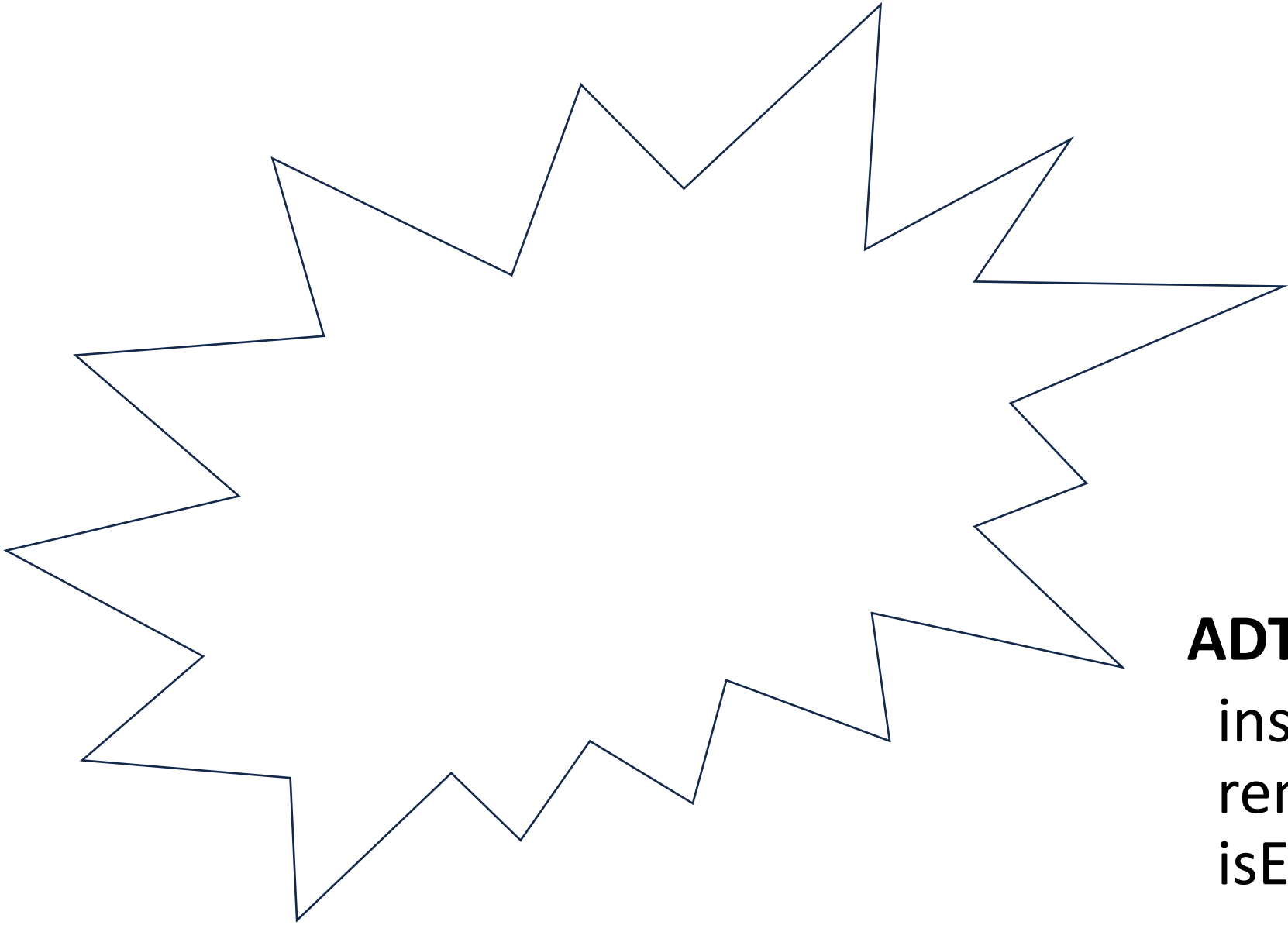
~~::lower_bound(key) → Iterator to first element ≤ key~~

~~::upper_bound(key) → Iterator to first element > key~~

::load_factor()

::max_load_factor(ml) → Sets the max load factor

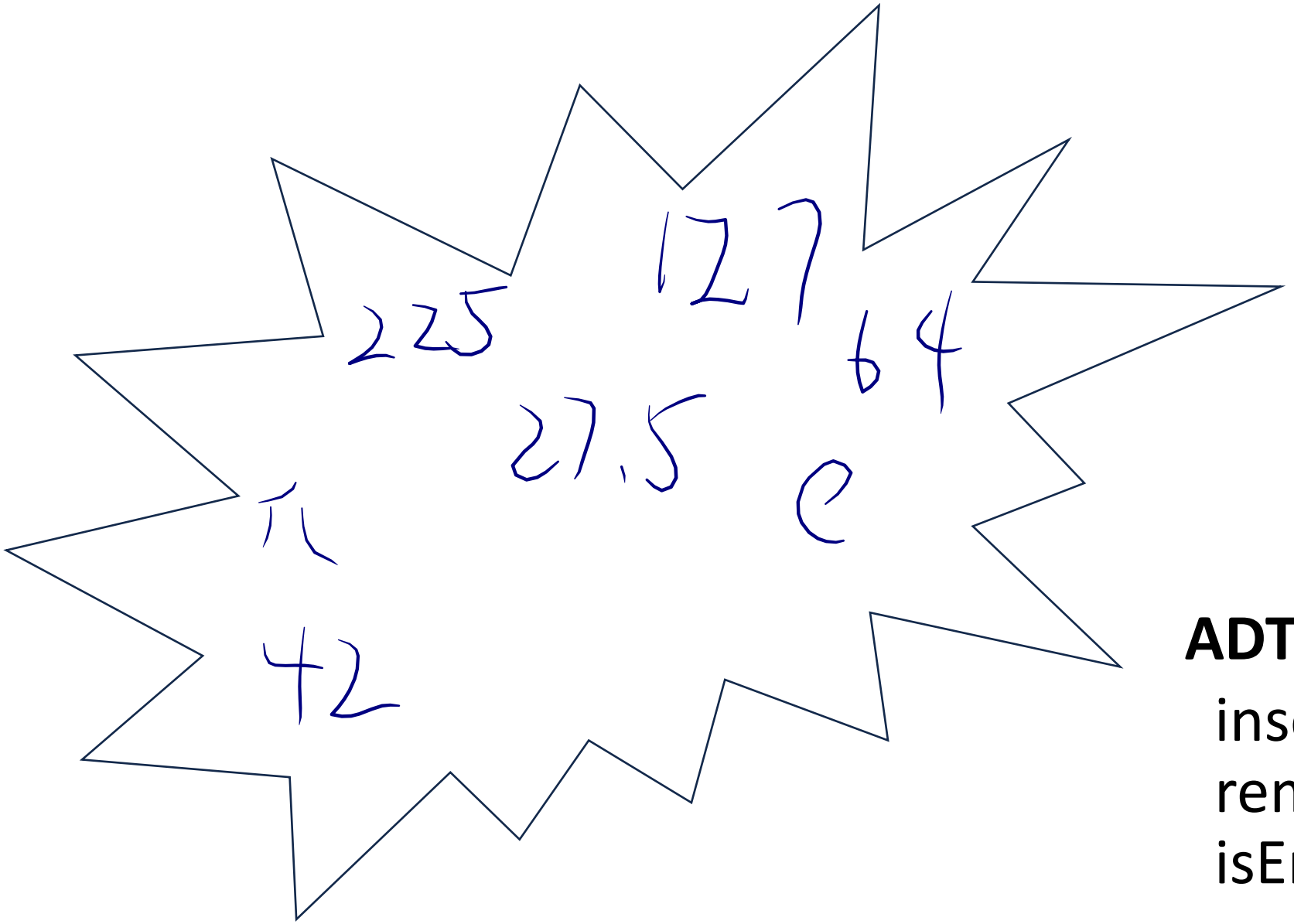
Secret, Mystery Data Structure



ADT:

insert
remove
isEmpty

Secret, Mystery Data Structure



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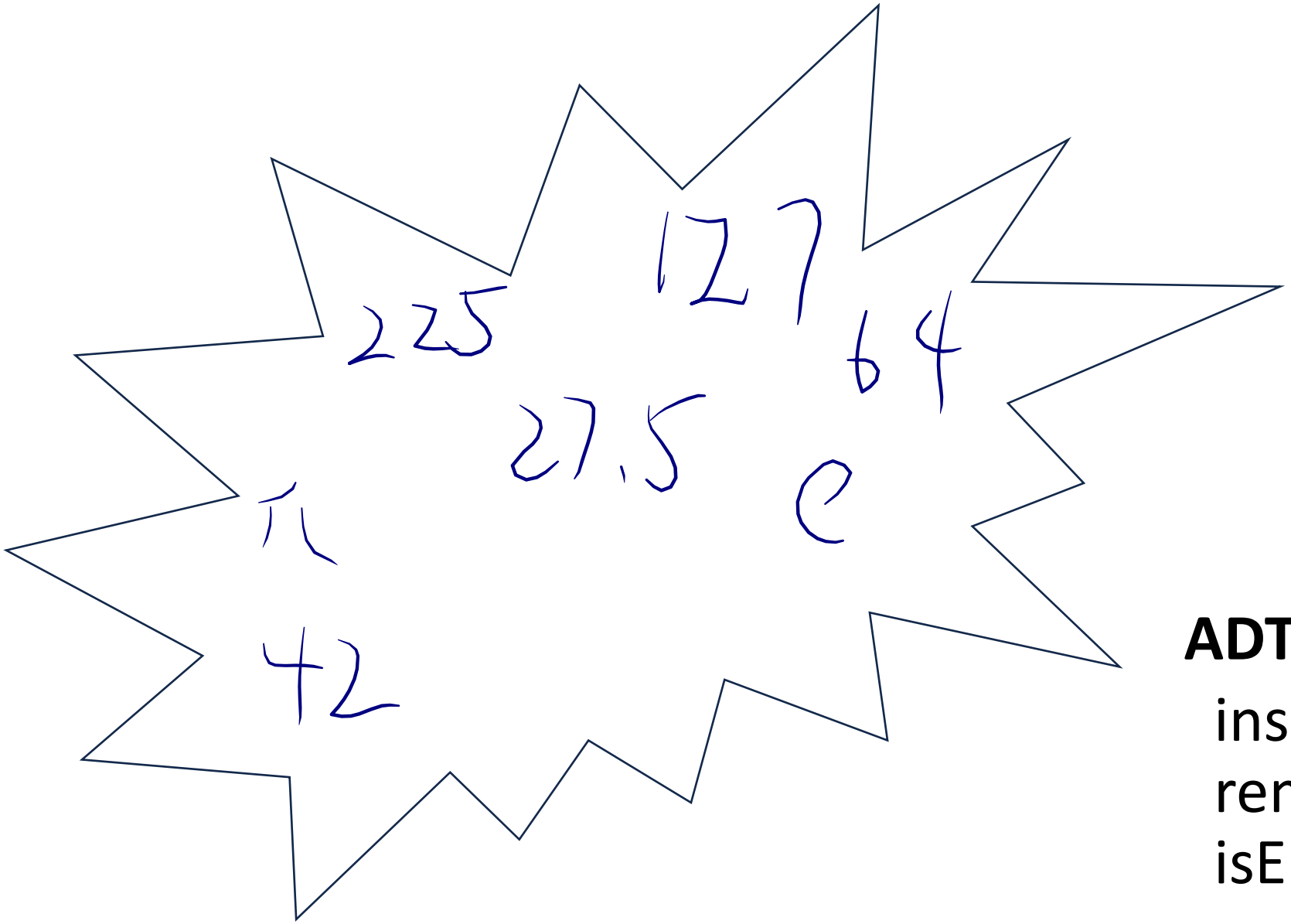


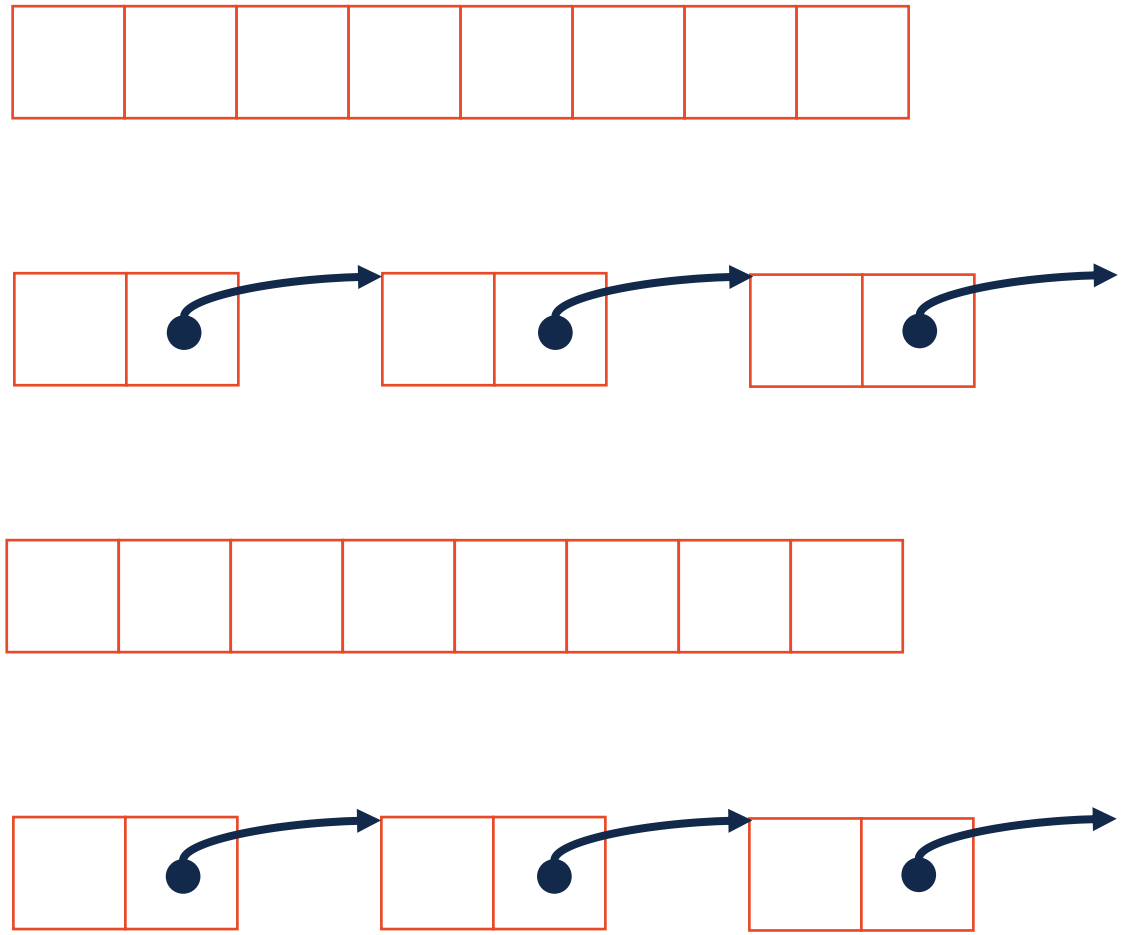
ADT:

insert
remove
isEmpty

min()

Priority Queue Implementation

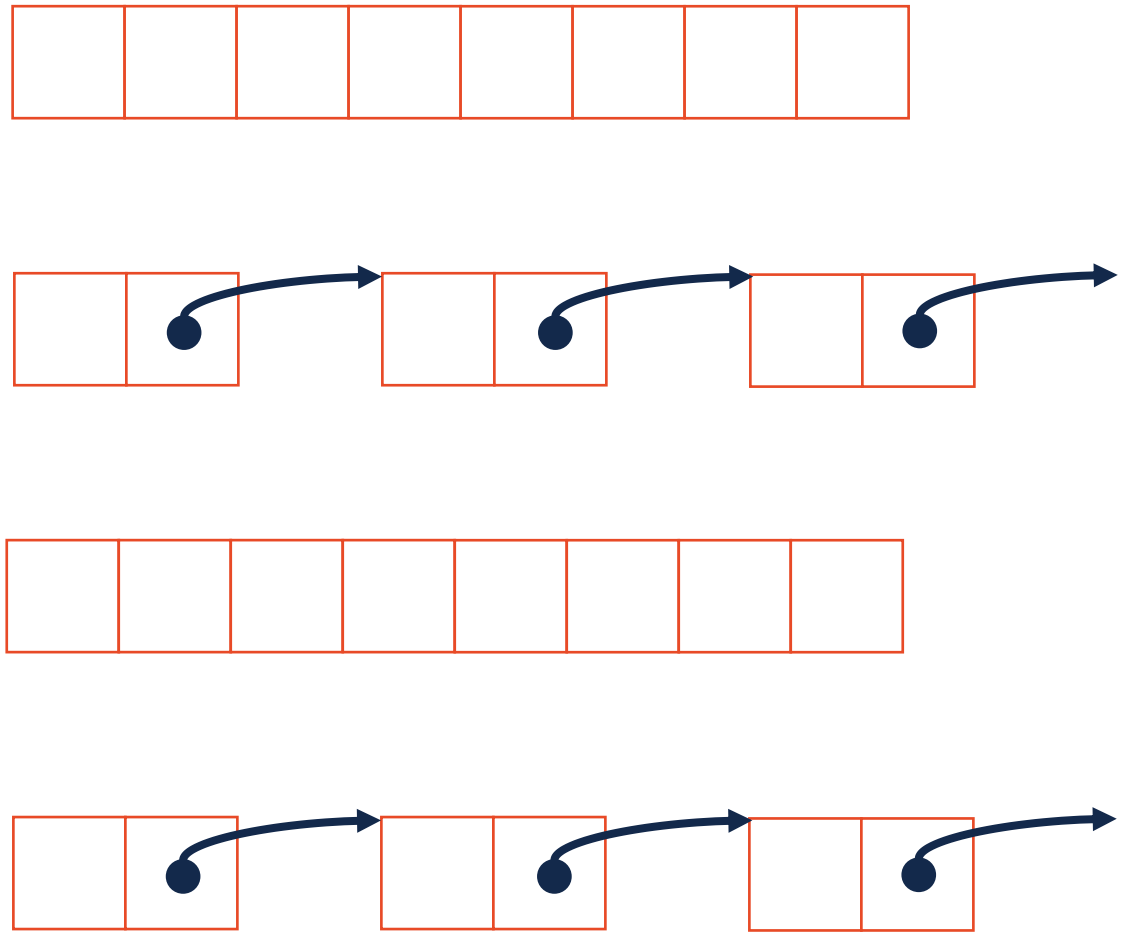
insert	removeMin
O(n)	O(n)
O(1)	O(n)
O(lg(n))	O(1)
O(lg(n))	O(1)



Priority Queue Implementation

insert	removeMin
O(n)	O(n)
O(1)	O(n)
~~O(lg(n))~~ O(4)	O(1)
O(lg(n))	O(1)

find O(lgn)
insert push

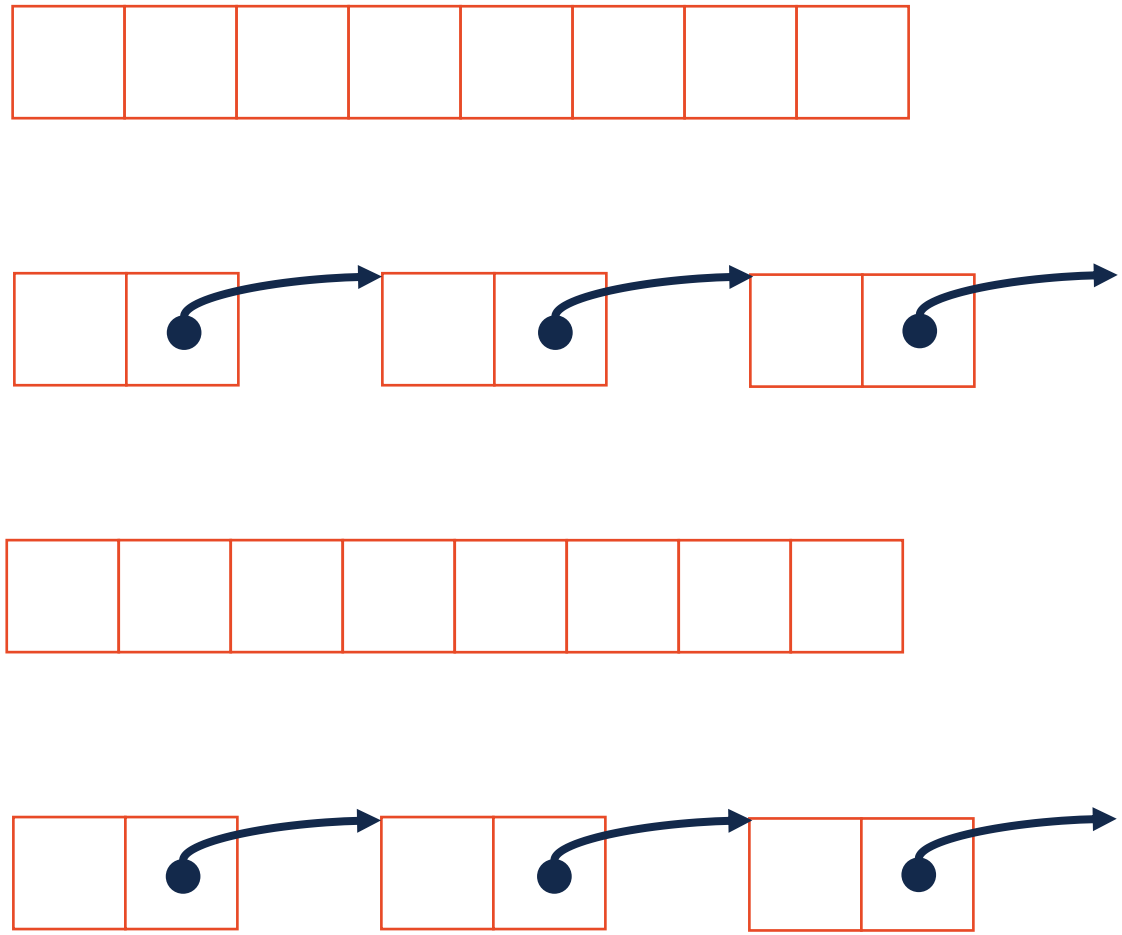


Priority Queue Implementation

insert	removeMin
O(n)	O(n)
O(1)	O(n)
~~O(lg(n))~~ O(n)	O(1)
~~O(lg(n))~~ O(n)	O(1)

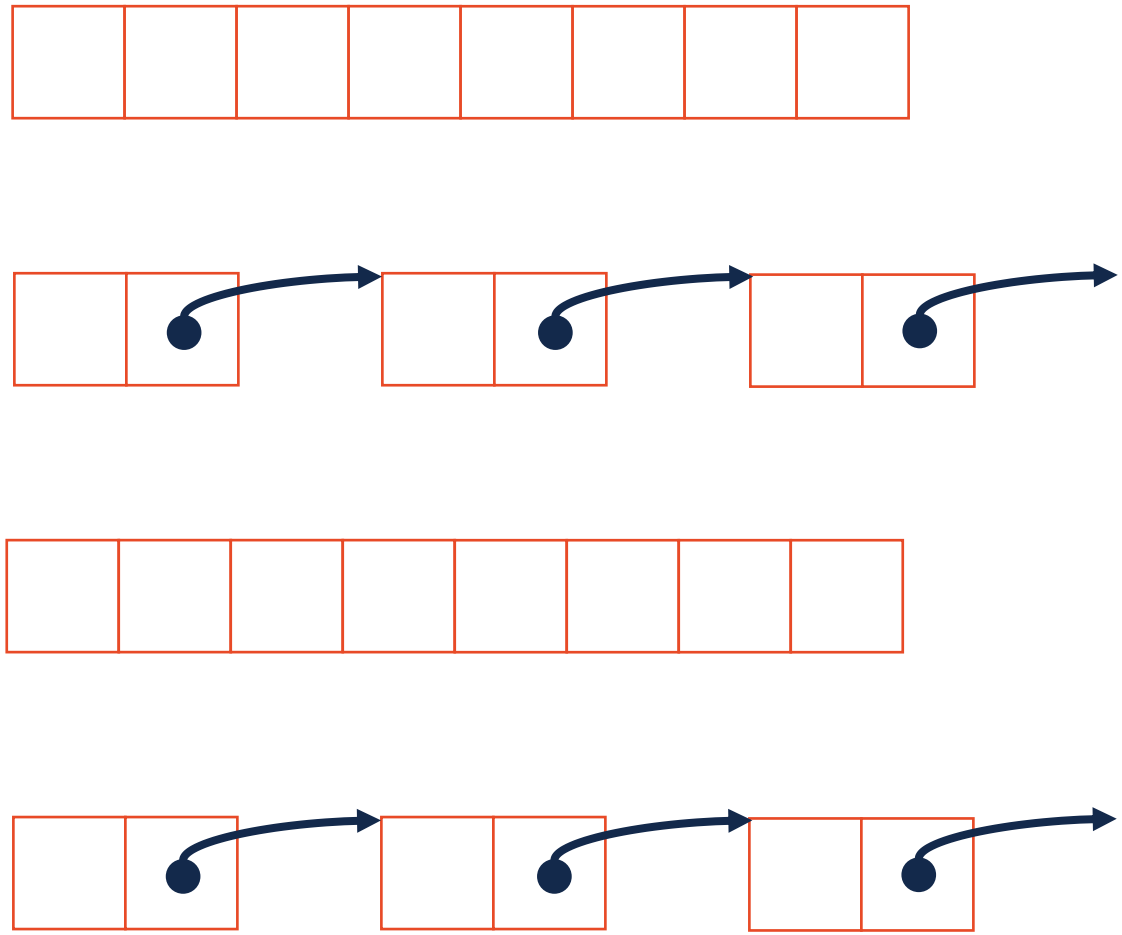
find O(lgn)
insert push O(1)

find O(n)
insert O(1)
+ O(n)



Priority Queue Implementation

insert	removeMin
~~O(n)~~ O(1)	O(n)
O(1)	O(n)
~~O(lg(n))~~ O(4) find O(lgn) insert push	O(1)
~~O(lg(n))~~ find O(n) insert O(1) + O(n)	O(1)



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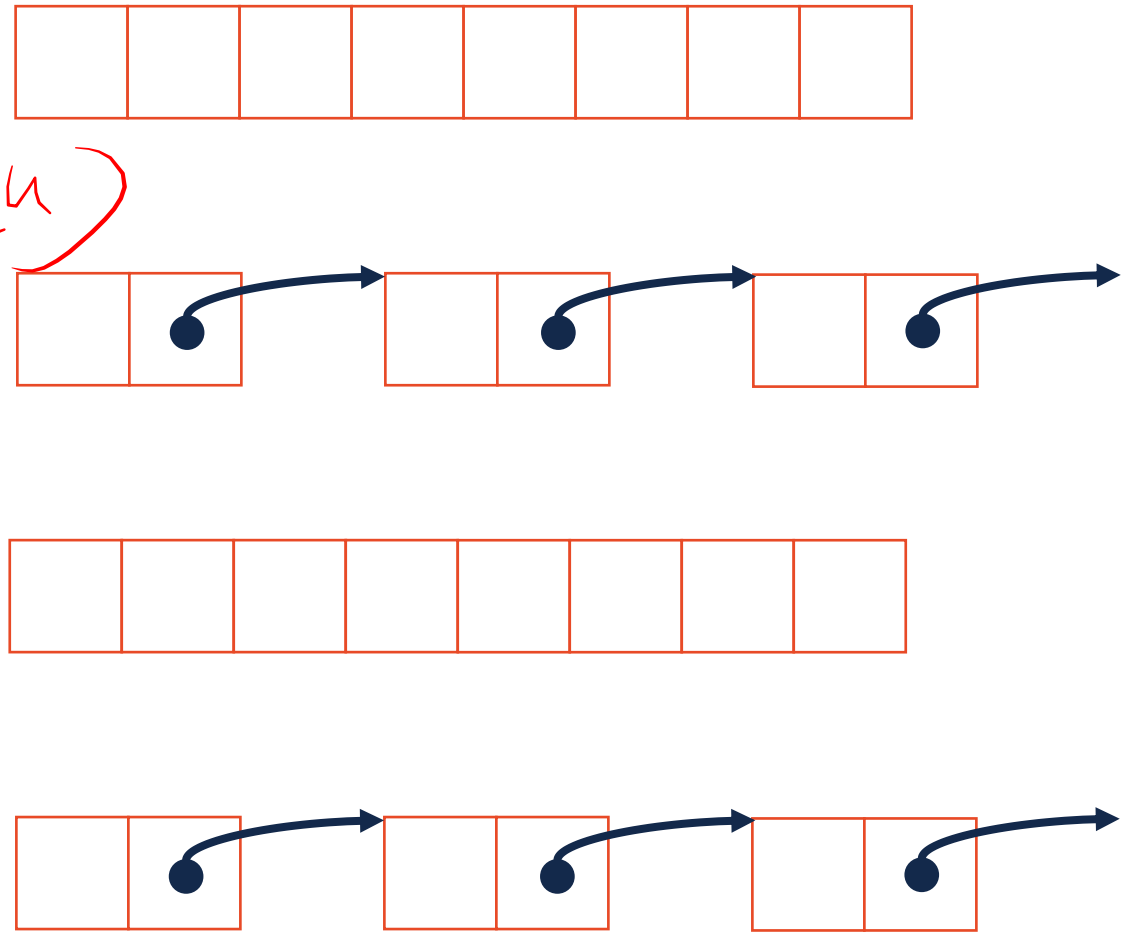
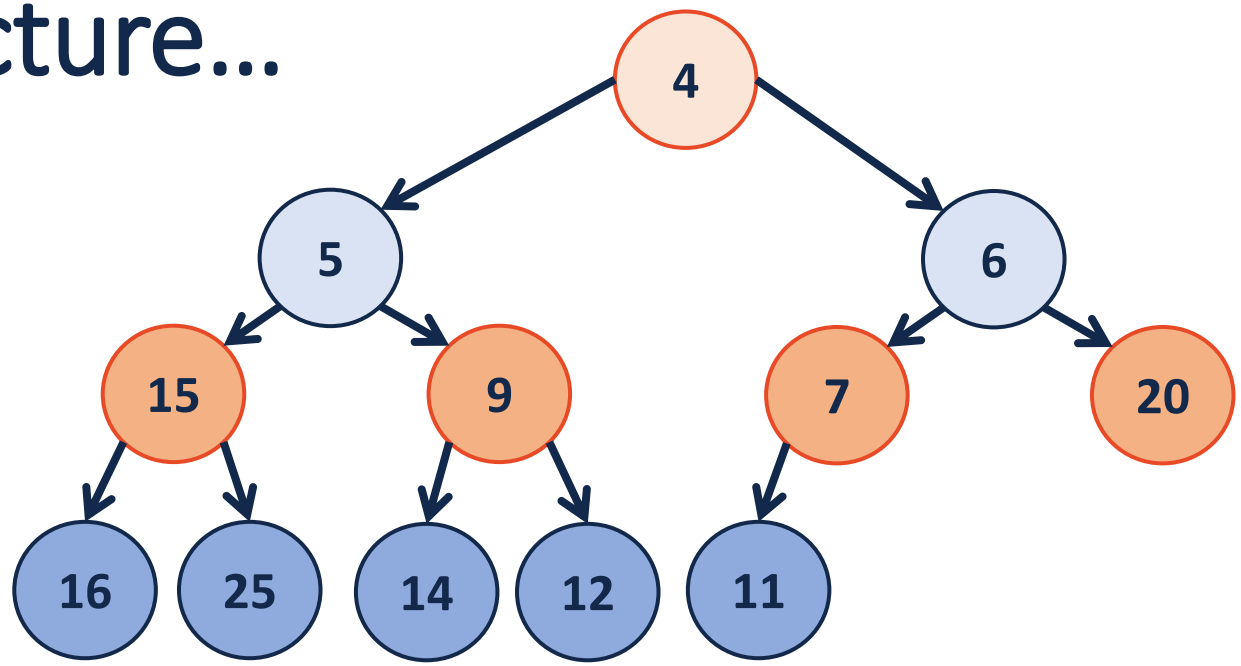
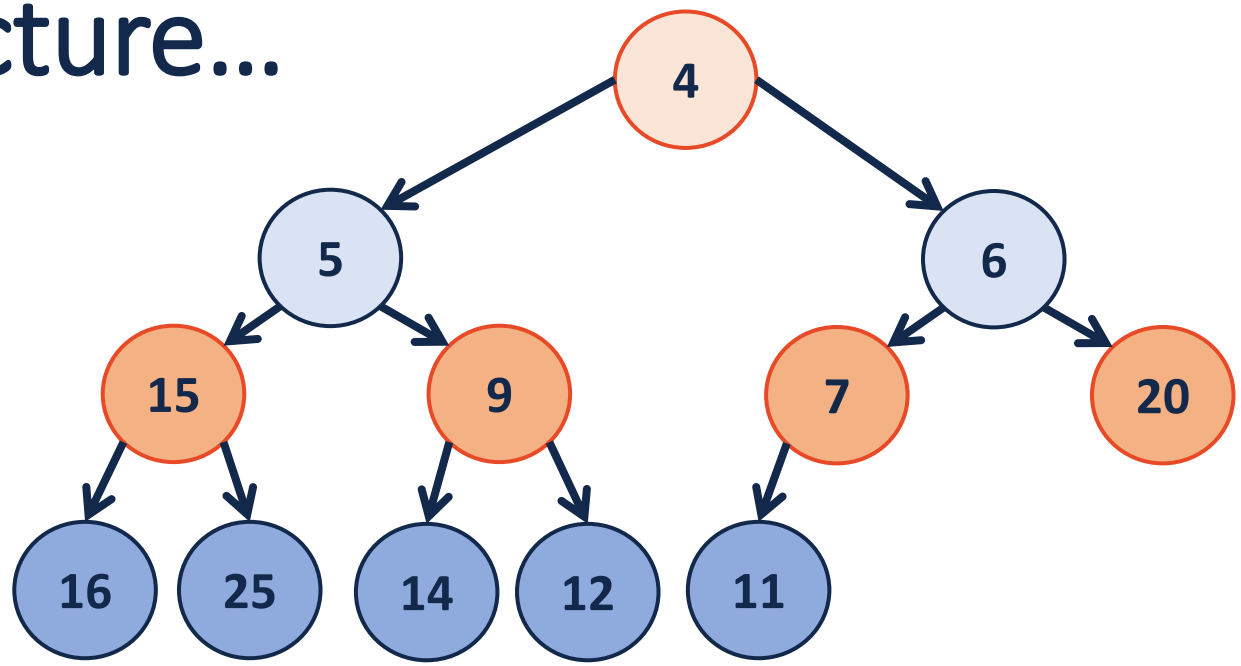


Another possible structure…



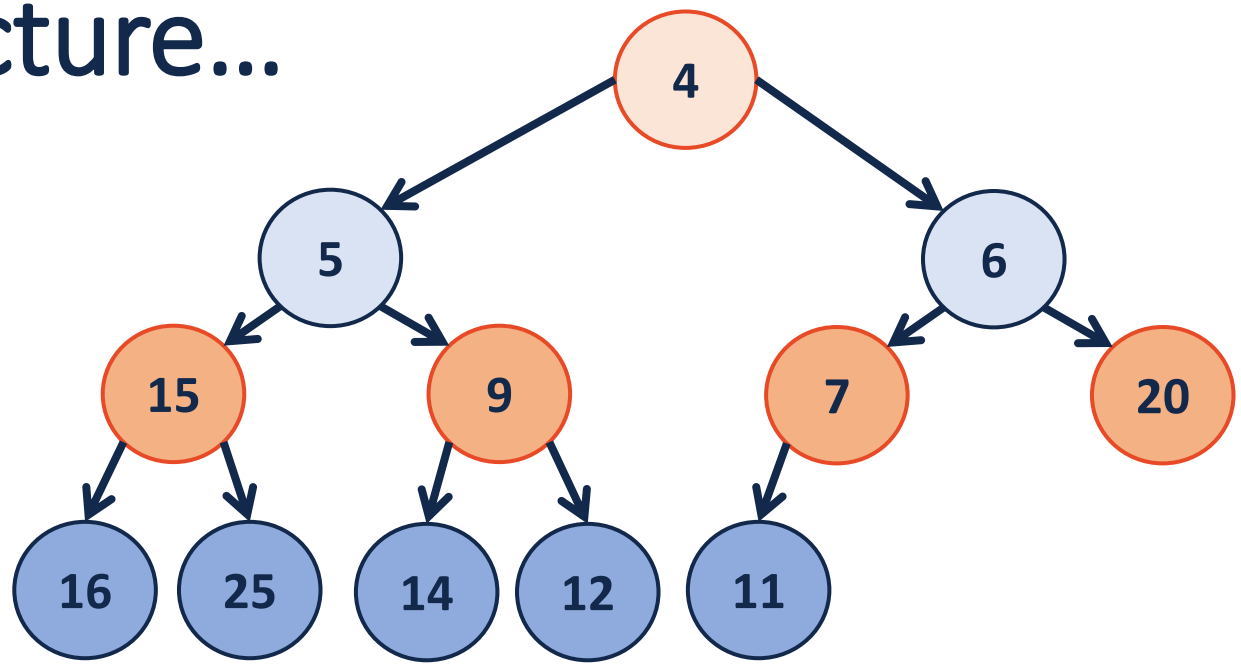
Another possible structure…

Binary Tree



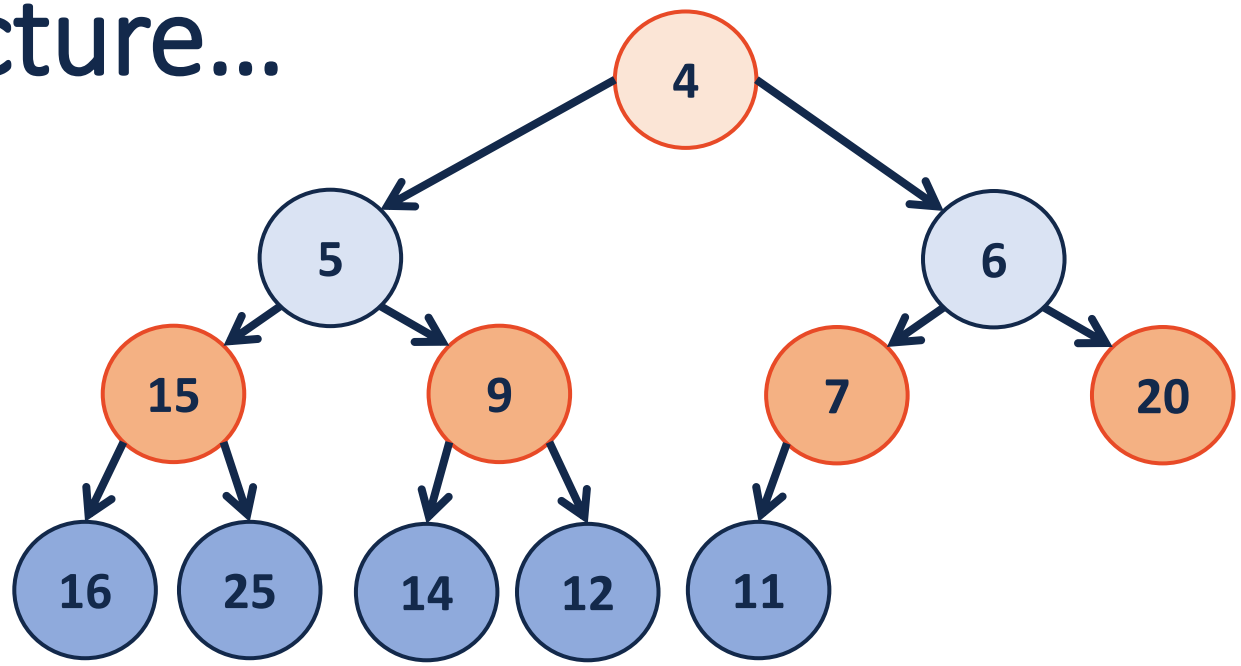
Another possible structure…

Binary Tree
- Complete



Another possible structure…

Binary Tree
- Complete
- !BST



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Binary Tree
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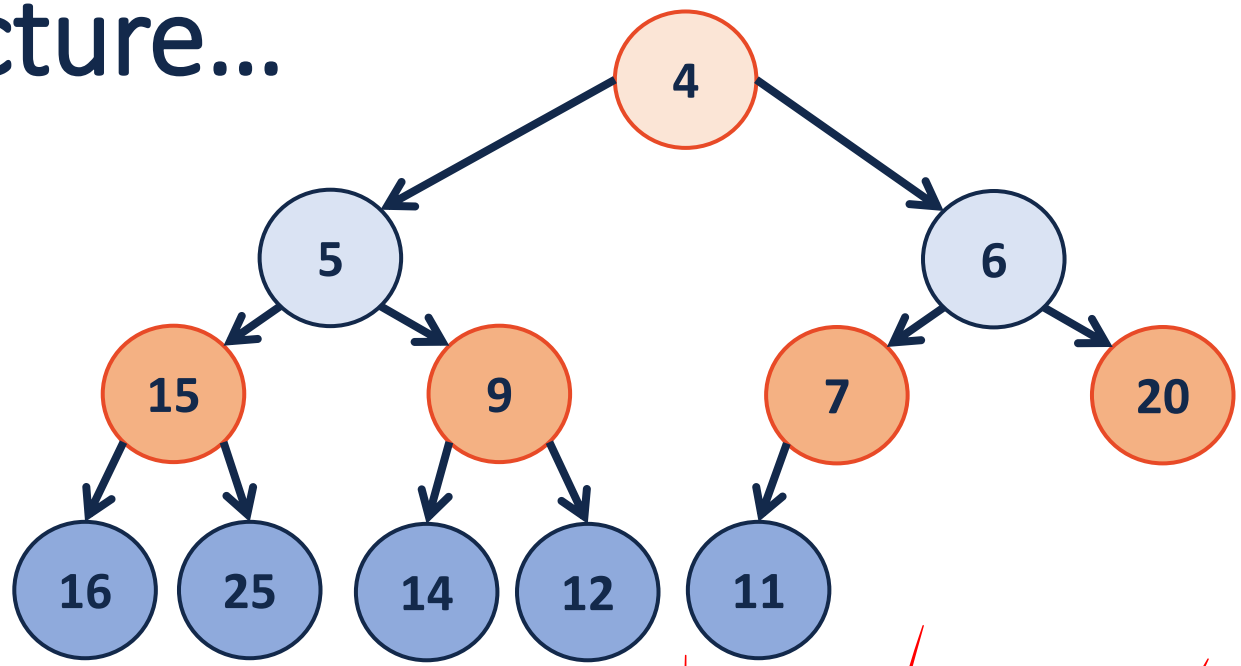

- All descendents of a node > the value of the node

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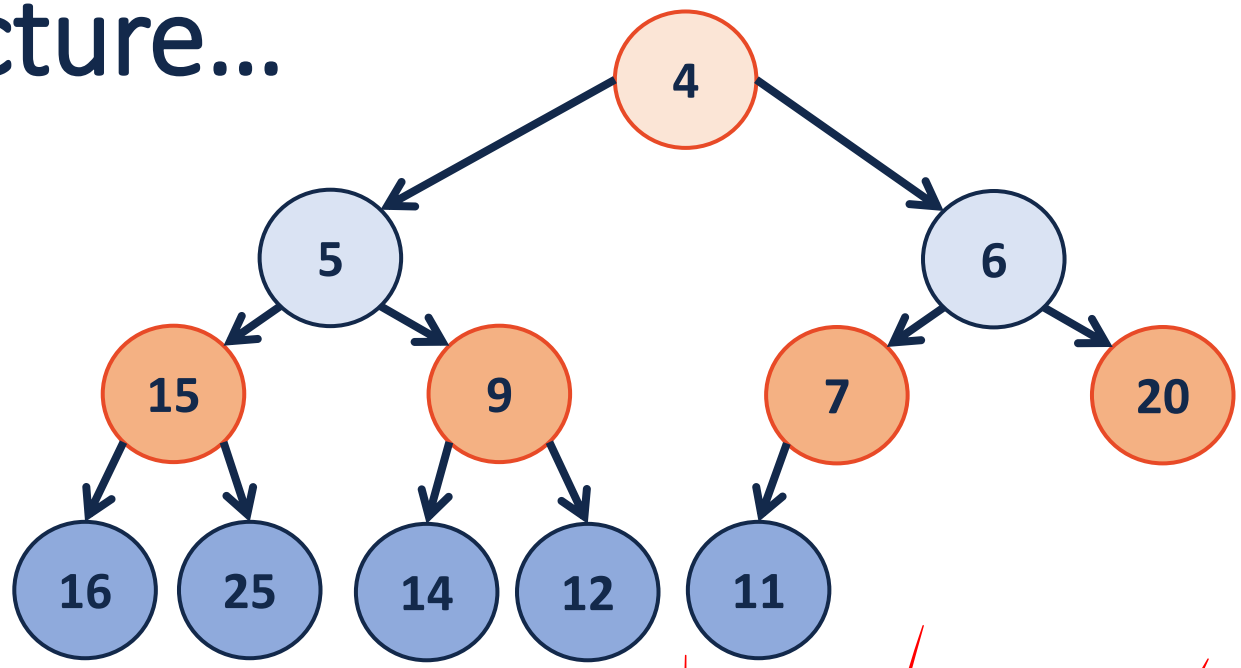


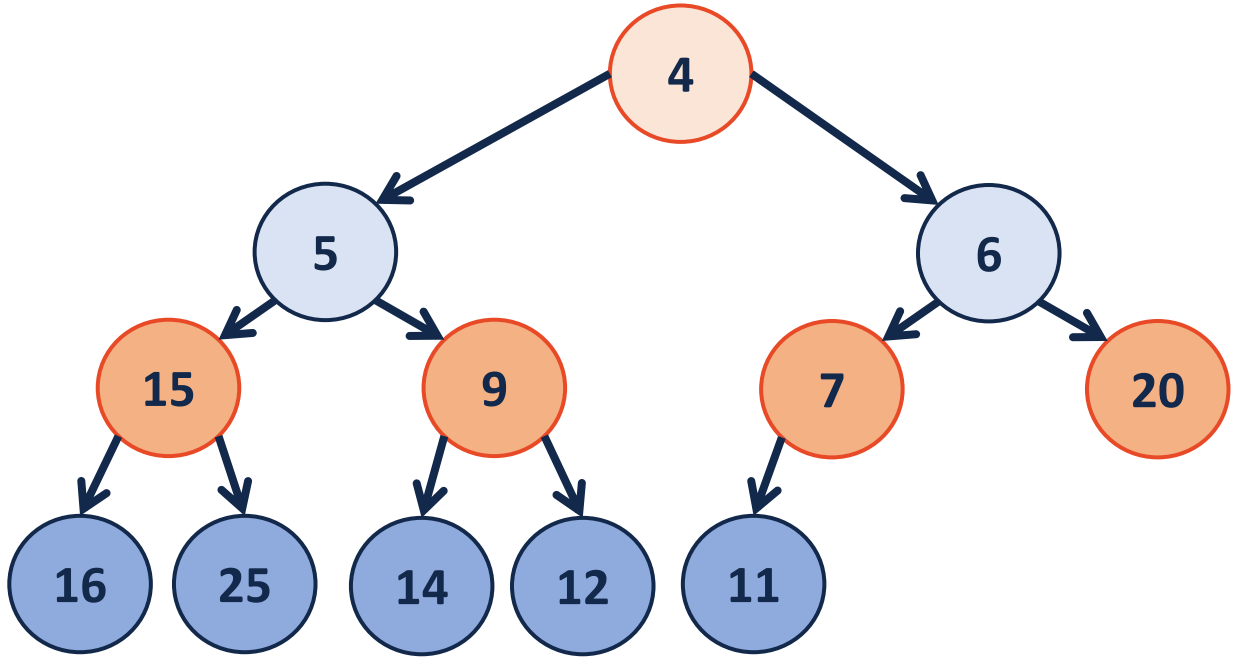
Binary Tree
- Complete
- !BST

- All descendents of a node > the value of the node
  - ↳ Root is the smallest element

(min)Heap

A complete binary tree T is a min-heap if:

- **T = {}** or
- **T = {r, T_L, T_R}**, where **r** is less than the roots of {T_L, T_R} and {T_L, T_R} are min-heaps.



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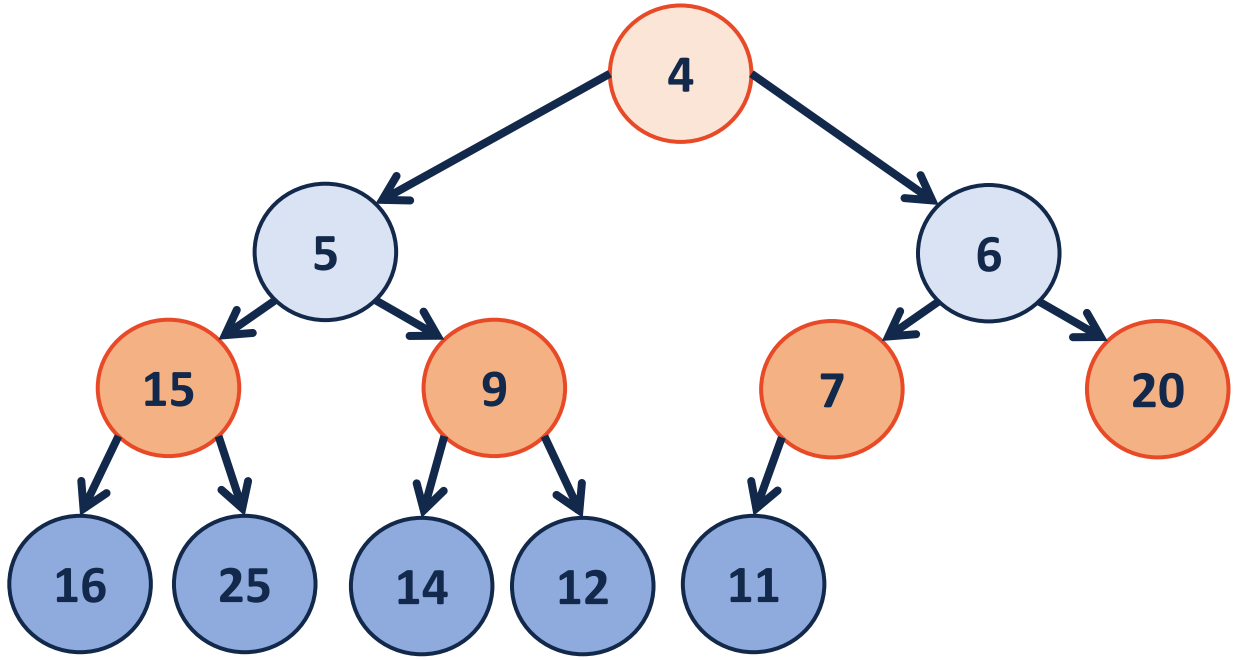
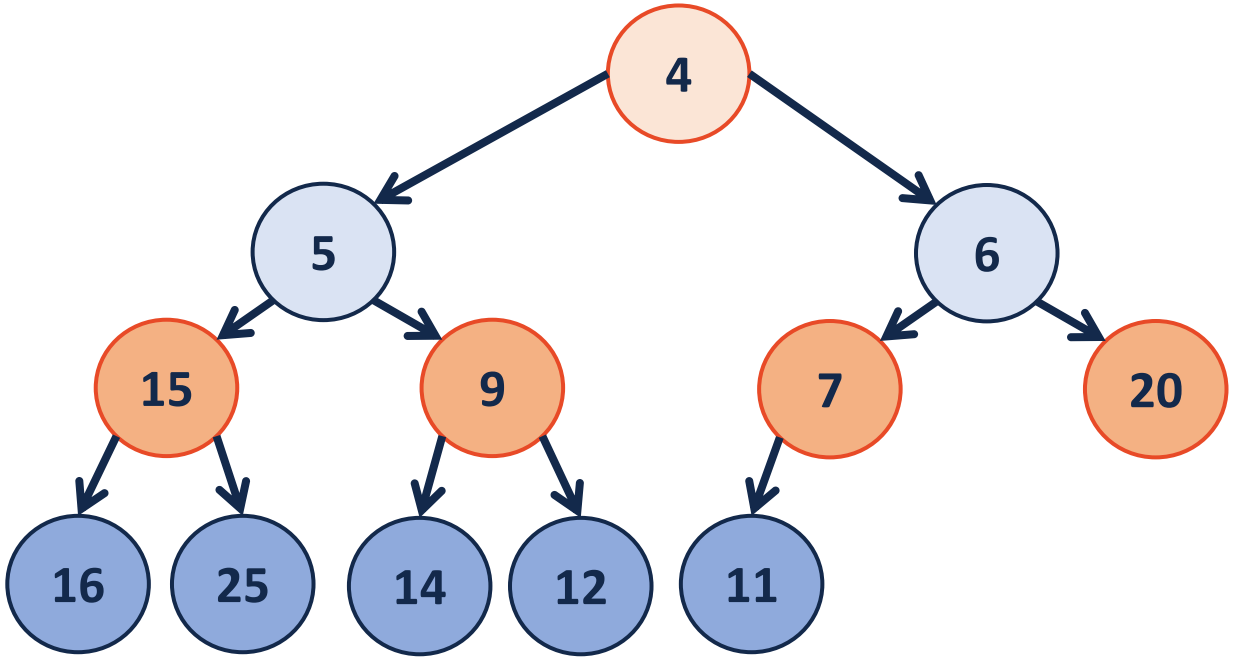


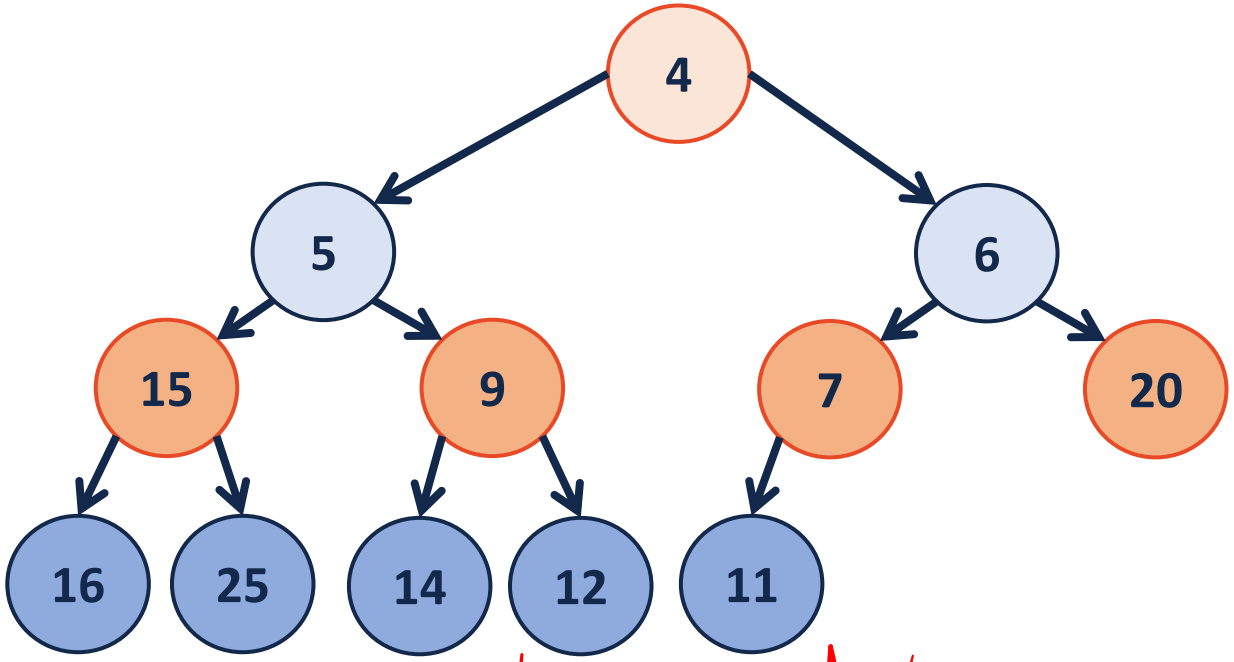
visual model of implementation

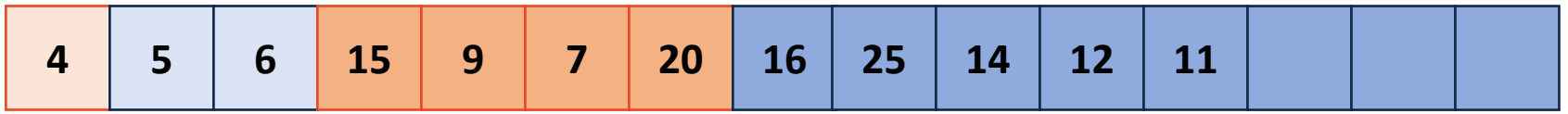
(min)Heap



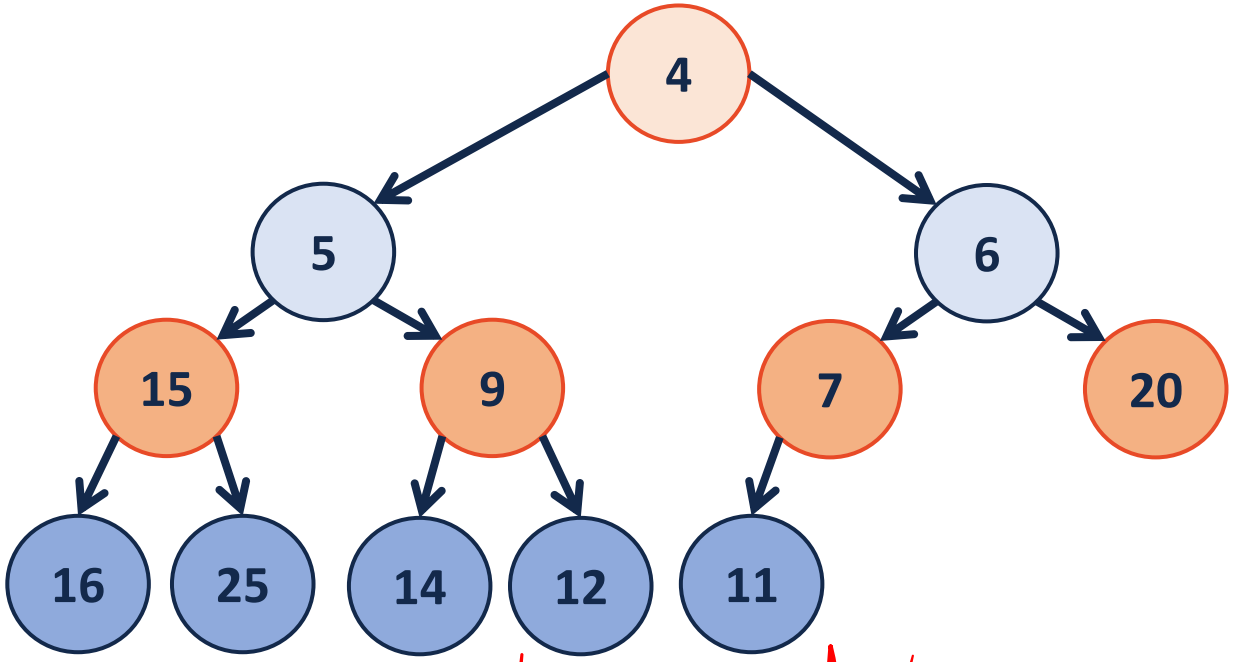
4	5	6	15	9	7	20	16	25	14	12	11			

(min)Heap





(min)Heap



visual model

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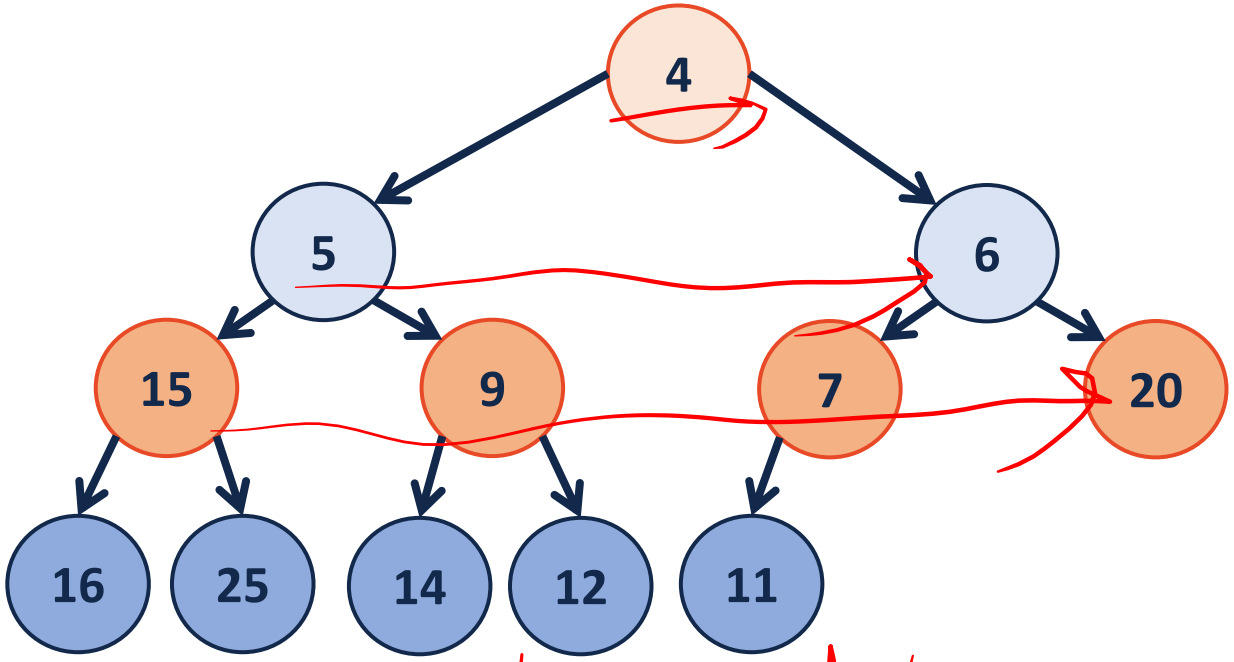
impl

(min)Heap



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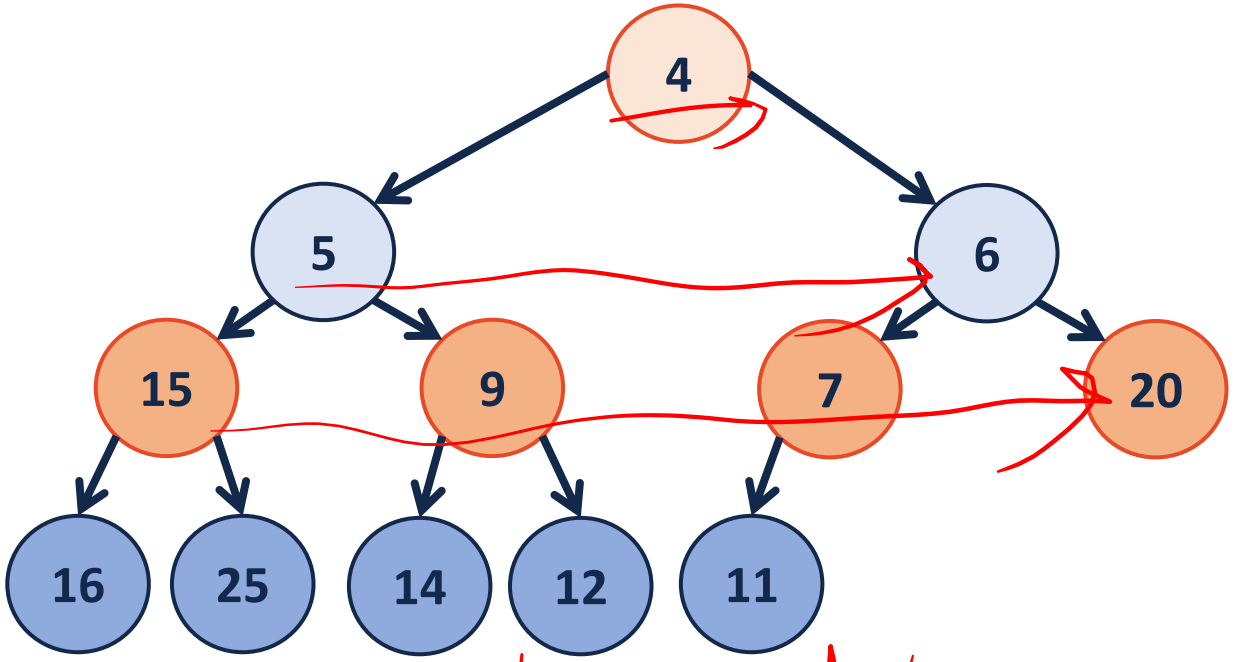
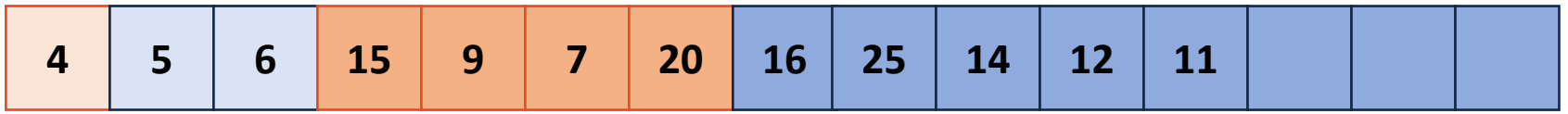
impl

(min)Heap

Level order TRAV



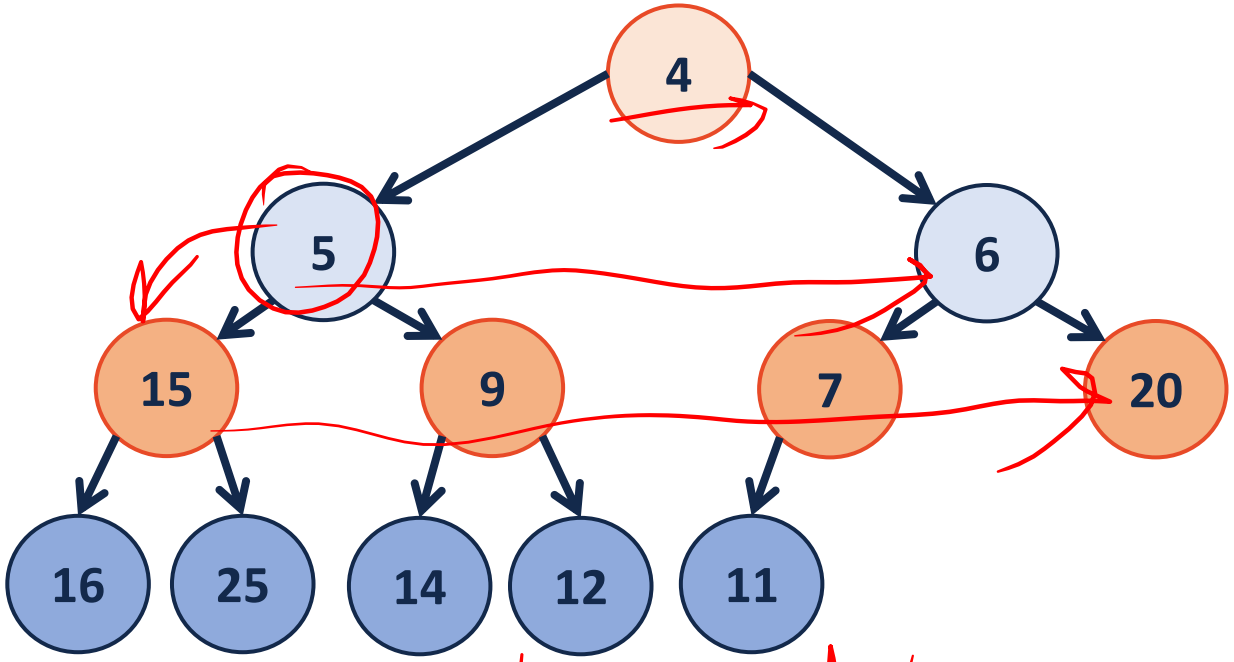
visual model



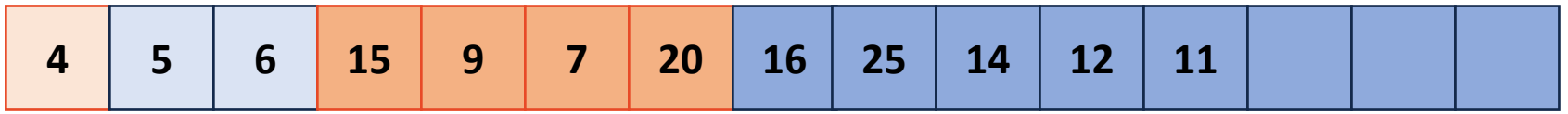
impl

(min)Heap

Level order TRAV



visual model



impl

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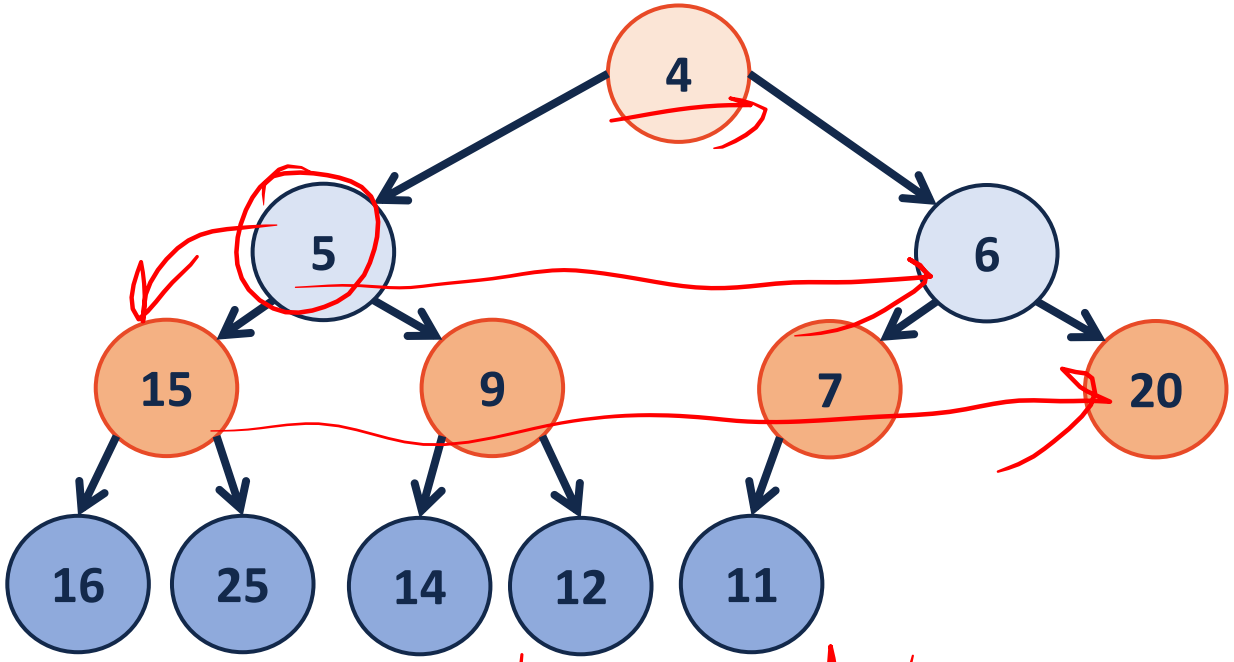
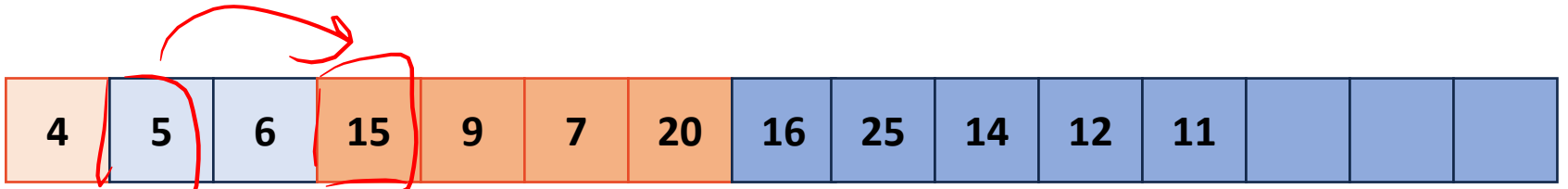
Level order TRAV



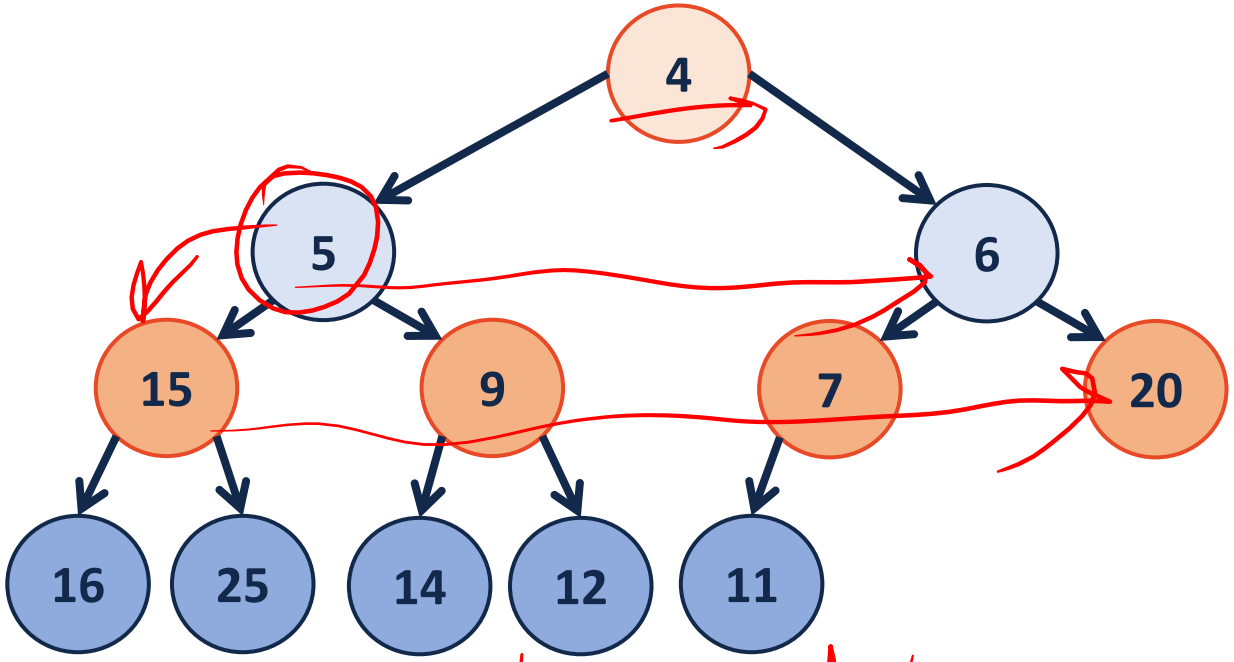
visual model



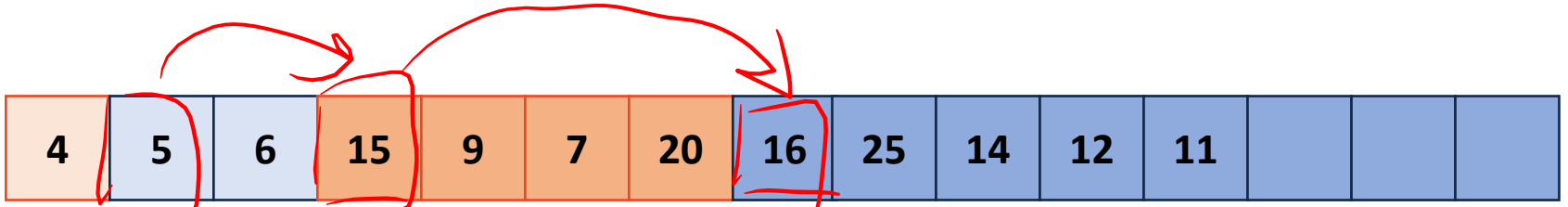
impl

(min)Heap

Level order TRAV



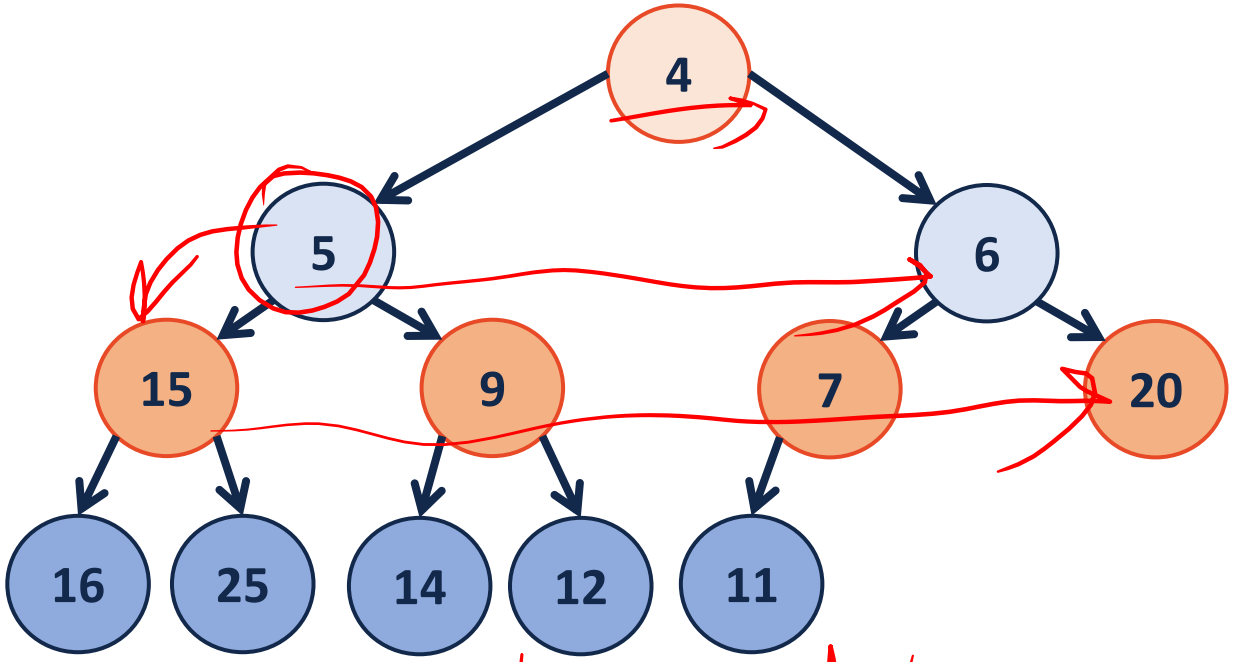
visual model



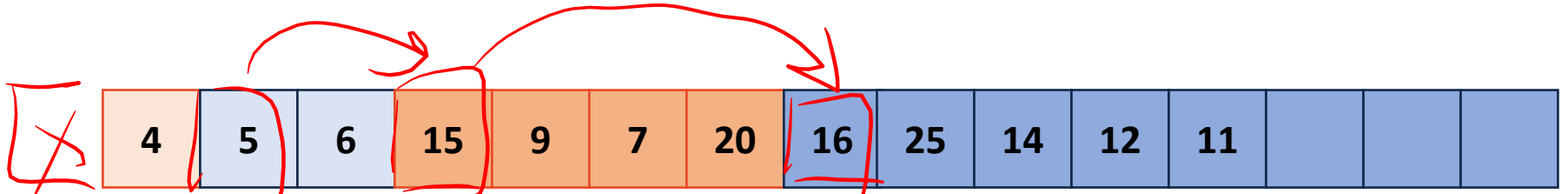
impl

(min)Heap

Level order TRAV



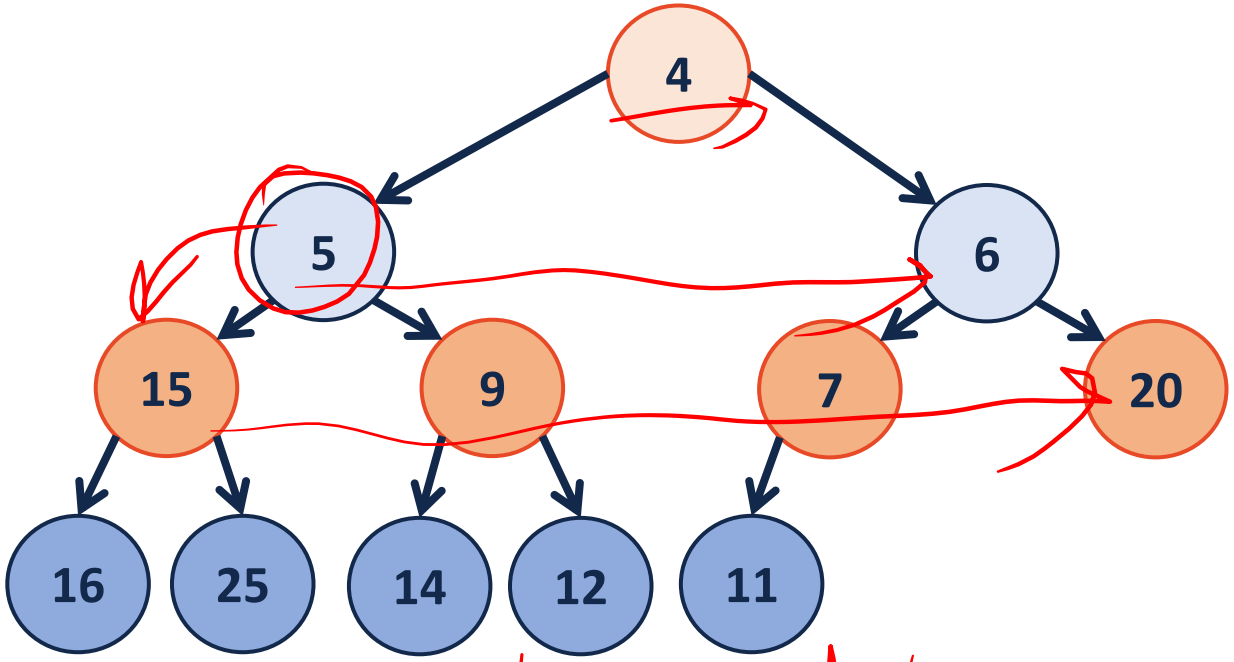
visual model



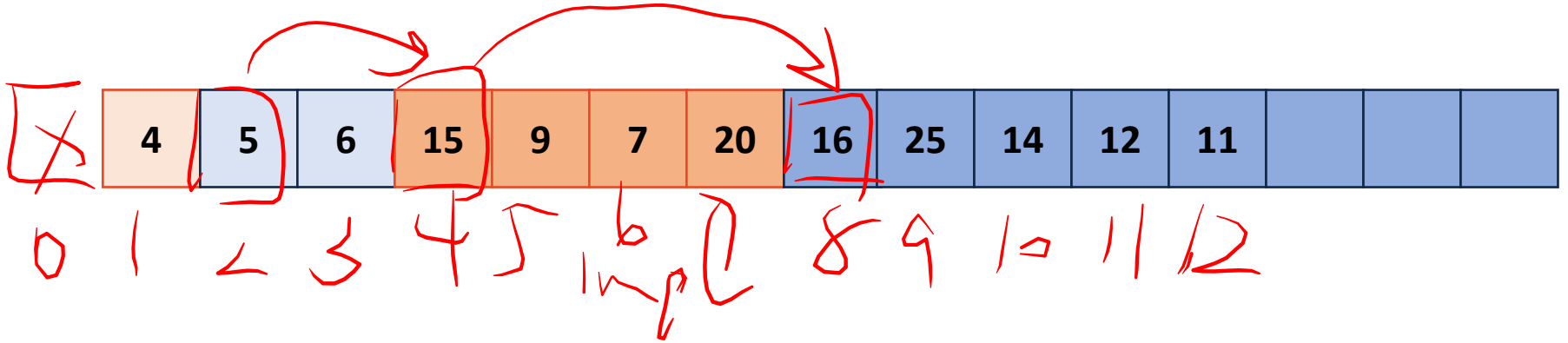
impl

(min)Heap

Level order TRAV



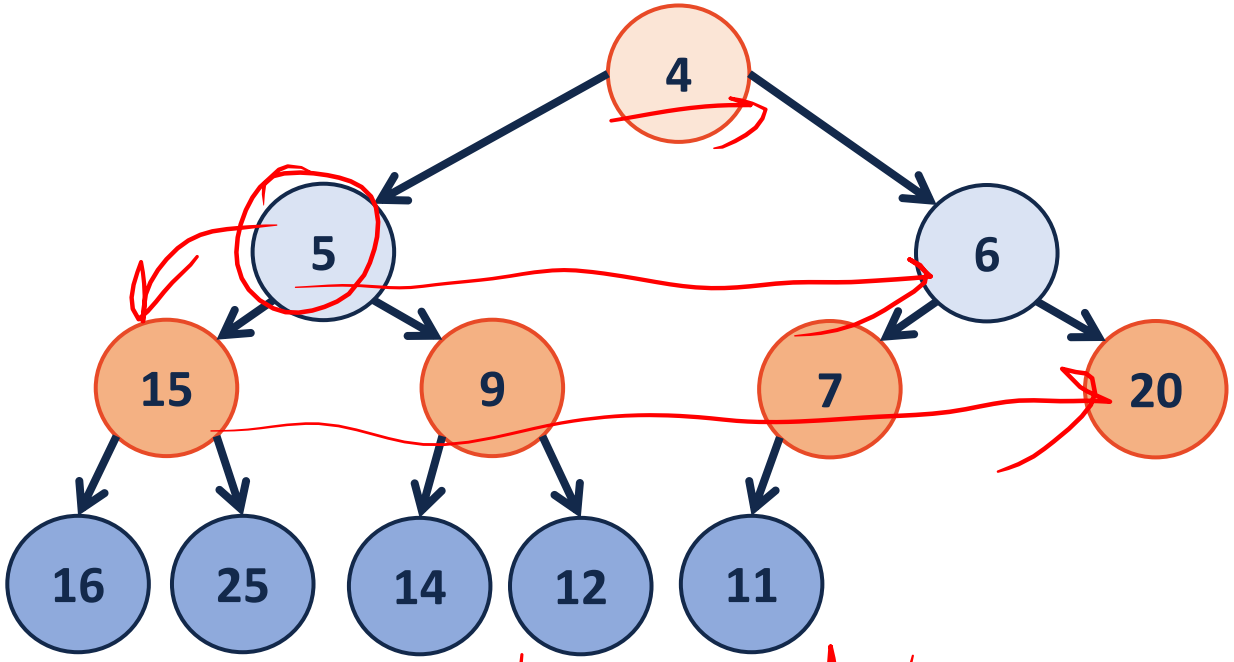
visual model



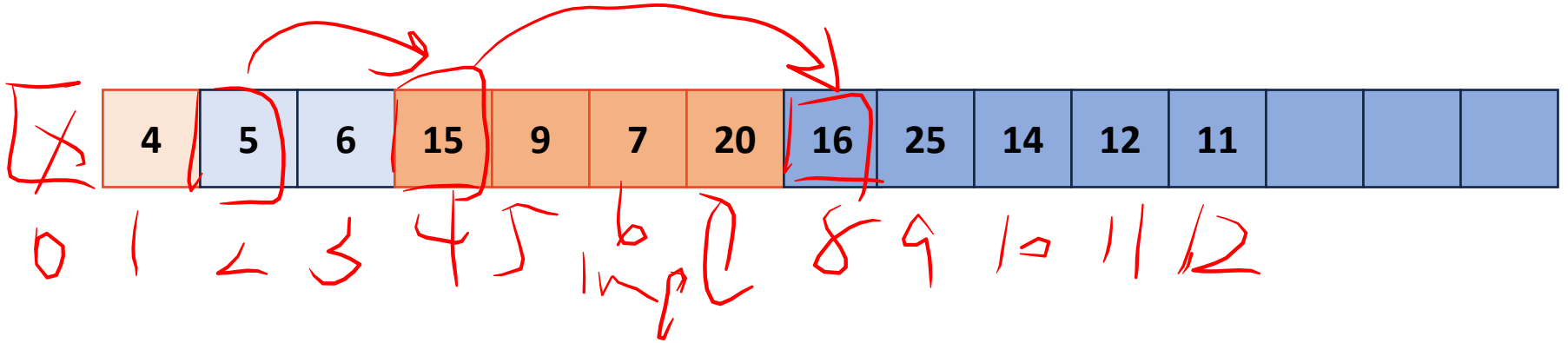
(min)Heap

Level order Trav

* In a heap impl [0] is not used



visual model

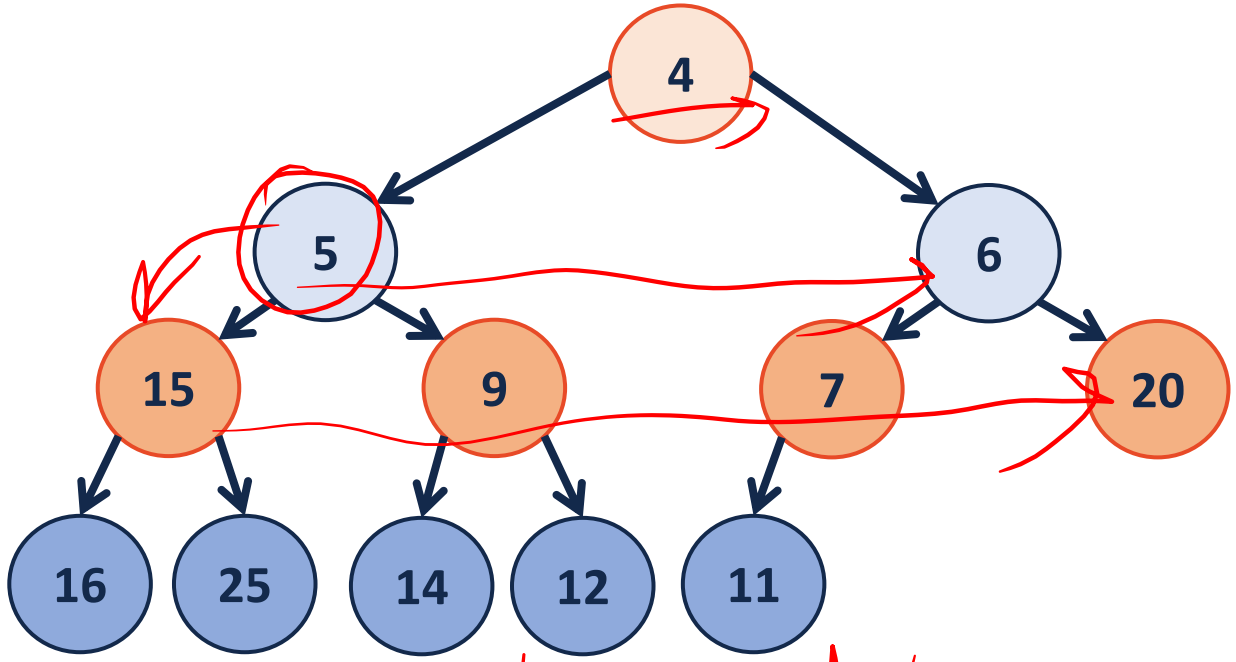


(min)Heap

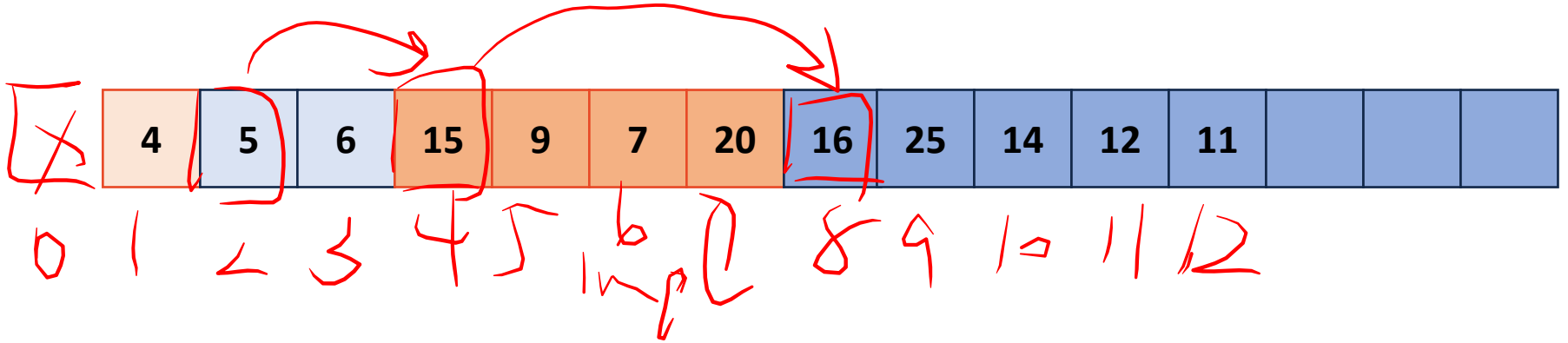
Level order Trav

* In a heap impl [0] is not used

Left child = 2 * index



visual model



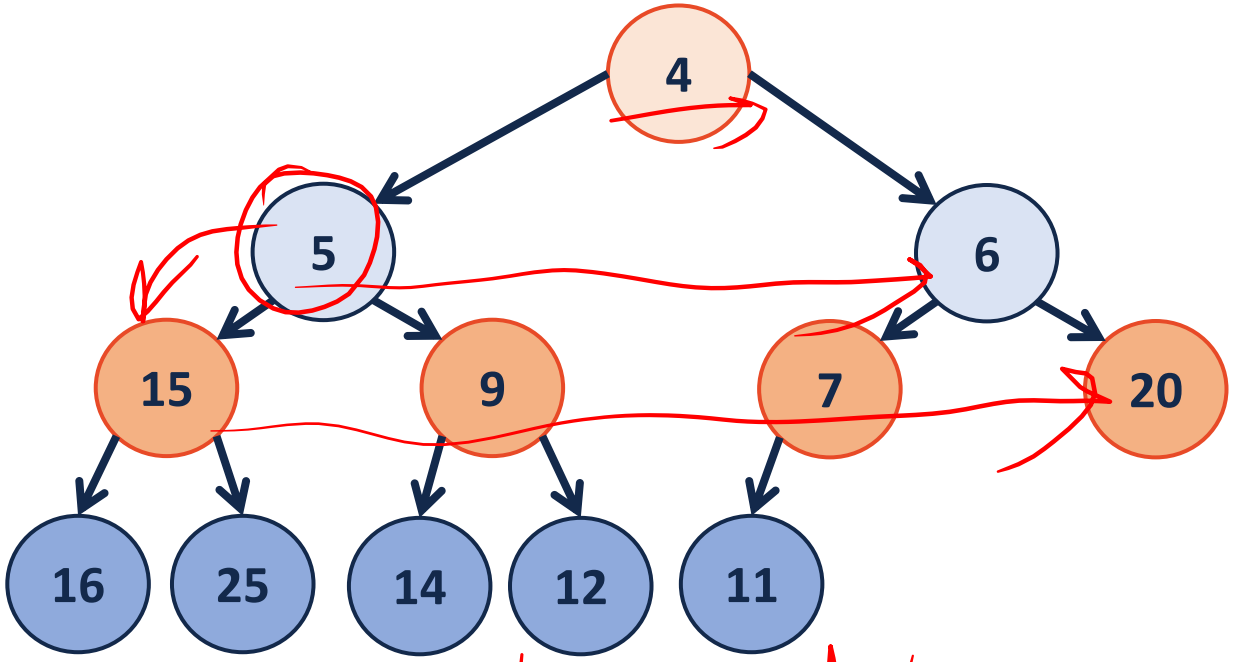
(min)Heap

Level order Trav

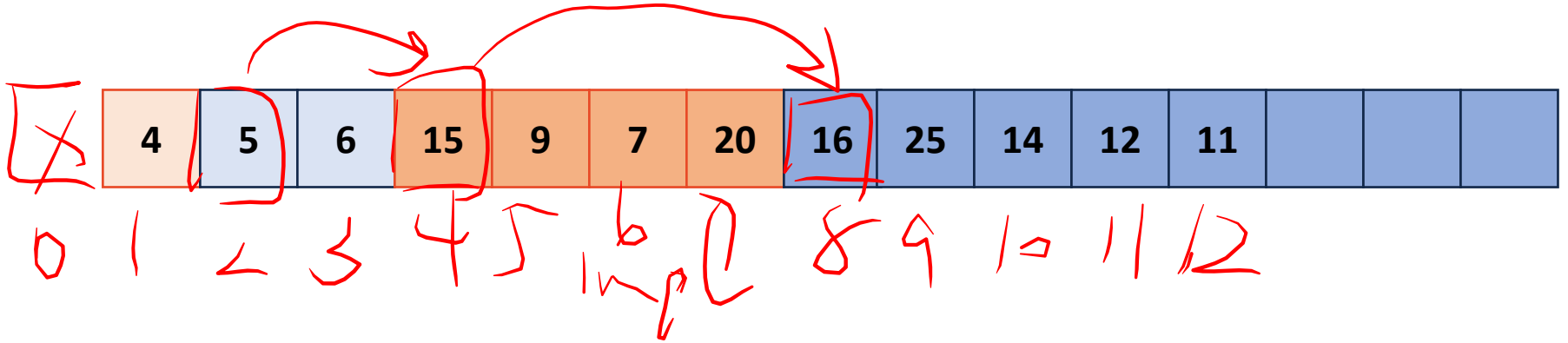
* In a heap impl [0] is not used

Left child = 2 * index

Right child = 2 * index + 1



visual model



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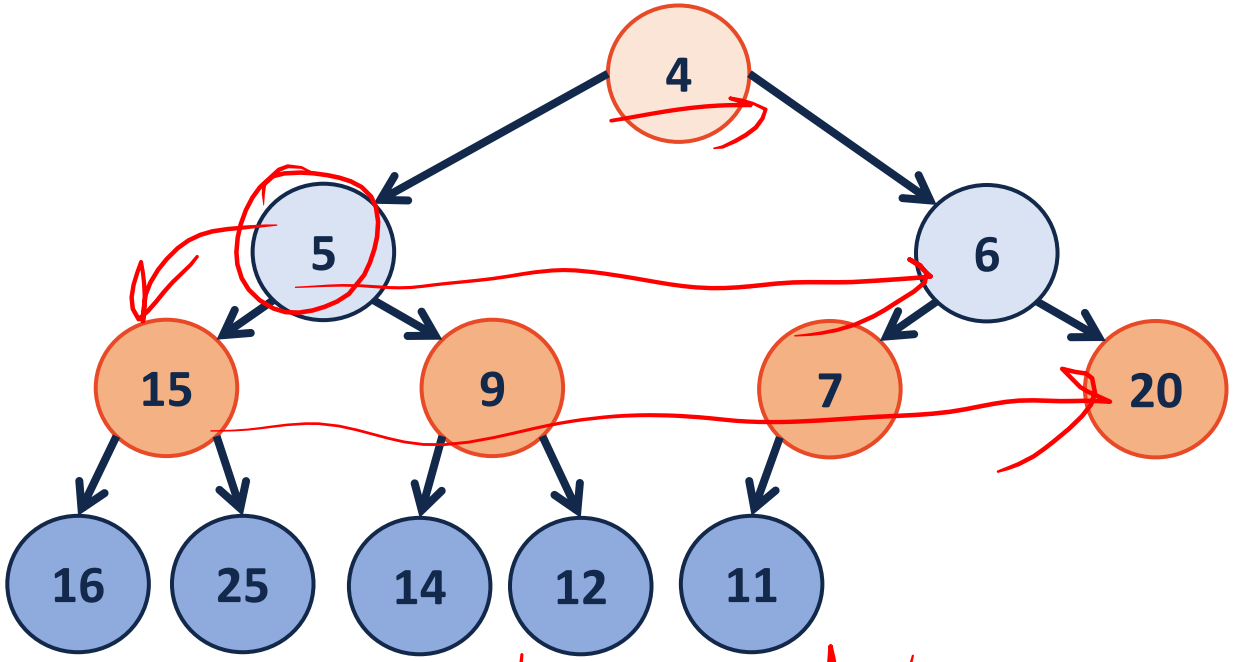
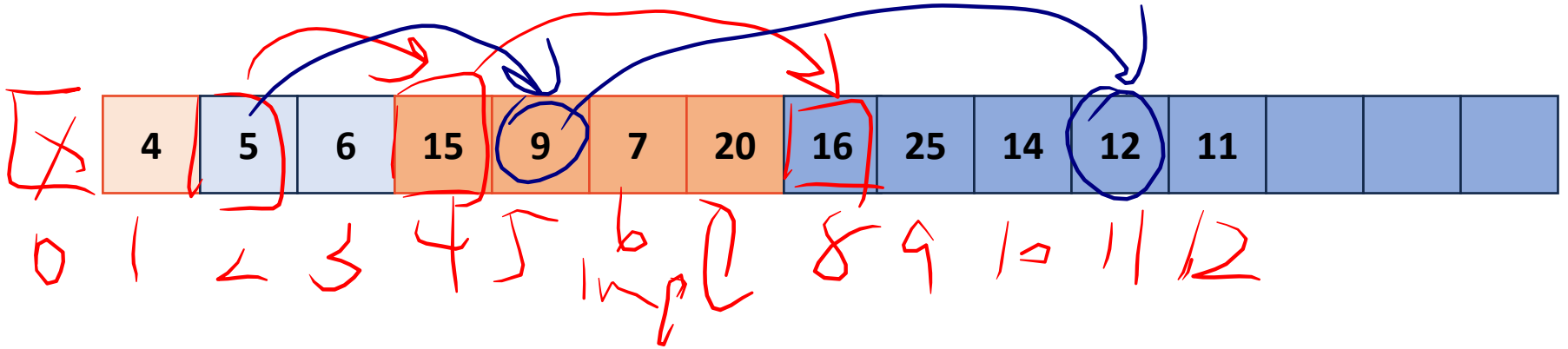


visual model



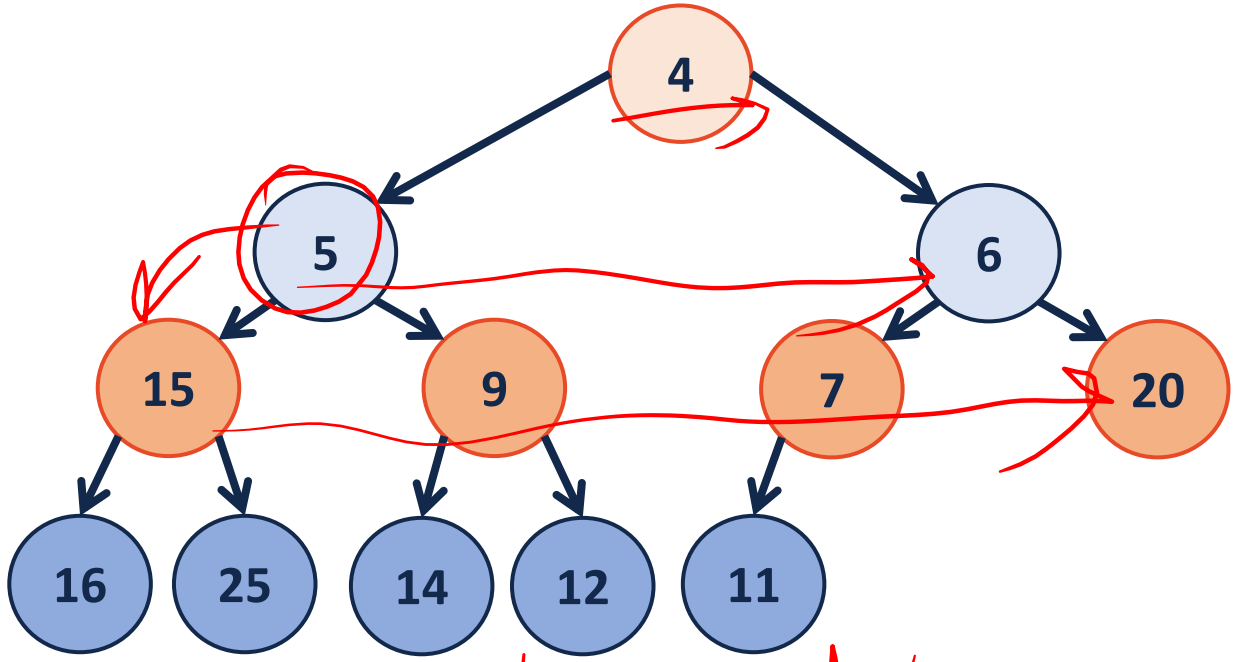
(min)Heap

Level order Trav

* In a heap impl [0] is not used

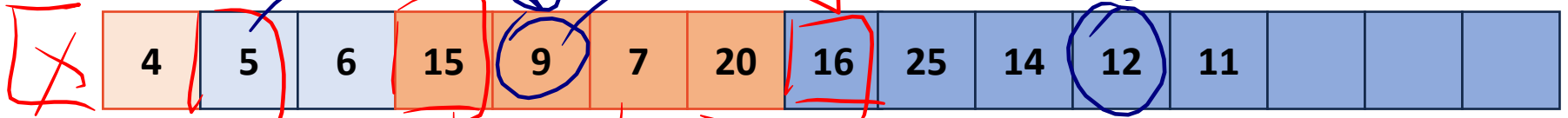
Left child = 2 * index

Right child = 2 * index + 1



Visual model

Parent = ⌊index/2⌋



0 1 2 3 4 5 6 7 8 9 10 11 12

impl

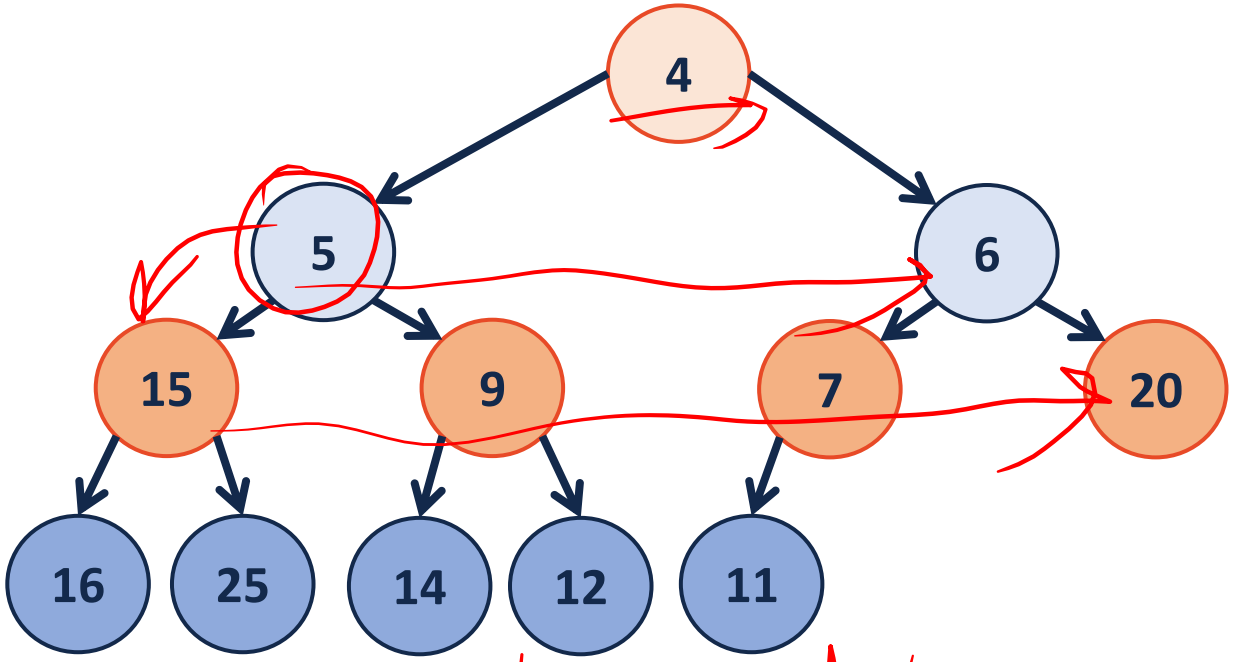
(min)Heap

Level order Trav

* In a heap impl [0] is not used

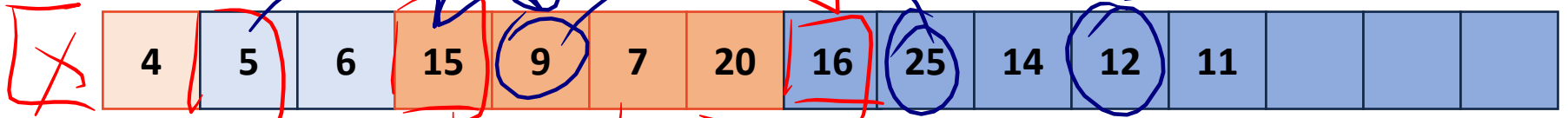
Left child = 2 * index

Right child = 2 * index + 1



Visual model

Parent = ⌊index/2⌋



0 1 2 3 4 5 6 7 8 9 10 11 12